# Efficiency, Risk and the Gains from Trade in Interbank Markets *

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**Abstract**

Bank-to-bank markets play a central role in liquidity provision. However, by propagating shocks between banks, they may be a source of aggregate risk. We develop a quantitative framework to capture the welfare trade-off between efficiency and risk accompanying interbank market integration. In the model, we derive analytical approximations for welfare that depend on features of the interbank network and few key elasticities that we estimate using microdata on bilateral asset positions for the population of German banks. Our findings indicate that the current level of integration improves welfare by .7% to 1%, mostly due to an increase in efficiency and dynamic benefits from openness and risk diversification.

**Keywords:** Bank Networks; Interbank Markets; Trade; Lender-of-last-resort

**JEL Codes:** E32, E44, E51, F19, G21

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# 1 Introduction

The financial crisis of 2007/2008 and the subsequent global recession highlighted the devastating effects of financial shocks propagating to the real economy. The crisis spread through the entire financial system in the US and abroad through financial linkages with a few large underwater financial institutions. Indeed, the level of interbank market integration – the degree to which banks are interconnected by bilateral lending and borrowing – emerges as a key determinant of whether shocks to large, "granular" banks lead to risk in the financial sector as a whole.

At the same time, interbank market *openness* also offers dynamic benefits. In the presence of small idiosyncratic funding shocks, interbank relationships enable banks to substitute between funding sources, resulting in less volatile funding costs and more stable financial conditions benefitting consumers and firms. In addition to this dynamic trade-off, market integration is accompanied by improvements in liquidity provision of banks, thereby lowering banks' funding costs and leading to static *efficiency gains* in terms of output and welfare.

Overall, the net effect of interbank market integration on aggregate volatility and welfare is ambiguous, and its size and direction depend on salient market features, such as bank heterogeneity, market concentration, and degree of participation. However, realistic, granular interbank markets are largely absent from quantitative macroeconomic models, limiting their ability to address this question. In contrast, network models of financial intermediation seldom incorporate the analysis of business cycles to study the welfare implications of interbank markets. From a policy maker's perspective, quantifying the gains or losses from market integration are crucial to better address market regulation and prevent future crises.

In this paper, we develop a quantitative framework that can account for a realistic interbank network while remaining tractable enough to study the implications of market integration for aggregate volatility and welfare. Specifically, drawing on recent models in international trade, our interbank network retains convenient aggregation properties despite featuring a large number of heterogeneous banks, funds as a homogeneous good, and various shocks. To address aggregate welfare, we embed the network into a New Keynesian DSGE model. We develop analytical expressions for the welfare gains from interbank trade that hold under arbitrary network structures and require only a small set of elasticities which we estimate using microdata of German banks.

A representative household may save through bonds and derives utility from holding real deposit balances across a discrete array of banks. Banks then transfer these savings to firms, which require a constant supply of bank loans (e.g, credit line at a fixed interest rate, agreed upon at the beginning of a period) to finance capital investments essential for production throughout each period. However, within each period (e.g., a month or quarter), the household's relative preferences concerning banks are subject to rapid, transient shocks. The constant loan demand by firms and fluctuating deposit supply over the period generate brief liquidity mismatches, subsequently motivating banks to trade homogeneous funds in the interbank market.[1] Consequently, our model can accommodate typical features of interbank

[1] Our model includes a central bank that can offer short-term funding to financially troubled banks as the financial system's lender-of-last-resort. This discount window reduces financial market volatility by imposing a limit on banks' funding expenses, even at a "penalty" rate above the market rate. We should note that this activity is distinct from more standard monetary policy.

markets, such as substantial gross debt positions between banks and structural net positions, without presuming that banks trade in differentiated goods.[2]

Across periods, banks encounter fluctuating funding costs due to idiosyncratic and potentially correlated shocks to transaction costs, loan demand, and deposit supply. Depending on market integration and concentration, such idiosyncratic shocks aggregate to a volatile, economy-wide interest rate spread over the bond rate. While we abstract from default and insolvency of banks for tractability – a very rare event in our data on the German banking sector – our model generates volatility on the intensive margin of bank size and interbank trade volume when banks adjust to transaction costs that stand in for a variety of more micro-founded frictions in banking such as monitoring costs, capital regulation, counterparty risk, intermediation spreads etc.. The remaining features of the DSGE component are well-studied (Calvo pricing, Taylor rule, etc.), resulting in a familiar dynamic system of equations. The primary distinction between our model and a workhorse DSGE model is that the natural interest rate depends on the now volatile interbank market spread over the bond rate, and its volatility influences the output gap, ultimately affecting aggregate volatility and welfare.[3]

At the core of our analysis lies the concept of gains from interbank trade, which refers to the static and dynamic welfare benefits or costs associated with varying degrees of interbank market integration.[4] First, we develop a sufficient-statistic welfare formula that only relies on the time series of aggregate interbank market openness and a few elasticities. Thus, the formula can be applied to alternative settings and does not impose steep data requirements. Under stronger assumptions on the stochastic properties of financial sector shocks, we derive a structural approximation to welfare that enables us to study the channels contributing to the gains from interbank trade. We decompose the gains or losses into a static and a dynamic component; the latter being driven by different financial sector shocks that interact with features of the interbank market such as the degree of market openness, bank concentration in the loan market, and bank-level Herfindahl indices for funding sources. In turn, the analytical approximation can be used to study the gains from interbank trade under various scenarios: changing the level of integration, counterfactual bank concentration, or alternative network structures.

We quantify our model using rich microdata from the German banking sector. To achieve this, we construct a database utilizing proprietary microdata from the Deutsche Bundesbank, which contains information on individual balance sheets and bilateral asset/debt positions for all banks domiciled in Germany. Our assessment of welfare gains is fundamentally dependent on the structure of the interbank market. First, a high concentration at the top of the size distribution suggests that shocks to large individual banks may propel aggregate economic fluctuations. Second, the German bank-to-bank market volume is substantial, with interbank liabilities constituting 29% of banks' balance sheets prior to 2007/08 and experiencing a 6 percentage point decline following the Great Recession. Our welfare formula establishes

[2] We draw upon Farrokhi (2020), who formulates a trade model encompassing homogeneous goods and bidirectional trade.

[3] We present the model in a New Keynesian setup for more generality; however, since financial market volatility enters as a real shock, our results easily extend to an RBC model.

[4] Arkolakis, Costinot, and Rodríguez-Clare (2012) demonstrate that the observed level of market openness functions as a sufficient statistic for the welfare gains from trade relative to autarky across a wide range of static models in international trade. Our paper not only replicates their insights regarding static welfare gains from trading funds but also extends their findings to encompass stochastic environments.

a direct connection between welfare gains and market integration, indicating potential welfare losses due to the reduction in market volume after 2008. Third, the interbank market displays a core-periphery structure, characterized by a small core of large banks with 100 or more connections and a periphery of smaller banks with no more than 10 interbank partners.[5] In the context of idiosyncratic liquidity shocks, the core-periphery structure restricts many banks' ability to substitute funding sources, while concurrently exposing the market to volatility from a limited group of large banks in the network's center.

During the 2007/08 US financial crisis, several large German banks suffered substantial losses in their US asset holdings and consequently reduced lending to domestic partner banks in the interbank network. We study this episode to provide reduced-form evidence for the propagation of shocks through the interbank network and to estimate the key elasticities of our framework. Viewed through the lens of the model, *indirectly exposed* borrowers encounter elevated funding costs, since liquidity must be financed through potentially more expensive means, such as deposits or equity. To formalize this mechanism, we devise a measure of indirect exposure of a bank's balance sheet to the crisis by interacting a bank's liabilities vis-à-vis directly exposed domestic lenders and their level of US bank asset holdings prior to the crisis. Following the turmoil of 2007/08, more affected banks increased interest rates on non-financial loans by approximately 23 basis points on average due to higher funding costs, and reduced lending to firms and consumers by up to 5% relative to less affected banks. Our findings indicate that banks with high indirect exposure significantly decreased interbank borrowing and financed a larger share of non-financial loans from their own sources (by around 6%). In order to credibly quantify our welfare formulas, we estimate the key demand and supply elasticities of our model utilizing the funding cost shock as plausibly exogenous variation in bank-level interest rates.

Within our framework, interbank trade is subject to transaction costs, which we model as "wedges".[6] We remain neutral regarding the origin of these wedges; instead, we recover them directly from the data as structural residuals for each quarter, interbank connection, and bank.[7] To achieve this, we reformulate the equilibrium relationships of the interbank market model such that the wedges are functions of the data. Combined with our estimated elasticities, we extract the set of wedges, ensuring that the model reproduces the data exactly in each quarter. We document that our recovered transaction costs are increasing during the financial crisis and that they are strongly correlated with measures of lender size and borrower risk. Subsequently, we compute the stochastic properties of the recovered wedges as components of our welfare formula. In the absence of wedges in interbank trade, the model approaches the free trade benchmark characterized by zero intermediation costs and minimal volatility. Infinite wedges result in financial autarky, wherein banks fund loans exclusively through volatile deposits, which increases funding costs due to a less efficient allocation of funds but avoids interbank market volatility. In practice, transaction costs lie between these extremes, giving rise to an efficiency-volatility trade-off in welfare terms.

Finally, we use our analytical welfare expressions to quantify the gains from interbank market trade.

[5] See Craig and Ma (2022) for additional details concerning the core-periphery structure of the German interbank market.

[6] We use the term wedges to refer to real barriers that cause differences in interest rates between banks. The other wedges in the model are deposit supply and loan demand parameters.

[7] Possible micro-foundations for such wedges proposed in the literature include asymmetric information (e.g., Babus and Hu (2017), Babus and Kondor (2018)), costly link formation (e.g., Craig and Ma (2022)), and intermediation spreads (e.g., Farboodi (2021)).

Moving the market from autarky to its current level of openness, we estimate welfare gains between 0.7% and 1% of quarterly consumption. Roughly 60% of these gains reflect a more efficient allocation of funds in the steady state (static gains), while the remaining 40% arise from banks using their interbank network to take advantage of temporary fluctuations in funding conditions, thereby reducing both funding costs and their volatility (dynamic gains). Under our baseline calibration, the diversification of funding sources made possible by an active interbank market outweighs the costs of risk exposure through interbank connections. However, dynamic gains can shrink substantially when the supply of funds in the interbank market is more elastic than firms' loan demand, bank concentration is high, or banks are only weakly diversified in their funding sources.

To better understand how salient market features shape these magnitudes, we compute the gains from trade under a range of counterfactual interbank market structures. Equalizing depositor preferences across banks generates substantially larger gains, whereas eliminating concentration in the loan market has only minor effects. A theoretical upper bound for the static gains is attained when steady-state transaction costs are set to zero. In that case, gains from trade can reach nearly 15% of quarterly welfare, with about 90% of the total accounted for by the static component. We also quantify the gains from trade under flexible prices, and find that the New Keynesian features of the model do not interact strongly with the volatility of the interest rate spread, which is a real shock analogous to a TFP shock in a standard RBC model.

The financial crisis in 2007/08 persistently reduced participation in interbank markets and lead to elevated credit spreads. Being such an outsized event for financial markets, we calculate the gains from trade allowing for two different steady states and volatilities before and after 2008. We estimate utility losses of approximately 0.28% of quarterly consumption due to structural changes in the interbank market since the Great Recession.

In our final counterfactual, we evaluate the benefits of the observed steady-state level of lender-of-last-resort intervention. In our model, central bank liquidity provision can mitigate short-term idiosyncratic shocks to banks' funding costs by effectively placing an upper bound on those costs. We estimate that the presence of this discount window raises welfare by 2.5% of consumption. This estimate should be interpreted with caution, however, as the model does not explicitly account for potential costs of central bank intervention, including misallocation, political costs, and other distortions.

Following earlier work by Bernanke and Gertler (1986), Kiyotaki and Moore (1997), and the financial accelerator of Bernanke, Gertler, and Gilchrist (1998), a substantial literature emerged to study how the financial system, in its role as an intermediary between household savings and firms' investment, generates credit frictions that amplify business cycle fluctuations. While the majority of these papers concentrate on the bank-to-firms or depositor-to-banks aspects of the financial channel, others, such as Cingano, Manaresi, and Sette (2016), Afonso, Kovner, and Schoar (2011) or Iyer, Peydró, da Rocha-Lopes, and Schoar (2014), empirically demonstrate that a significant portion of the decline in bank lending observed during the Great Recession can be attributed to the shrinking of interbank markets. Our paper offers a theoretical framework to analyze these channels jointly.

Our paper is also related to a recent literature on the spatial organization of banking. Oberfield, Rossi-

Hansberg, Trachter, and Wenning (2024) study banks' expansion across locations and branch-location choices, Morelli, Moretti, and Venkateswaran (2025) analyze geographic funding risk and market power in deposit markets, and Corbae and D'Erasmo (2024) study how geographic expansion affects the bank lending channel across time and space. These papers focus on branch geography, local deposit markets, and geographic diversification. By contrast, our paper studies the bilateral interbank network connecting banks and how its structure shapes liquidity allocation, aggregate volatility, and welfare. In one of the exercises below (see Table 4 and the surrounding discussion), we also briefly examine how the structure of the interbank network relates to spatial distance and other bank characteristics.

Among the few papers that investigate the impact of the interbank market on the macroeconomy, compromises are made in favor of tractability: Gertler, Kiyotaki, and Prestipino (2016) simplify the problem by assuming two types of banks, retail banks that obtain deposits from households and wholesale banks that borrow from retail banks. Piazzesi, Rogers, and Schneider (2019) and De Fiore, Hoerova, and Uhlig (2018) build on search models that presume a continuum of atomistic banks differentiated solely by the magnitude of the liquidity shock they receive each period. In this paper, we propose an alternative approach by adapting the Eaton and Kortum (2002) model of international trade to the banking sector. Trade models are naturally well-suited for this task, as they often feature a discrete number of heterogeneous agents and straightforward expressions for trade volumes and cost structures.

Hence, our paper relates to research in international finance and macroeconomics that studies financial markets with imperfect substitution in asset demand. Koijen and Yogo (2020) solve for bilateral asset holdings across countries using an asset demand system that takes a logit-form similar to our interbank model. Kleinman, Liu, Redding, and Yogo (2023) propose a quantitative, open-economy growth model in which assets are subject to extreme valued-distributed productivity shocks leading to imperfect substitution in asset demand. Our interbank framework shares this property, however, we propose a micro-foundation for imperfect substitution in funds demand based on short-term liquidity mismatches.

Drawing on earlier theoretical work by Allen and Gale (2000) and more recently Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015), Afonso and Lagos (2015), Ladley (2013), another extensive literature on banking networks delineates the conditions under which interbank market relationships emerge and yield a trade-off between an efficient allocation of funds and an increased risk of contagion (or default, volatility, etc.).[8] The paper closest to ours is Craig and Ma (2022) who document similar facts about the German interbank market. They then develop a network model of financial intermediation that features an endogenous, potentially volatile core-periphery structure emerging from scale effects in monitoring costs. However, the focus of this literature remains on the banking system itself. Instead of attempting to explain why the interbank market developed its current structure, we accept it as given and explore how this structure contributes to the efficiency and volatility of the aggregate economy, as well as how lender-of-last-resort policy can alleviate its adverse effects on welfare.

The remainder of this paper is structured as follows. Section 2 presents the data. Section 3 highlights some key aspects of the German banking system. Section 4 introduces the model. Section 5 derives the welfare approximation to market integration and lender-of-last-resort intervention. Section 6 empirically

[8] See, for example, Farboodi (2021), Babus and Hu (2017), Babus and Kondor (2018), Chang and Zhang (2021), Üslü (2019)

estimates the effects of the 2007/08 financial crisis on the German interbank market and calibrates the model. Section 7 discusses the quantification exercises. Finally, Section 8 offers concluding remarks.

## 2 Data

In this section, we describe how we combine several proprietary and confidential datasets provided by the Deutsche Bundesbank, the German central bank (within the system of Eurozone central banks) and supervisory entity for the German financial market. The resulting database contains detailed, quarterly information on the balance sheets of monetary financial institutions (MFI), borrowing and lending connections with other German MFIs and other information such as the type of banking group for the universe of German MFIs covering 2004-2016. Our dataset contains 1552 distinct MFIs.[9,10]

For the construction of this database, we start with the MFI Masterdata (MaMFI[11]), which includes information such as MFI type and headquarter location on a monthly basis and allows us to account for mergers and acquisitions in all other datasets.[12] Next, we add the Monthly Balance Sheet Statistics (BISTA[13]) that covers broad balance sheet positions of the universe of German domestic MFIs at the end of each month. To better account for each MFI's business model, we complement the broad loan categories in the BISTA with a more detailed breakdown of loans by sectors, borrower type and maturities from the Quarterly Borrowers' Statistics (VJKRE[14]).

Finally, the Credit Register of Loans (Millionenkreditregister) provides MFI-level supervisory information on all loans that exceeded 1 million Euro (1.5 million Euro before 2014) in each quarter. The dataset contains various loan types, but most importantly, it covers the vast majority of loans at all maturities between domestic MFIs.[15] Due to this censoring, we scale individual positions between two MFIs such that the total MFI lending in this dataset is consistent with domestic MFI loans in the BISTA data. Bilateral loan amounts allow us to capture the interbank market network in its full granularity over many years throughout our empirical and model-based analysis.

We complement our main database with interest rate data for a sample of 200 to 240 MFIs representing

[9] Monetary and financial institutions are defined by the European Central Bank as "financial institutions whose business is to receive deposits and/or close substitutes for deposits from entities other than MFIs and, for their own account (at least in economic terms), to grant credits and/or make investments in securities", https://www.ecb.europa.eu/stats/financial_corporations/list_of_financial_institutions/html/index.en.html. However, for the remainder of the paper we will use the terms "MFI" and "bank" interchangeably.

[10] There are 9 different banking groups in the Deutsche Bundesbank statistical definition. The largest among them are credit banks, state banks, savings banks, mortgage banks and cooperative banks.

[11] See Stahl (2018) for a description of the MaMFI dataset.

[12] In order to avoid sudden discontinuities in the balance sheet size and its subcategories, we treat MFIs before and after a merger or acquisition as a single entity and add up the relevant categories for the MFIs participating in the M&A.

[13] See Beier, Krueger, and Schaefer (2017) for a description of the BISTA dataset.

[14] See Beier, Krueger, and Schaefer (2018) for a description of the VJKRE dataset.

[15] We find that a comparison of liabilities and assets with MFIs in the balance sheet data and aggregated loans in the credit registry line up very tightly. This suggests that the reporting threshold of 1 million Euro (1.5 million Euro before 2014) is not a serious concern for lending between MFIs.

65% to 70% of total lending activities in the banking sector.[16] The Monthly Interest Rate Statistics (ZISTA[17]) reports average interest rates on loans and deposits vis-à-vis firms and households and their respective volumes. For most of our analysis, we use the average interest rate on outstanding loans for each MFI that we calculate as the average interest rate across all maturities and borrower types weighted by their respective loan volumes.[18]

Lastly, we measure the exposure of each domestic MFI to the US financial market before the Great Recession using the assets and liabilities of German banks vis-à-vis US residents (including banks) available in the External Position of MFIs (AUSTA[19]). This dataset contains the gross foreign positions by partner country of the 80 largest German banks and their foreign branches on a monthly basis, covering 90% of the value of all foreign positions involving a German MFI.

## 3 The German Interbank Market

Here, we use our database to introduce several facts about the structure of the German interbank market that inform our model choices in Section 4 and the quantification of our welfare formula in Section 5. A few of these facts coincide with work by Craig and Ma (2022). However, we extend the time frame of their analysis and show some new facts. First, banks serve as the central players in the German financial market. As of December 2016, their combined balance sheet of 7.8 trillion Euros (approximately 250% of GDP) exceeds the sizes of investment funds (1.9 trillion in assets), insurers (2.2 trillion in assets), and other financial services providers by a wide margin.

Second, interbank activity is an important source of banks' short-term and long-term funding. Panel (a) of Figure 1 reveals that the combined liabilities in the interbank market as a share of the aggregate balance sheet fluctuates between 20% and 30% throughout the sample period. Over two-thirds of bank-to-bank liabilities have maturities longer than overnight, challenging the common perception that interbank markets primarily serve for very short-term funding. Third, the 2007/2008 crisis had persistent effects on interbank market activity. Not only did aggregate interbank liabilities fall persistently by around 6 percentage points, as shown in panel (a), banks compensated this shortfall by relying more heavily on own resources. Panel (b) displays the share of loans to firms and consumers that banks can fund with deposits or equity, as opposed to borrowing on the interbank market.[20] Our welfare formula in Section 5 suggests that this increase in the own share leads to persistent welfare losses, which we quantify in Section 7.

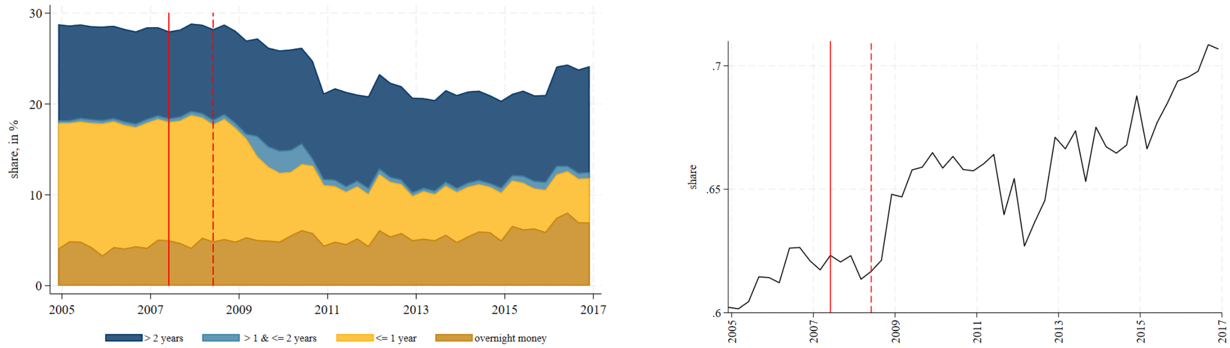
[16]The sample is designed to be representative and, at the same time, capture a large share of the financial sector. The first stratification criterion is a combination of state and banking group in order to capture regional and institutional heterogeneity. Within each of the strata, the largest banks in terms of lending were selected. Throughout our analysis we tried to address this selection bias whenever possible.

[17]See Beier and Bade (2017) for a description of the ZISTA dataset.

[18]Another adjustment to our data comes from the 2010 German Accounting Modernization Law (see Bundesbank (2010) for a description of the Accounting Modernization Law), that, among other changes for firms, caused a break in banks' balance sheet sizes. Generally, the most prominent change was the introduction of a fair value of the trading portfolio, partly adapting to the International Financial Reporting Standards (IFRS). While this change affects only larger banks with a trading book, it was left to their discretion at what point in the course of 2010 they applied the new rules in monthly balance sheet statistics (BISTA). We mitigate this circumstance by deducting the derivative exposures of the trading book from total assets.

[19]See Gomolka, Munzert, and Stahl (2019) for a description of the AUSTA dataset.

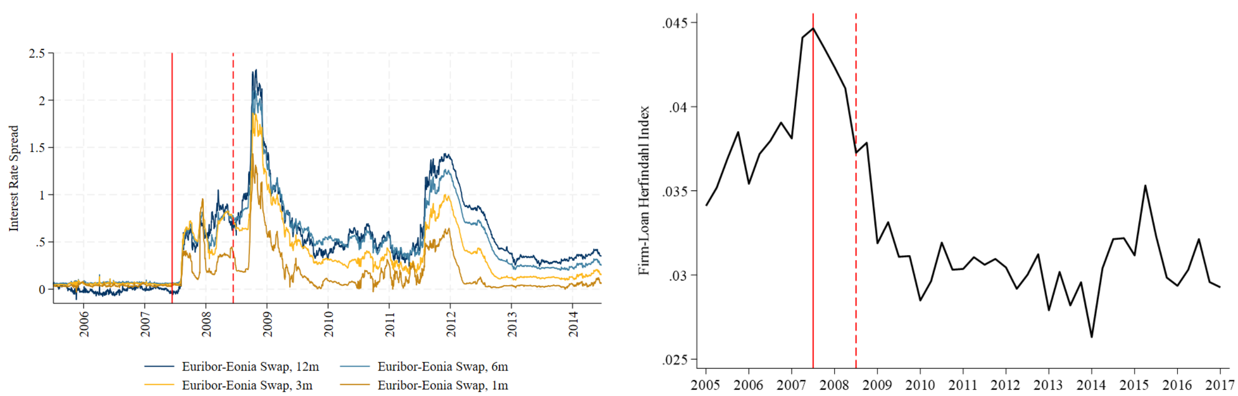
[20]The aggregate own share in panel (b) is not the same as one minus the interbank liability share. The reason is that the denominator in panel (b) captures all assets excluding interbank assets, a measure of loans to the real economy.




(a) Aggregate Interbank Liabilities

(b) Aggregate Own Share

Figure 1: (a) Share of interbank liabilities in total liabilities by maturity. (b) Aggregate Own Share defined as $1 - \frac{\text{Interbank Liabilities}}{\text{Assets}-\text{Interbank Assets}}$. Red lines indicate 2007$Q$2 and 2008$Q$2 as the starting period for the financial crisis. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, BISTA, 2004m12 - 2018m12, own calculations.



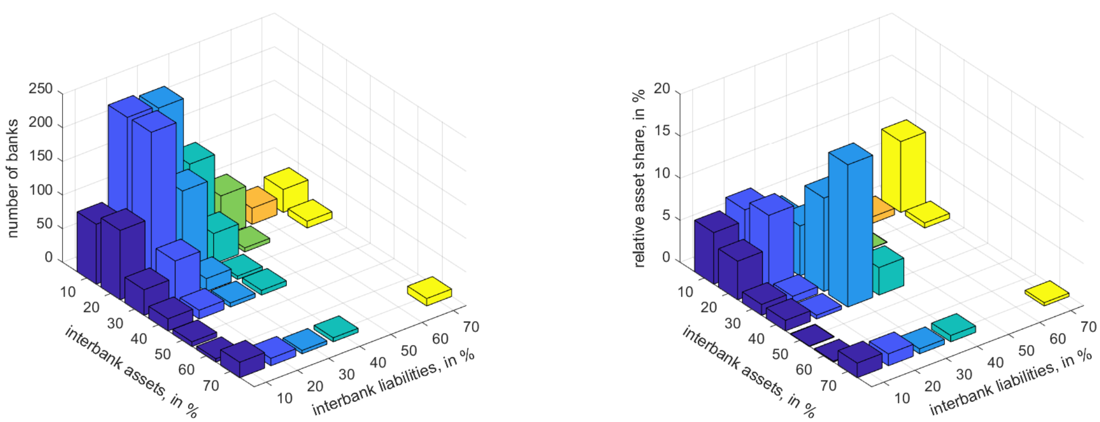

(a) Interbank Spread

(b) Herfindahl Index of Loan Concentration

Figure 2: Figure (a) shows the interbank credit spread at different maturities, computed as the difference between the Euribor rate and the EONIA swap index. The Euribor is an average of the unsecured interbank rate at which Eurozone banks are willing to lend funds to each other. The EONIA Swap is a financial instrument commonly used to hedge against overnight moves of the unsecured interbank rate. Figure (b) shows the Herfindahl index of bank concentration in loans over our sample period. Red lines show 2007$Q$2 and 2008$Q$2 as the starting period for the financial crisis. Source: Deutsche Bundesbank

In addition, interbank credit spreads remain elevated after the crisis. Figure 2a displays the evolution of the spread at various maturities, calculated as the difference between Euribor and Eonia Swap rate. After the turmoil of the 2007 financial crisis and the European debt crisis, the spread stabilizes, but it does not return to pre-crisis levels, particularly at longer maturities. Through the lens of our model, we interpret the fluctuations in credit spreads as (correlated) shocks to counter-party risk or monitoring costs, which we will quantify as stochastic transaction costs in our welfare exercise.

Fourth, a prominent feature of the German banking market is its high degree of concentration. The three largest banks control approximately 30% of all assets (incl. interbank assets), and the 200 largest banks (out of more than 1500) control around 80%.[21] Figure 2b plots the Herfindahl index of bank concentration in the German loan market over our sample period. In the run-up to the financial crisis, loan concentration rose dramatically but dropped significantly during the crisis, suggesting that mostly large lenders contracted lending in this period. Accounting for the high degree of granularity in our model is key to contagion risk, as shocks to large lenders can create aggregate volatility in the financial sector, as in Huber (2018), and similarly to how large firms drive economic fluctuations in Gabaix (2011).




(a) Bank Count

(b) Asset Share

Figure 3: Share of interbank assets and liabilities on the balance sheet, expressed in percentages. The vertical axes in Figure 3a display the number of banks within each bin. Figure 3b presents the share of total MFI assets represented by banks within the bin. Bins with fewer than three observations are not reported due to confidentiality requirements. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, BISTA, December 2016, own calculations.

Fifth, banks not only hold large and persistent gross positions as both lenders and borrowers simultaneously, but also sizable structural deficits or surpluses in the interbank market. Figure 3a categorizes all MFIs into bins based on their share of interbank assets and liabilities in their balance sheets and reports the number of entities within each, while Figure 3b displays the share of total assets contained in each bin. We first observe that most banks participate in the market as borrowers and lenders suggesting that the interbank market serves to cover temporary liquidity shortfalls. However, as first noted by Craig and Ma (2022), numerous banks adopt a net lender or borrower position in the market (mass outside the

[21] Figure A.1 illustrates the cumulative share of total assets held by the n-largest banks as of December 2016.

horizontal diagonal), with net positions exceeding 10% being common. Consequently, the interbank market not only facilitates short-term liquidity provision but also enables banks to cover structural funding deficits and allocate structural funding surpluses. A comparison between panels (a) and (b) of Figure 3 also suggests that a few large MFIs play a central role in the market (see bin on the 40% assets, 30% liabilities position). Large banks have access to a diversified pool of funding sources, with close to 150 unique lenders, whereas smaller banks typically have no more than 20 connections.[22]

Table 1: Spearman rank correlation tests

| Months | 3 | 6 | 12 | 24 |
|---|---|---|---|---|
| Interbank asset share | .922 (.001) | .887 (.001) | .846 (.002) | .768 (.002) |
| Interbank liability share | .970 (.001) | .955 (.001) | .934 (.001) | .888 (.001) |
| Interbank net position | .960 (.001) | .940 (.001) | .915 (.001) | .852 (.002) |

We construct the table by ranking the interbank market share of each bank and estimating the correlation with the ranking m-Months ahead. The first row contains the correlation of the interbank asset share, the second row displays the correlation of the liability share, and the third row presents the correlation in the interbank balance. All coefficients are statistically significant at a threshold of < 1%. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, BISTA, December 2004 to December 2018, own calculations.

To demonstrate that such funding gaps are a stable feature of the market, we provide estimates of the Spearman rank correlation for the shares of interbank assets and liabilities, as well as their difference in Table 1. The high correlation at horizons of up to two years indicates that banks' positions are indeed very persistent over time. A market for structural funding is also consistent with the non-trivial share of medium and long-term borrowing observed in Figure 1. The distinction between the two roles of interbank markets is relevant, as a market freeze, such as the one experienced after 2007, can result in stronger effects on the real economy if banks are unable to cover their structural positions, thereby being forced to cut back lending to the non-financial sector.[23] Despite funds being treated as a homogeneous good, our model can accommodate temporal liquidity (large gross positions) and structural borrowing (net positions) in the interbank market and adequately capture the relative importance of each role for the aggregate economy.

## 4 Model

Our model features a standard New-Keynesian economy augmented with a banking sector composed of $N$ distinct banks. Figure 4 illustrates the various components of the model.

[22] Figure A.2 plots the average number of distinct borrowing connections by deciles over the sample period. Deciles in panel (a) of Figure A.2 are based on the number of connections, while those in panel (b) are based on bank balance sheet size. The similarity between the two graphs indicates that central positions in the market are highly correlated with bank size and very persistent. On the asset side, a similar pattern emerges, with large banks acting as lenders to the rest of the system (available upon request).

[23] In section 6, we show that net borrowers indirectly exposed to the US crisis indeed experienced larger increases in interest rates, cut lending more, and relied more heavily on own funds compared to net lenders after the crisis.

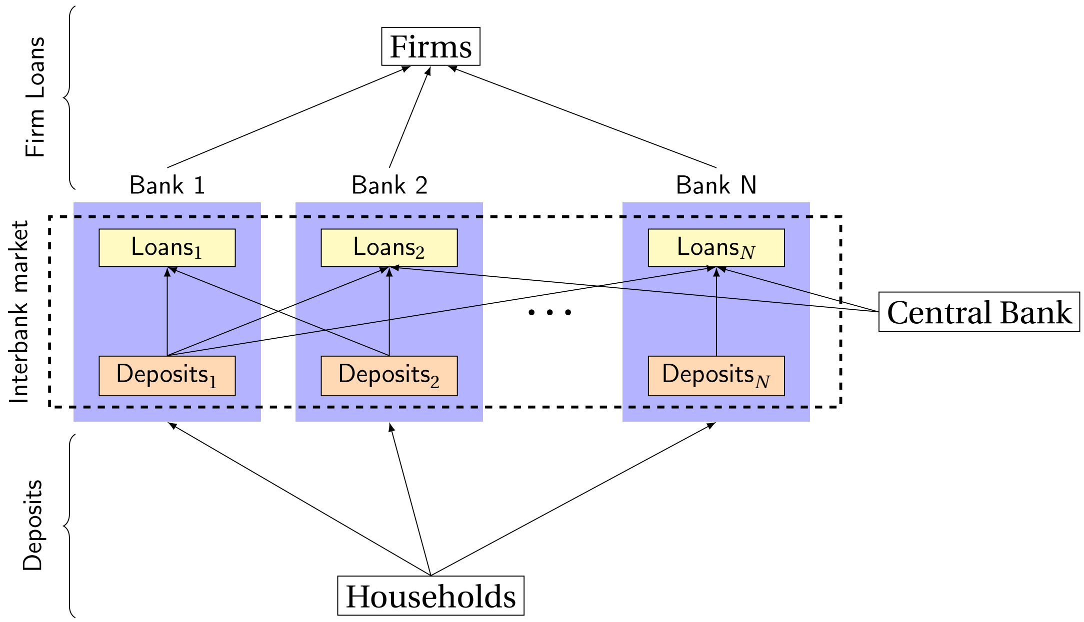



Figure 4: Components of the financial channel: Households allocate savings across banks in the form of deposits. Banks lend these funds to firms, which utilize them to finance capital investments. Mismatches between bank deposits and firm loan demand are resolved in the interbank market. A central bank supplies lender-of-last-resort (LoLR) credit to the system banks. Arrows indicate the flow of funds between agents.

Funds enter the banking system through household savings deposited in demand accounts, which are then lent to firms that borrow to build capital stock for production. Households determine how much to deposit and in which banks based on the returns offered by each institution and the convenience utility derived from holding deposits in specific banks. There are no time restrictions on households' withdrawal of funds from their demand accounts, and shocks to either the returns on deposits or the convenience utility cause households to reallocate their funds across banks or withdraw them from the banking system in the short term.

Loans provided by each bank are tied to the financing of $N$ distinct types of differentiated and partially substitutable capital used by firms. Firms and banks meet at the beginning of each period to agree on a contract with a fixed interest rate and loan conditions, similar to a credit line. Importantly, the terms of the loan contract are non-renegotiable until the next period, and banks cannot pass short-term fluctuations in the cost of funds or funds availability onto firms. This leads to short-term mismatches between loan demand and deposits, creating the need for an interbank market and providing a rationale for a central bank's lending facility to offer emergency credit to the system.

In the interbank market, banks engage in lending to address funding shortfalls arising from deposit fluctuations and to meet broader structural funding needs for their loan operations. A real wedge exists between the cost at which a bank can raise funds from its own depositors and the rate at which it can safely lend to counterparties in the interbank market, which we refer to as volatile transaction costs. This wedge can be understood as a broad array of frictions that impede the smooth flow of funds across banks, including physical transaction costs, collateral posting, enforcement and monitoring costs, and perceived default risk, among others. These wedges vary bilaterally and may be asymmetric within bank pairs, reflecting the absolute and relative advantage of each bank as a provider or borrower of funds. These wedges can also fluctuate over time and exhibit flexible correlation patterns across pairs, reflecting the varying challenges of conducting business in the interbank market at specific times, such as during the U.S. financial crisis, or with particular counterparties. Consequently, an interbank network emerges from the profit-maximization decisions of banks, shaping both the efficiency of credit allocation across the entire banking system and the extent to which the network dampens or amplifies the volatility of credit in response to shocks affecting individual banks.

This section presents the main assumptions and components of the model. A more detailed derivation of the model solution and equilibrium conditions can be found in Online Appendix 2.

## 4.1 Notation and Timing Conventions

The model features discrete quarters indexed by $t$, and each quarter is divided into a $[0,1]$ continuum[24] in which agents take actions, such as consumption, employment, or saving decisions. We assume that all agents in the economy have perfect foresight within the quarter and rational expectations between quarters. A moment within the continuum is indexed by $\tau$, and the pair $\{t,\tau\}$ uniquely identifies a moment $\tau$ within quarter $t$.

## 4.2 Representative Household

Households obtain positive utility from the consumption of a final good and supply labor to the firms producing it. They also derive utility from holding real deposit balances in banks, which captures a reduced-form preference for liquidity.[25] The representative household maximizes the following objective function:

$$\max \quad \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[ \log\left(X_{t+j}\right) - \left(\frac{\eta}{\eta+1}\right) \int_0^1 N_{t+j,\tau}^{1+1/\eta} \, \mathrm{d}\tau \right] ,$$

where $N_{t,\tau} = \left[\int_0^1 N_{t,\tau}(v)^{1+1/\eta} \, \mathrm{d}v\right]^{\frac{\eta}{\eta+1}}$ is the aggregate labor index and $N_{t,\tau}(v)$ labor supplied to intermediate industry $v$, $\eta$ is the Frisch labor supply elasticity, and variable $X_t$ is a composite of consumption and bank deposit balances. In particular,

$$X_t = C_t + \sum_{n=1}^{N} \int_0^1 \left(1 - T_t^n z_{t,\tau}^n\right) \frac{D_{t,\tau}^n}{P_t} \, \mathrm{d}\tau \, , \tag{1}$$

where $C_t = \int_0^1 C_{t,\tau} \, \mathrm{d}\tau$ is aggregate consumption in $t$, $P_t$ is the aggregate price index of the economy, $D_{t,\tau}^n$ are one-period nominal deposits at bank $n$ paying a (gross) return $R_{t,\tau}^{D,n}$, $(1 - T_t^n)$ is the average utility of deposits at bank $n$, and $z_{t,\tau}^n$ is an exogenous shock to those deposit preferences. We model $z_{t,\tau}^n$ as a

[24]The continuum can be interpreted as a smooth approximation to the days that comprise a quarter.

[25]Alternatively, utility from real deposit balances can be interpreted as capturing the usefulness of money in the completion of consumption transactions.

Weibull-distributed shock with mean one and shape parameter $\kappa$ controlling its volatility, and assume the shocks to be i.i.d. across banks $n$, quarters $t$, and time continuum $\tau$. In addition to bank deposits, we assume that households also have access to a one-period bond with return $R_t^B$, which is in zero net supply in equilibrium.

The first-order equilibrium condition for deposit rates is:

$$R_{t,\tau}^{D,n} = R_t^B T_t^n z_{t,\tau}^n , \qquad \forall n . \tag{2}$$

Movements in the return of bonds $R_t^B$ have a proportional impact on the deposit rates paid by all banks, while shocks to individual bank preferences $z_{t,\tau}^n$ alter the relative costs of attracting deposits and lead to a reallocation of deposits across banks at each moment $\tau$.

## 4.3 Firms

A mass-one of differentiated intermediate goods indexed by $\nu$ is in monopolistic competition with the following production function employing capital and labor:

$$Y_{t,\tau}(\nu) = \left(\frac{K_t(\nu)}{\alpha}\right)^{\alpha} \left(\frac{N_{t,\tau}(\nu)}{1-\alpha}\right)^{1-\alpha} ,$$

Intermediate producers have sticky prices à la Calvo (1983) and reset prices at the beginning of the quarter with probability $1-\theta$. A representative, perfectly competitive firm combines intermediates into a final good $Y_t$ via a CES aggregator with $\epsilon > 1$ elasticity of substitution across varieties.

We assume that firms must employ a constant level of capital throughout the quarter, $K_{t,\tau}(\nu) = K_t(\nu)$, $\forall \tau$, so all production adjustments within the quarter occur through the labor margin. Aggregate capital is a CES composite of $N$ distinct types of capital:

$$K_t(\nu) = \left[ \sum_{n=1}^{N} \left(a_t^n\right)^{1/\sigma} K_t^n(\nu)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} ,$$

where $\sigma > 1$ is the elasticity of substitution between types, and $a_t^n$ is a demand shock, normalized such that $\sum_{n=1}^{N} a_t^n = 1$ for all $t$. Capital investment requires one unit of the final good, and without loss of generality to the qualitative results of the paper, we assume full depreciation and instant build-up of capital from investment. Firms finance their investment with credit from $N$ distinct banks,[26] each specialized in the provision of one-period loans $L_{t,\tau}^n(\nu)$ at gross interest rate $R_t^{F,n}$ for the purchase of a distinct type of capital. Solving the firm's optimization problem and adding loan demands across firms, we obtain an expression for bank $n$'s aggregate demand as:

$$L_t^n = a_t^n \left(\frac{R_t^{F,n}}{R_t^F}\right)^{-\sigma} L_t , \tag{3}$$

where $L_t$ and $R_t^F$ are aggregate loan and interest rate indices, respectively.

[26] We do not consider self-financed firm investment, but such distinction would not affect the qualitative results of the model.

## 4.4 Banking Sector

Each bank performs three activities: they obtain deposits from the representative household, provide credit to firms, and trade funds with each other in the interbank market. For expositional purposes, we assume that each bank is composed of two divisions, each one responsible for a different set of tasks. The Loan Division provides credit to firms and secures the necessary funding through internal funds or interbank loans. The Deposit Division procures deposits from the representative household and distributes them to the Loan Divisions.

### 4.4.1 Loan Division

Loan Division $n$ is subject to the following constraints limiting the creation of firm loans:

$$L^n_{t,\tau} \le M^n_{t,\tau}\,, \quad M^n_{t,\tau} \ge 0\,, \tag{4}$$

where $M^n_{t,\tau}$ is the amount of internal and/or interbank funding available at time $\tau$. The first constraint restricts banks' credit provision by the amount of available funds and holds with equality in equilibrium. Bank funds are perfect substitutes, and Loan Divisions obtain them as one-period interbank loans (or internal transfer) from the bank that offers the lowest rate. Formally,

$$\begin{aligned} M^n_{t,\tau} &= M^{in}_{t,\tau}, \quad i^n_{t,\tau} = \arg_j \min\left\{R^{I,jn}_{t,\tau}\right\}\,, \\ R^{I,n}_{t,\tau} &= R^{I,in}_{t,\tau}\,, \end{aligned} \tag{5}$$

where $M^{in}_{t,\tau}$ are the interbank funds lent by Deposit Division $i$ to Loan Division $n$ and $R^{I,in}_{t,\tau}$ is the gross interbank rate at which bank $i$ is willing to lend to bank $n$. Banks know their individual firm loan demands given by equation (3) and act as monopolistic competitors, taking the aggregate index $R^F_t$ as given. Banks and firms meet at the beginning of the quarter and agree on an interest rate that will prevail throughout the period.[27] This results in a constant firm loan demand throughout the quarter, which banks have to finance while experiencing a varying capacity to attract funds due to depositor preference shocks $z^n_{t,\tau}$, forcing them to borrow from the interbank market or, in the absence of trading opportunities, reduce loan demand ex-ante by charging higher interest rates to firms. However, interbank rates are renegotiated at each instant $\tau$ and reflect the shifting capacity to provide funds by the emitting bank.

A natural question is whether banks would respond to the possibility of future interbank stress by accumulating additional deposits in advance. In our framework they do not, because banks are modeled as risk-neutral intermediaries that choose current funding to minimize current expected costs, and therefore have no precautionary motive to raise deposits merely to leave them idle.[28] We view this as a

[27] An alternative assumption with equivalent results would be that firm interest rates are sticky *within* the continuum and can only be reset at the beginning of each quarter. Sørensen and Werner (2006) provide supporting empirical evidence for the stickiness of retail interest rates.

[28] This connects to the broader critique in the financial-frictions literature that forward-looking borrowers may partially self-insure against future financing difficulties by accumulating internal funds; see, for example, Moll (2014), Midrigan and Xu (2014). In a banking context, however, limited liability, deposit insurance, and related agency distortions may weaken private incentives to self-insure through liquidity holdings; see Keeley (1990), Hellmann, Murdock, and Stiglitz (2000). This is also the logic behind modern liquidity regulation; see Basel Committee on Banking Supervision (2013, 2014). More generally, even

parsimonious approximation for the mechanism of interest. Consistent with this assumption, Section 3 documents large and persistent net lender and borrower positions in the data, suggesting that in practice banks do not hold liquidity buffers large enough to eliminate structural funding imbalances.

Solving the maximization problem of the Loan Division, we obtain the aggregate interest rate on firm loans as a constant mark-up over the average cost of funds:

$$R_t^F = \left(\frac{\sigma}{\sigma-1}\right) R_t^I , \qquad R_t^I = \left[\sum_{n=1}^{N} a_t^n \left(R_t^{I,n}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}} ,$$

where $R_t^{I,n} \equiv \int_0^1 R_{t,\tau}^{I,n}\, \mathrm{d}\tau$ is the average interbank rate paid by bank $n$ in quarter $t$.

#### 4.4.2 Deposit Division

The Deposit Division obtains deposits from the household and converts them into internal funding or interbank loans. The amount of funds that $n$ can provide is given by

$$M_{t,\tau}^{nn} + \sum_{i\neq n} d_t^{ni} M_{t,\tau}^{ni} = D_{t,\tau}^n ,$$
$$\text{subject to:} \quad M_{t,\tau}^{ni} \geq 0 , \quad D_{t,\tau}^n \geq 0 , \quad \forall n, i ,$$

where $d_t^{ni} \geq 1$ are transaction costs incurred for the transfer of funds from Deposit Division $n$ to Loan Division $i$. We interpret these costs as capturing screening, enforcement, capital regulation, collateral posting or other costs related to an interbank transaction.[29] We normalize the transaction costs between divisions of the same bank to one, $d_t^{nn} = 1$, $\forall n$, and assume that all other transaction costs are exogenously given to the banks or, alternatively, represent the lowest possible cost arrangement between any trading pairs at time $t$.

Additionally, we assume that the payment of the transaction cost ensures the full repayment of any interbank loan under the specified contract terms (e.g., as in the case of adequately collateralized loans). Consequently, we do not explicitly model scenarios of bank insolvency and default, which are relatively rare in the context of the German banking system. Instead, we implicitly incorporate their effect on credit through the costs $d_t^{ni}$, which represent the expense of guaranteeing a safe loan contract with a counterparty.[30] We assume that the markets for interbank loans and deposits are perfectly competitive, with banks acting as price takers.[31]

Solving the optimization problem of the Deposit Division and using equation (2), we obtain an

if one introduced precautionary motives, the relevant benchmark would be a finite buffer-stock of liquidity rather than full self-insurance against large or persistent funding shocks; see Carroll (1997, 2011).

[29]For example, uncertainty surrounding the value of mortgage-backed securities (and related assets) following the 2007 financial crises can be interpreted through the lens of the model as an increase in the $d_t^{ni}$ costs of collateral screening.

[30]Explicitly modeling default risk in our framework would multiplicatively scale the credit costs in equation (6) by $\left[1-\mathbb{E}_t\left[Pr(Default_{i,t+1})\right]\right]^{-1}$, where $Pr(Default_{i,t+1})$ denotes the default probability of borrower $i$. Therefore, when default risk is not modeled explicitly, it can instead be interpreted as one component of the transaction cost shocks $d_t^{ni}$ that multiplicatively scales the other transaction costs within the pair. In Table 4, we document that borrower $i$'s default risk is positively correlated with our recovered transaction costs $\log \hat{d}_t^{ni}$, but not with the default risk of lender $n$.

[31]Alternatively, modeling the interbank market under Bertrand competition also provides a tractable solution but significantly complicates the welfare analysis.

expression for the interbank loan rate offered by bank $n$ to bank $i$,

$$R_{t,\tau}^{I,ni} = R_t^B d_t^{ni} T_t^n z_{t,\tau}^n \,. \tag{6}$$

Given that the loan division of bank $n$ receives funds from the lowest rate lender in every moment $\tau$ (see equation (5), we obtain an expression for bank $n$'s expected interbank borrowing credit spread in period $t$ as[32]

$$\mathbb{E}_t \left[ \min_{i \in \{1,\dots,N\}} \left\{ R_{t,\tau}^{I,in} \right\} \right] / R_t^B = \left[ \sum_{i=1}^{N} \left( d_t^{in} T_t^i \right)^{-\kappa} \right]^{-1/\kappa} \equiv \Phi_t^n \,,$$

where $d_t^{in} T_t^i$ is the average spread at which bank $i$ is willing to lend funds to bank $n$. It reflects the bilateral transaction costs $d_t^{in}$ and efficiency $T_t^i$ with which bank $i$ attracts funds from its own depositors.

### 4.5 Central Bank

The central bank provides direct credit to banks via its lending facilities[33] and other lender-of-last-resort (LoLR) interventions. We assign subindex zero to the central bank and model it as an additional bank within the system with some unique characteristics. Namely, the central bank does not raise deposits from households, has the capacity to freely create money, and therefore can arbitrarily set the interest rate at which it offers funds to the system banks. Consistent with most historical discount window policies, we model the lending rate charged by the central bank as a penalty rate over the average cost-of-funds at which each bank $n$ is able to borrow from the rest of its funding sources at time $t$. Formally, this relationship is given by

$$R_{t,\tau}^{I,0n} = \underbrace{e^{\varpi_1} z_{t,\tau}^0}_{\equiv \text{penalty}_{t,\tau}^n} \Phi_t^n R_t^B \,, \tag{7}$$

where $\varpi_1$ is a parameter governing the steady-state level of the penalty, and $z_{t,\tau}^0$ is a policy shock, introduced for analytical convenience, which is assumed to follow a Weibull distribution with mean one and shape parameter $\kappa$.[34] We also considered an extended version of this penalty-rate rule that targets cyclical deviations of the bank's average interbank credit spread, $\Phi_t^n$, from its steady-state value. Since this extension has limited quantitative or qualitative effect on the results in the following sections, we relegate it to the Online Appendix.

Finally, the central bank also determines the nominal risk-free rate $R_t^B$ of the economy through conventional open market operations. We assume that it follows a Taylor rule of the form

$$R_t^B = R^B \left( \frac{\Pi_t}{\Pi} \right)^{\gamma_\pi} \left( \frac{Y_t}{Y_t^n} \right)^{\gamma_y} \,,$$

[32] To derive this expression, we are using the property that the minimum of a Weibull-distributed variable is itself distributed Weibull.

[33] Examples include the ECB's marginal lending facility or the Fed's discount window.

[34] More generally, $\text{penalty}_{t,\tau}^n$ may also capture other costs of accessing central bank credit, such as discount window stigma.

where $\Pi_t \equiv P_t/P_{t-1}$ stands for gross inflation and $Y_t^n$ is output under flexible prices.

### 4.6 Banking Sector Aggregation & Equilibrium Conditions

Plugging equations (6) and (7) into (5), we obtain the following expression for the average credit spread paid by bank $n$:

$$\widetilde{R}_t^{I,n} \equiv \frac{R_t^{I,n}}{R_t^B} = \Phi_t^n \left(1-\xi_t^{0n}\right)^{1/\kappa} , \tag{8}$$

where $\xi_t^{0n} \equiv \left[1+e^{\kappa\varpi_1}\right]^{-1}$ denotes the share of total funding obtained through the central bank lending facility, implying a constant quarterly probability of access to central bank credit in equilibrium. Because banks borrow at each instant $\tau$ from the cheapest available source at the rates implied by equations (6) and (7), and because the minimum of Weibull draws is itself Weibull, aggregation over the within-quarter continuum yields the following closed-form expression for the quarterly bilateral funding share from each bank source, excluding central bank's lending facility funding:

$$\lambda_t^{ni} = \left(\frac{d_t^{ni}\,T_t^n}{\Phi_t^i}\right)^{-\kappa} , \quad \text{and} \quad M_t^{ni} = \left(1-\xi_t^{0i}\right)\lambda_t^{ni} L_t^i , \qquad \forall n \in \{1,\dots,N\} . \tag{9}$$

Intuitively, $d_t^{ni}T_t^n$ is the average spread above the risk-free rate at which funds from bank $n$ are *offered* to bank $i$ over the quarter, whereas $\Phi_t^i$ is the average interbank credit spread at which bank $i$ *effectively* borrows. The larger the difference between these two spreads, the less likely bank $n$ is to be the lowest-cost supplier of funds to bank $i$ at a given moment $\tau$. As a result, the quarterly transaction volume from $n$ to $i$ is smaller, both in absolute and relative terms. Finally, it is also worth noting that the depositor-preference parameter $\kappa$ can equivalently be interpreted as the elasticity of interbank funding supply in equation (9).[35]

**Definition 1** (Aggregate funding shares)**.** We define the aggregate own-funding share (*own share*) and the aggregate central bank funding share, respectively, as

$$\lambda_t^{Own} = \left[\sum_{n=1}^{N} s_t^n \left(\lambda_t^{nn}\right)^{\frac{\sigma-1}{\kappa}}\right]^{\frac{\kappa}{\sigma-1}} \quad \text{and} \quad \xi_t^0 = \sum_{n=1}^{N} s_t^n \xi_t^{0n} , \tag{10}$$

where $s_t^n \equiv \frac{R_t^{F,n}L_t^n}{R_t^F L_t}$ denotes bank $n$'s share in aggregate gross loan claims (i.e., principal plus interest).

**Definition 2** (Autarky Economy)**.** The autarky economy is the counterfactual equilibrium without interbank trade. Banks can neither borrow from nor lend to one another and therefore finance their loan portfolios using only their own deposits and, when needed, credit from the central bank lending facility. Formally, autarky is obtained by taking bilateral interbank transaction costs between distinct banks to infinity:

$$d_t^{ni} \to +\infty \qquad \forall\, t,\, \forall\, n,\, \forall\, i \neq n\, .$$

[35] The link between these two interpretations of $\kappa$ is intuitive: when depositor preferences are less volatile (higher $\kappa$), it becomes less likely that temporary deposit inflows offset differences between the offered spread $d_t^{ni}T_t^n$ and the borrowing spread $\Phi_t^i$.

Variables evaluated in this counterfactual equilibrium carry the superscript $AU$.

Definition 2 formally defines the autarky (AU) counterfactual economy as the economy without interbank trade. This does not eliminate all financial reallocation. Depositors can still move funds across banks in response to preferences and deposit rates, and firms can still borrow from any bank in the system in whatever proportions they find optimal. In our context, autarky refers specifically to the absence of interbank trade, rather than to a broader segmentation of the banking system or firm-bank relationships.

To highlight the role of interbank trade in reducing steady-state borrowing costs, compare the credit spread in autarky and with an active interbank market:

$$\widetilde{R}^{I,AU} = \left(1-\xi^0\right)^{1+1/\kappa} \left[\sum_{n=1}^{N} a^n \left(T^n\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}} \quad \text{and} \quad \widetilde{R}^I = \left(\lambda^{Own}\right)^{1/\kappa} \widetilde{R}^{I,AU} .$$

In autarky, the spread $\widetilde{R}^{I,AU}$ is determined by two components: the central bank funding share $\xi^0$ (capturing LoLR support) and a weighted average of households' deposit-preference shifters, $\{T^n\}_{n=1}^N$. A higher LoLR share lowers the spread by capping banks' marginal funding costs and compressing the wedge between those costs and the risk-free rate. Stronger household demand for deposits (lower $T^n$) also reduces the spread by lowering the deposit rate households require, thereby narrowing the deposit–risk-free rate differential. With interbank trade, the additional factor $\left(\lambda^{Own}\right)^{1/\kappa}$ captures the further reduction in borrowing costs relative to autarky: for a given autarky spread $\widetilde{R}^{I,AU}$, a lower own-funding share – that is, greater reliance on interbank funding—implies a lower credit spread $\widetilde{R}^I$.[36]

Next, we log-linearize the equilibrium conditions around the deterministic steady state. Lowercase letters denote logs, and hats denote deviations from steady state. A first-order approximation yields the New Keynesian block, composed of a Phillips curve and a dynamic IS equation:

$$\begin{aligned} \pi_t &= \Omega\, \hat{\tilde{y}}_t + \beta \mathbb{E}_t\left[\pi_{t+1}\right] , \\ \hat{\tilde{y}}_t &= -\left[1+\alpha\left(\frac{\eta}{\eta+1}\right)\right]\left[r_t^B - \mathbb{E}_t\left[\pi_{t+1}\right] - r_t^n\right] + \mathbb{E}_t\left[\hat{\tilde{y}}_{t+1}\right] , \end{aligned}$$

where $\tilde{y}_t \equiv y_t - y_t^n$ is the output gap, $\Omega$ is a constant, and $r_t^n \equiv -\log(\beta) + \mathbb{E}_t\left[x_{t+1}^n - x_t^n\right]$ is the natural (flexible-price) real interest rate. Here $-\log(\beta) > 0$ is the subjective discount rate, and $x_t^n$ denotes the natural counterpart of the (log-) utility bundle in (1). The term $r_t^B - \mathbb{E}_t[\pi_{t+1}] - r_t^n$ is the real-rate gap (IS wedge). The coefficient $\left[1+\alpha\left(\frac{\eta}{\eta+1}\right)\right] > 1$ governs the sensitivity of the output gap to this wedge.[37]

Interbank market conditions affect this system through the natural real interest rate $r_t^n$ in the IS wedge. This channel operates through the growth rate of the natural utility bundle, $\mathbb{E}_t\left[x_{t+1}^n - x_t^n\right]$, which depends on both consumption, and therefore production, and real deposit balances. Both margins are shaped by credit conditions: borrowing costs affect the user cost of capital, and thereby output

[36] Figures 1 and 2a provide suggestive evidence consistent with this channel, as the aggregate own share and the Euribor credit spread both rise persistently after 2007.

[37] This coefficient equals one in standard New Keynesian models. In our setting, the extra sensitivity arises from non-separable preferences for deposits in households' utility. See Fisher (2015) for how non-separable preferences for assets driven by liquidity and/or safety motives modify the dynamic IS equation in New Keynesian models.

and consumption, while funding spreads affect incentives to hold deposits within the banking system. Expressing $\mathbb{E}_t[x^n_{t+1} - x^n_t]$ in terms of credit conditions yields

$$r^n_t = -\log(\beta) - \left(\frac{\alpha}{1-\alpha}\right)\mathbb{E}_t\left[\tilde{r}^I_{t+1} - \tilde{r}^I_t\right].$$

Hence, expected changes in the aggregate credit spread $\tilde{r}^I_t$ directly affect $r^n_t$ and, in turn, the IS wedge.

# 5 Welfare

This section derives welfare gains from interbank trade and from lender-of-last-resort (LoLR) policy, each defined relative to its corresponding no-intervention counterfactual (no interbank trade; no LoLR credit provision).

## 5.1 Gains from Trade

We measure the gains from interbank trade as the consumption-equivalent change in expected household utility relative to autarky:

$$\mathbb{J} \equiv \mathbb{E}\left[\frac{U_t - U^{AU}_t}{U_X X}\right], \tag{11}$$

where $U_X \equiv \mathrm{d}U/\mathrm{d}X$ and the normalization $U_X X$ expresses welfare in consumption-equivalent units. Unlike standard business-cycle welfare measures, which isolate the cost of fluctuations around a fixed steady state, $J$ compares the integrated interbank economy with the autarky counterfactual.[38] It therefore captures not only second-order effects from volatility, but also level effects associated with a more efficient allocation of funding across banks and the resulting changes in average capital and consumption.

We present three approximations for $\mathbb{J}$ with increasing structural content: (i) a static expression at the shockless steady state, (ii) a sufficient-statistic approximation to (11) based on observable moments of the own trade share $\lambda^{Own}_t$, and (iii) a structural approximation that decomposes gains from trade into components associated with different shocks.

### 5.1.1 Static Gains from Trade

We define the static gains from trade, $\mathbb{J}^{ss}$, as (11) evaluated at the shockless steady state. Online Appendix 3 shows that this yields the following closed-form expression:[39]

$$\mathbb{J}^{ss} = -\left(\frac{\alpha}{1-\alpha}\right)\frac{1}{\kappa}\log\left(\lambda^{Own}\right) \geq 0\,. \tag{12}$$

Interbank integration is summarized by the own-funding share $\lambda^{Own}$: deeper integration (lower $\lambda^{Own}$) raises welfare monotonically by improving the allocation of funds across banks. The gains are larger when

[38] See Lucas Jr (2003) for the classic welfare cost of business-cycle fluctuations, Arkolakis, Costinot, and Rodríguez-Clare (2012) for the gains-from-trade interpretation of changes in trade frictions, and Barlevy (2004) for the broader point that welfare effects can be much larger when frictions or policy affect average levels rather than only volatility.

[39] Note the similarity of this expression with the gains from trade formula for static economies in the international trade literature, e.g., Arkolakis, Costinot, and Rodríguez-Clare (2012).

funding sources are less substitutable across interbank connections (lower supply elasticity $\kappa$) and when capital is more important for production (higher capital share $\alpha$).[40]

### 5.1.2 Sufficient Statistic Approximation

The gains-from-trade expression in (11) accounts for the presence of shocks in the stochastic equilibrium and therefore generally differs from (12). Online Appendix 3 derives the following second-order approximation:

$$\mathbb{J} = -\left[\left(\frac{\alpha}{1-\alpha}\right)\frac{1}{\kappa}\mathbb{E}\left[\log\left(\lambda_t^{Own}\right)\right] + \frac{\Lambda_{rr}}{2\kappa^2}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}^2\right] + \frac{\Lambda_{rr}}{\kappa}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\hat{\hat{r}}_t^{I,AU}\right]\right] + \text{h.o.t.}\,, \tag{13}$$

where $\Lambda_{rr} > 0$ is a positive constant and the term h.o.t. denotes higher-order terms omitted from the approximation.

The first term, $-\left(\frac{\alpha}{1-\alpha}\right)\frac{1}{\kappa}\mathbb{E}\left[\log\left(\lambda_t^{Own}\right)\right] \geq 0$, captures the level effect of interbank openness. When the mean of $\log\left(\lambda_t^{Own}\right)$ is close to its steady-state value, this term essentially coincides with the static gains in (12). The second term, $-\frac{\Lambda_{rr}}{2\kappa^2}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}^2\right] \leq 0$, captures the welfare cost of credit-spread volatility: interbank transaction-cost shocks generate fluctuations in funding costs, which translate into more volatile consumption and output. This implies a welfare loss due to the concavity of household utility. In the presence of nominal rigidities, volatile credit spreads can also generate additional welfare losses through increased price dispersion.The third term, $-\frac{\Lambda_{rr}}{\kappa}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\hat{\hat{r}}_t^{I,AU}\right] \lesseqgtr 0$, captures whether the interbank network amplifies or dampens fluctuations in the autarky credit spread, which is driven by loan-demand shocks ($a^n$) and depositor-preference shocks ($T_t^n$) across all $n$.

Equation (13) is not directly estimable because it depends on the counterfactual autarky credit spread $\hat{\hat{r}}_t^{I,AU}$. To obtain a sufficient-statistic expression in observables, we impose one identifying assumption and one moment approximation. First, we assume that $\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\,\hat{\hat{r}}_t^{I,AU}\right] \approx 0$. This is plausible if the forces driving the autarky spread differ from those driving variation in interbank integration. Under autarky, $\hat{\hat{r}}_t^{I,AU}$ is driven by loan-demand and depositor-preference shocks, whereas fluctuations in $\lambda_t^{Own}$ also reflect interbank transaction cost shocks. If these interbank shocks account for most of the variation in $\lambda_t^{Own}$ and are weakly correlated with loan-demand and depositor-preference shocks, then $\widehat{\log\left(\lambda_t^{Own}\right)}$ is approximately orthogonal to $\hat{\hat{r}}_t^{I,AU}$, so the covariance term is small.[41] Second, we use the approximation $\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}^2\right] \approx Var\left(\log\left(\lambda_t^{Own}\right)\right)$, which is appropriate when the unconditional mean is close to the steady state. Under these two restrictions, (13) simplifies to:

$$\mathbb{J} = -\left[\left(\frac{\alpha}{1-\alpha}\right)\frac{1}{\kappa}\mathbb{E}\left[\log\left(\lambda_t^{Own}\right)\right] + \frac{\Lambda_{rr}}{2\kappa^2}Var\left(\log\left(\lambda_t^{Own}\right)\right)\right] + \text{h.o.t.}\,, \tag{14}$$

which depends only on observable moments of $\lambda_t^{Own}$ and can be taken directly to the data. Equation (14) is therefore an *ex-ante* welfare measure: it does not require knowledge of the model's underlying structure (including the interbank network) or of the autarky counterfactual.

[40] The $1-\alpha$ term in equation (12) follows from an input-output multiplier effect between loan supply and output, resulting in the creation of additional capital investment.

[41] Figures 1 and 2a provide anecdotal support for this assumption: the main fluctuations in credit spreads and own-trade shares in our sample coincide with the 2007 financial crisis and the euro crisis, episodes that the literature typically associates with distortions in interbank market conditions rather than with changes in commercial banks' depositor base.

### 5.1.3 Structural Approximation and Decomposition

Finally, we obtain a structural decomposition by combining the model's equilibrium relationships with distributional assumptions on the fundamental shocks to recover the autarky counterfactuals. For analytical tractability, we assume that $u_t^{a,n} \equiv \widehat{\log a_t^n}$, $u_t^{T,n} \equiv \widehat{\log T_t^n}$, and $u_t^{I,ni} \equiv \widehat{\log d_t^{ni}}$ follow $AR(1)$ processes with a common persistence parameter $\rho$ across the three shock blocks, but potentially correlated innovations. Section 6.5 later reports separate estimates for $\rho_T$, $\rho_a$, and $\rho_I$, and shows that they are quantitatively very similar supporting this simplifying assumption. For the variance-covariance structure of $u_t^y$, $y \in \{a, T, I\}$, we make the following assumptions:

*1. Loan Demand Shifters $a_t^n$:*

$$\mathbb{E}\left[u_t^{a,i} u_t^{a,n}\right] = \begin{cases} \sigma_a^2 & \text{if } n = i\,, \\ \zeta_a \sigma_a^2 & \text{otherwise}\,. \end{cases}$$

*2. Depositor Preferences $T_t^n$:*

$$\mathbb{E}\left[u_t^{T,i} u_t^{T,n}\right] = \begin{cases} \sigma_T^2 & \text{if } n = i\,, \\ \zeta_T \sigma_T^2 & \text{otherwise}\,. \end{cases}$$

*3. Bilateral Transaction Costs $d_t^{ni}$:*

$$\mathbb{E}\left[u_t^{I,ji} u_t^{I,kn}\right] = \begin{cases} 0 & \text{if } j = i \text{ or } k = n\,, \\ \sigma_I^2 & \text{if } k = j,\ n = i\,, \\ \zeta_{I,B} \sigma_I^2 & \text{if } k \neq j,\ n = i\,, \\ \zeta_{I,L} \sigma_I^2 & \text{if } k = j,\ n \neq i\,, \\ \zeta_{I,X} \sigma_I^2 & \text{otherwise}\,. \end{cases}$$

*4. Zero Cross-Correlation:*

$$\begin{aligned} \mathbb{E}\left[u_t^{I,ji} u_t^{a,k}\right] &= \mathbb{E}\left[u_t^{I,ji} u_t^{T,k}\right] \\ &= \mathbb{E}\left[u_t^{a,j} u_t^{T,k}\right] = 0\,, \quad \forall j, i, k\,. \end{aligned}$$

We assume that each shock $u_t^y$ is correlated between banks with parameter $\zeta_y$. In particular, we allow the volatility of transaction costs on a given connection to affect all other connections, either at the lender ($\zeta_{I,L}$), at the borrower level ($\zeta_{I,B}$) or any other connection ($\zeta_{I,X}$). For tractability, we assume that the different shocks are not cross-correlated.

This approach avoids the auxiliary restrictions underlying the sufficient-statistic approximation above and, more importantly, delivers a decomposition of the gains from trade by shock type. Formally, we can write:

$$\mathbb{J} = \mathbb{J}^{ss} + \frac{1}{2}\left[\sigma_T^2 \mathfrak{J}^T + \sigma_a^2 \mathfrak{J}^a + \sigma_I^2 \mathfrak{J}^I\right] + \text{h.o.t.}\,, \tag{15}$$

where $\sigma_T^2$, $\sigma_a^2$, and $\sigma_I^2$ denote the variances of depositor-preference, firm-loan-demand, and transaction-cost shocks, respectively. The multipliers $\mathfrak{J}^y$ summarize how market integration changes the welfare cost generated by each shock: positive (negative) values indicate that integration reduces (increases) the welfare cost associated with shocks of type $y$.

To link these multipliers to features of the banking system, we summarize concentration using the following indices:

$$H_t^F = \sum_{n=1}^{N} \left(s_t^n\right)^2 , \quad \text{and} \quad H_t^{I,n} = \sum_{j=1}^{N} \left(\lambda_t^{jn}\right)^2 , \forall n ,$$

where $H_t^F$ denotes the Herfindahl-Hirschman index (HHI) of concentration in the provision of loans to firms, while $H_t^{I,n}$ denotes the HHI of concentration in the funding sources of bank $n$. These indices summarize concentration on the output side (loans) and the input side (funding). For example, the multiplier associated with transaction-cost volatility, $\mathfrak{J}^I$, can be written as

$$\mathfrak{J}^I = \underbrace{\Theta_0 \sum_{n=1}^{N} s^n \left[1 - \lambda^{nn}\right]}_{\text{Openness}} + \underbrace{\Theta_0 \left[\left(\frac{\sigma-1}{\kappa}\right) - \left[1 + \Theta_1 H^F\right]\right] \sum_{n=1}^{N} \omega^n \left[H^{I,n} - \left(\lambda^{nn}\right)^2\right]}_{\text{Diversification}} + \text{covariance terms} , \qquad (16)$$

where $\Theta_0 \equiv \left(\frac{\alpha}{1-\alpha}\right)\kappa$ and $\Theta_1 \equiv \left(\frac{\sigma-1}{\kappa}\right) + \frac{\Lambda_{rr}}{\Theta_0}$ are positive constants, and $\omega^n$ are weights that sum to one.

The first term captures a pure *openness* effect. Holding everything else fixed, a bank with greater access to the interbank market, as measured by its share of external funding $1 - \lambda^{nn}$, is better able to substitute across quarters toward *temporarily* cheaper funding sources. This is related to, but conceptually distinct from, the first-order effect captured by the static gains from trade (12), which reflects access to funding sources with lower within-quarter borrowing costs. By allowing banks to exploit transitory funding opportunities across quarters, openness lowers both the average borrowing cost over time and the volatility of that cost, thereby unambiguously raising welfare.

The second term captures the effect of *diversification* across external funding sources. The relevant sufficient statistic is

$$H^{I,n} - \left(\lambda^{nn}\right)^2 \geq 0,$$

that is, the Herfindahl concentration of interbank funding net of own funding. The contribution of this term is ambiguous because diversification has both a benefit and a cost.

On the benefit side, for any given level of *openness*, $1 - \lambda^{nn}$, a more diversified bank can exploit a broader set of temporary low-cost funding opportunities within its network. Because fluctuations in banks' costs of supplying interbank credit are not perfectly correlated, a bank with more diversified connections is more likely to have at least one counterparty that is temporarily able to provide funds cheaply. These gains are larger when $\kappa$ is low, that is, when depositor preferences are more volatile. In that case, relative funding costs vary more across banks at a given moment $\tau$, increasing the value of access to multiple counterparties and making it more likely that a well-connected bank can tap a particularly cheap source of funds.

On the cost side, participation in the interbank market exposes banks to transaction-cost shocks that are absent in autarky.[42] These shocks propagate through the network and affect funding conditions. When banks borrow from only a limited set of counterparties, they cannot fully offset such disruptions by

[42] Recall that autarky transaction costs with oneself are constant, since we normalize $d_t^{nn} = 1$ for all $n$ and $t$.

switching to other lenders.[43] These disruptions are especially costly when firms cannot easily substitute across funding sources, as captured by a low elasticity $\sigma$, or when loan provision is concentrated in a small number of banks, as captured by a high loan Herfindahl index $H^F$.

The sign of the *Diversification* term is therefore governed by

$$\left(\frac{\sigma-1}{\kappa}\right)-\left[1+\Theta_1 H^F\right].$$

When this expression is positive, diversification amplifies the gains from trade because the option value of access to alternative funding sources dominates. When it is negative, diversification reduces the gains from trade because exposure to transaction-cost shocks dominates.

This trade-off is especially transparent in the extreme case $H^F = 1$, in which firm lending is fully concentrated. In that case, the coefficient above is negative, so greater interbank diversification reduces welfare gains. Firms are then completely unable to substitute across sources of capital, and additional transaction-cost volatility mainly raises funding costs. More generally, when $H^F < 1$, as in the data (see Figure 2b), the sign depends on $\left(\frac{\sigma-1}{\kappa}\right)$. A higher $\sigma$ makes firms' funding sources more substitutable, while a lower $\kappa$ increases dispersion in banks' funding conditions. Both forces raise the value of diversification in the interbank network. When $\left(\frac{\sigma-1}{\kappa}\right) > 1+\Theta_1 H^F$, this benefit dominates, and diversification generates additional gains from integration.

The remaining covariance terms capture how comovement in transaction-cost shocks across counterparties contributes to the gains from trade. The multipliers $\mathfrak{J}^T$ and $\mathfrak{J}^a$ reflect analogous mechanisms operating through depositor-preference and firm-loan-demand shocks. Since these components contribute only modestly to the gains from trade in the quantification exercises of Section 7, we do not discuss them here and relegate the full derivations to Online Appendix 3.B.

## 5.2 Gains from the Lender-of-Last-Resort (LoLR)

Similar to the gains from trade, we quantify the welfare gains from LoLR policy by comparing equilibrium welfare to the counterfactual equilibrium without LoLR credit provision:

$$\mathbb{G} \equiv \mathbb{E}\left[\frac{U_t - U_t^{no-LoLR}}{U_X X}\right].$$

The steady-state share of credit supplied by the central bank is

$$\xi^0 = \frac{1}{1+e^{\kappa\varpi_1}},$$

where $\lim_{\varpi_1\to+\infty}\xi^0 = 0$ corresponds to the counterfactual scenario without LoLR credit. In our model, the central bank mitigates intra-quarter interbank volatility generated by the reshuffling of deposits across banks, which creates temporary funding shortfalls. By placing an upper bound on bank funding costs, parameter $\varpi_1$ makes funding within a quarter both more stable and, on average, cheaper.

[43] As discussed in Section 3, small and medium-sized banks typically have fewer than 20 distinct borrowing links; see Figure A.2 in the Online Appendix.

The gains from LoLR are given by

$$\mathbb{G} = -\left(\frac{\alpha}{1-\alpha}\right)\left(1+\frac{1}{\kappa}\right)\log\left(1-\xi^0\right) \geq 0\,, \tag{17}$$

with the derivation provided in Online Appendix 3. The gains from LoLR are therefore monotonically increasing in steady-state central bank participation, $\xi^0$, which summarizes the benefits of lower intra-quarter funding volatility and lower average funding costs. This result should be interpreted with caution however, as high levels of steady-state reliance on LoLR credit are likely to involve misallocation, political, or other costs not captured by the model. Accordingly, (17) is best interpreted as a measure of the gains from LoLR at low steady-state levels of intervention—typically observed in most advanced economies—where these concerns are less likely to be quantitatively important.

# 6 Estimation and Calibration

In this section, we take the model to data on Germany's banking sector introduced in Section 2. First, by examining the propagation of the US financial crisis to German banks through their interbank connections, we underscore the key mechanism in the model, specifically, that a bank's interbank market access is a crucial driver of its lending decisions, interest rates, and funding choices. Using domestic banks' *indirect exposure* to the crisis as a plausibly exogenous funding cost shock, we then estimate the key elasticities $\sigma$ and $\kappa$ which enter the welfare formulas in Section 5. Lastly, we recover and analyze the model-implied "wedges" (i.e., loan demand parameters $\hat{a}_t^n$, deposit supply parameters $\hat{T}_t^n$, and bilateral transaction costs $\hat{d}_t^{ni}$) for each bank and quarter using our data and estimated elasticities from the previous step.

## 6.1 Measuring Banks' Indirect Exposure to the US Financial Crisis

Prior to 2007/2008, several German banks were heavily invested in loans (broadly defined) with banks domiciled in the US. With the onset of the US financial crisis, German banks *directly* exposed to US bank assets experienced serious liquidity problems, as highlighted by Huber (2018) in the case of the two largest German banks, and significantly reduced their lending activity in both the German real economy and the interbank market.

To emphasize the role of interbank market access in transmitting the US financial crisis to the German banking sector and the real economy, we focus our analysis on banks that borrow from *directly* exposed banks in the domestic interbank market prior to the crisis.[44] Specifically, we construct a measure of each bank's *indirect* exposure to the US financial market prior to the Great Recession according to:

$$Exposure_{t_0}^{US,n} = \frac{1}{L_{t_0}^n + \sum_{n\neq i} M_{t_0}^{ni}} \sum_{i\neq n}^{N} \frac{M_{t_0}^{in}}{\sum_{i'\neq n}^{N} M_{t_0}^{i'n}} \mathscr{M}_{t_0}^{i,US}. \tag{18}$$

The last term, $\mathscr{M}_{t_0}^{i,US}$, represents the value of assets that bank $n$'s lender $i$ reports with US banks in $t_0$ in the

[44] Iyer, Peydró, da Rocha-Lopes, and Schoar (2014) and Acabbi, Panetti, Sforza, et al. (2020) use similar exposure measures to study the effect of the interbank market freeze of 2007/08 on firm outcomes in Portugal.

External Position of MFIs (AUSTA) dataset.[45] We weigh each lender $i$'s direct exposure by bank $n$'s liabilities $M^{in}$ with lender $i$ out of $n$'s total interbank liabilities in the initial period $t_0$ and sum over all possible lenders of $n$. Banks with higher *indirect* exposure either borrow heavily from *directly* exposed lenders or have lenders that are heavily invested in the US banking sector. Finally, we divide bank $n$'s *indirect* exposure in Euros by bank $n$'s balance sheet $(L^n_{t_0} + \sum_{n \neq i} M^{ni}_{t_0})$ to capture the importance of exposure in total funding needs.

As the base period $t_0$, we choose 2006Q1, six quarters before the first signs of the financial crisis in 2007Q2, which we set as our event date for the onset of the US financial crisis, and ten quarters before the collapse of Lehman Brothers in 2008Q3. Generally, selecting the exact event date for the financial crisis proves to be somewhat ambiguous. Hence, we opt to present an event study with 2007Q3 as the single event date but exclude 2007Q3-2008Q2 as the event "period" in the model estimation below. We conclude the sample in 2011Q4 to avoid confounding the effect of the US financial crisis with the subsequent Euro-crisis.

Table 2: Summary Statistics of Exposure Instruments

| Exposure IV | Mean | Median | St Dev | 25th | 75th | 5th | 95th | Corr w/ own exposure | Corr of IVs | Obs |
|---|---|---|---|---|---|---|---|---|---|---|
| US MFI 2006Q3 in % | 0.777 | 0.762 | 0.335 | 0.551 | 0.978 | 0.281 | 1.371 | -0.235 | | 163 |
| PIIGS Non-MFI 2006Q3 in % | 3.276 | 3.147 | 1.983 | 1.708 | 3.863 | 1.086 | 7.471 | -0.073 | | 163 |
| US MFI vs PIIGS Non-MFI | | | | | | | | | 0.447 | 163 |

Table reports summary statistics of exposure instruments for 2006Q3. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2011m12, own calculations.

The first row of Table 2 summarizes our exposure instrument. In our sample of 163 unique banks, the mean indirect exposure is .78%, .55% at the 25th percentile, and 1.37% at the 95th percentile.[46]

### 6.2 The Impact of the US Financial Crisis on the German Interbank Market

Now, we examine the effect of indirect exposure to the US financial crisis on various bank-level outcomes, such as interest rates, lending decisions, and funding choices. Our model predicts that banks will reduce interbank borrowing and raise loan interest rates in response to the adverse funding cost shock, subsequently leading to a decrease in loan demand relative to unaffected banks. Furthermore, if banks are able to partially substitute now more expensive interbank sources with deposits and equity, a decrease in interbank borrowing should surpass the reduction in loans. This substitution is evidenced by an increase

[45] We have to restrict lenders to the 80 banks present in the External Position of MFIs dataset. We assume all other banks have zero direct exposure. However, these 80 banks cover 90% of all foreign assets held by the German financial sector.

[46] Since we are interested in the effect of exposure to the US financial crisis on interest rates, we are restricted to banks present in the interest rate sample (ZISTA). However, the exposure term in equation (18) is constructed from data on the universe of German banks.

in the proportion of final loans financed from a bank's own sources, or "own share" in funding.

To empirically evaluate these model predictions, we investigate the impact of indirect exposure on bank $n$'s outcome variable $y_t^n$ (e.g., final loan interest rate) over time using the following event-study design:

$$\log y_t^n = \rho_n + \mu_t + \sum_{\tau=2004Q4}^{2011Q4} \delta_\tau \left( Exposure_{2006Q1}^{US,n} \times \mu_\tau \right) + X_t^n \beta + u_t^n \,,$$

where $\rho_n$ denotes a bank fixed effect, and $\mu_t$ represents a fixed effect for each quarter in our sample. We include a vector of time-varying controls $X_t^n$: shares of various loan products in bank $n$'s aggregate loans (e.g., different borrower types or maturities) to eliminate the potential for capturing variation in the average loan rate that arises from adjustments in the types of loan products; bank $n$'s direct exposure to US MFI assets to mitigate the possibility of a spurious correlation with its indirect exposure;[47] leverage and repayment rates on banks' loan portfolio to account for changes in risk. Importantly, our findings remain almost identical when excluding these controls (see Figure A.3). We employ cluster robust standard errors at the bank group-quarter level.

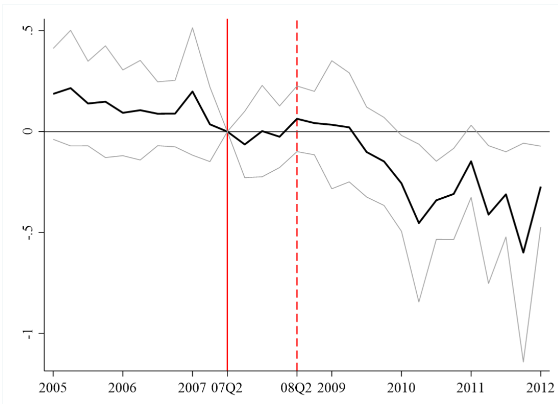
In Figure 5, we present estimates of $\delta_\tau$ and 95% confidence intervals for four distinct bank variables. To demonstrate that exposed banks experience limited access to the interbank market following the shock, we plot the coefficients for the log value of domestic interbank borrowing in a). After the crisis, banks reduce their liabilities with domestic banks by up to 30% at mean exposure, and this level is maintained thereafter. Importantly, we observe no significant pre-trend in interbank borrowing prior to the crisis.

In line with our interest rate predictions, the point estimates in b) indicate that interest rates on outstanding loans to firms and consumers rose sharply after the crisis, increasing by up to 30 basis points per one percent of indirect exposure and remaining elevated over the sample period. Our estimated effects are substantial: on average, a bank with the mean indirect exposure of .75% contracts at an interest rate approximately 23 basis points higher than an otherwise comparable bank with zero exposure. Reassuringly, there is no significant difference in interest rates between more and less exposed banks before the crisis.

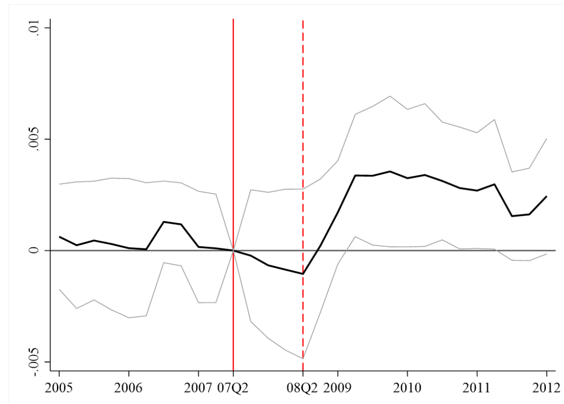
Figure c) investigates the impact of indirect exposure on outstanding loans to firms and consumers. We identify a significant and persistent decline of around 7% per one percent exposure immediately after the crisis, with a small but insignificant pre-trend. The funding cost shock originating from banks directly exposed to faltering US assets results in a reduction in credit to the real economy of approximately 5% by banks with mean indirect exposure.

We show the regression coefficients for the log "own share" in d). If banks were to reduce their own funds in the same proportion as interbank borrowing in order to fund the reduced loan demand, we should not observe any impact of indirect exposure on own share. Our estimates suggest that banks gradually raised their own share by up to 8% per one percent of exposure 4 years after the crisis, or 6% at mean exposure. Consistent with this estimate, between 2008 and 2011 the aggregate own share rose by approximately 6-7%, as shown in Figure 1.

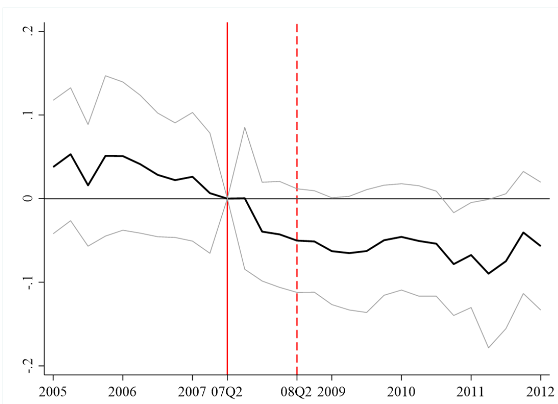
[47] As shown in Table 2, the indirect exposure instrument is negatively correlated with direct exposure with a correlation of −.235.



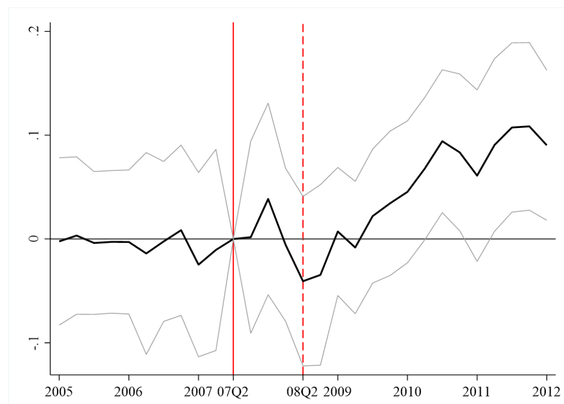

(a) Interbank borrowing




(b) Interest rate on outstanding final loans




(c) Outstanding final loans




(d) Own share

Figure 5: This event-study examines the impact of indirect exposure to the US financial crisis on borrowing from other German MFIs (upper left), interest rates of final loans (upper right), the quantity of final loans (lower left), and the share of funding from own sources (lower right). Each figure presents coefficients on $Exposure^{US}_{2006Q1} \times Quarter - FE$ and 95% confidence intervals. Solid red vertical lines indicate 2007$Q$2, the event quarter immediately preceding the crisis, while dashed red lines represent 2008$Q$2, just before the collapse of Lehman Brothers. The regression incorporates quarter fixed effects, bank fixed effects, direct asset exposure to the US, leverage and repayment rates, and loan shares of non-MFI and households. These loan shares are further divided into maturities of less than 1 year, between 1 to 5 years, and more than 5 years, as well as separate shares for secured and unsecured mortgages. Robust standard errors are clustered at the bank group-quarter level. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m1 - 2012m1, own calculations.

Since our loan and interest rate measures are weighted averages over different maturities, we investigate the effect of indirect exposure to the US bank assets on rates and volumes of loans with maturity under 1 year, between 1 and 5 years, and over 5 years, using the ZISTA sample. Figure A.4 shows that interest rate responses are very similar but more volatile at low maturities. In Figure A.5, we plot the response of the respective loan volumes which exhibit a less clear picture. Although short and very long maturities do not show significant declines, medium-term loans fall. Furthermore, our model predicts that banks that face funding shortages on the interbank market can increase their own funds by raising deposit rates to attract more deposits. The figures on the right bottom of A.4 and A.5 confirm this model prediction.

### 6.3 Estimation of Banking Sector Elasticities $\sigma$ and $\kappa$

To estimate the demand elasticity $\sigma$, we start by taking the logarithms of both sides of bank $n$'s loan demand in equation (3) and replacing aggregate, time-varying variables with a time-fixed effect,

$$\log L_t^n = \mu_t - \sigma \log R_t^{F,n} + \log a_t^n \,, \tag{19}$$

where $\log a_t^n$ captures the time-varying demand shifter for MFI $n$ at time $t$. First, we assume $\log a_t^n = \rho_n + \beta' X_t^n + \epsilon_t^n$, where $\rho_n$ is the bank fixed effect and $X_t^n$ represents time-varying controls, as discussed previously. Second, the causal identification of $\sigma$ requires exogenous variation in interest rates for bank $n$ at time $t$. We argue that our identification strategy in the preceding section provides variation in loan interest rates that is uncorrelated with demand shocks, $\epsilon_t^n$.

We proceed analogously with the estimation of supply elasticity $\kappa$. By substituting equation (8) into (9) for the probability that bank $n$ obtains funds from its own deposits, we establish a structural relationship between interest rates and the "own" share of bank $n$, i.e., the ratio of $M_t^{nn}$ and loans $L_t^n$ at time $t$. Specifically,

$$\frac{M_t^{nn}}{L_t^n} = \left(d_t^{nn}\right)^{-\kappa} \left(T_t^n\right)^{-\kappa} \left(\frac{R_t^{F,n}}{R_t^B}\right)^{\kappa} \left(\frac{\sigma-1}{\sigma}\right)^{\kappa}.$$

Taking logarithms, assuming $d_t^{nn} = 1, \ \forall t$, and collecting all constant and aggregate, time-varying variables into a time-fixed effect, we arrive at

$$\log\left(\frac{M_t^{nn}}{L_t^n}\right) = \zeta_t + \kappa \log R_t^{F,n} - \kappa \log T_t^n. \tag{20}$$

Equation (20) states that bank $n$ resorts to funding through its own funds with elasticity $\kappa$ when the cost of funds, including funding from the interbank market, increases. We capture the cost of funds (or interest rate spread) as the interest rate on loans to the real economy net of the bond rate.[48] However, the funding costs of a bank are correlated with unobservable shocks to own funding $T_t^n$ by construction. We assume

[48]While our model implies a direct mapping between the rate on final loans and the rate with which banks finance their operations, $R_t^{F,n} = \left(\frac{\sigma}{\sigma-1}\right) R_t^{I,n}$, the final loan rate in the data likely includes time-varying risk of the loan portfolio. Since we cannot directly observe the interbank rate of individual banks, we control for this type of risk by including the repayment rates on the loan portfolio and bank leverage in the regressions.

$\log T_t^n = \xi_n + \gamma' X_t^n + v_t^n$, where $\xi_n$ is a bank fixed effect and $X_t^n$ represents time-varying bank-level controls. We interpret $v_t^n$ as an own funds supply shock likely correlated with $R_t^{F,n}$. Equation (20) resembles equation (19), with the difference that we use the own share as the dependent variable. We make the analogous identification assumption as above, specifically that indirect exposure to the US financial crisis is uncorrelated with bank-level deposit shocks $v_t^n$, and captures variation in $R_t^{F,n}$ coming through the part of bank $n$'s funding costs related to interbank funds. Consequently, we estimate equations (19) and (20) with 2SLS using the interaction of indirect exposure in equation (18) and the post-crisis dummy as an instrument for loan rates $R^{F,n}$.

Table 3: Estimation of $\sigma$ and $\kappa$

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Panel A: | Loan Rate | Loans | Own Share | Loan Rate | Loans | Own Share |
| $Exposure_{t_0} \times Post_t$ | 0.003*** | -0.095*** | 0.055*** | 0.002*** | -0.091*** | 0.063*** |
| | (0.001) | (0.028) | (0.017) | (0.001) | (0.017) | (0.018) |
| R-squared | 0.926 | 0.992 | 0.911 | 0.939 | 0.994 | 0.915 |
| Mean Exposure | 0.757 | 0.757 | 0.757 | 0.768 | 0.768 | 0.768 |
| Panel B: | *2SLS: Instrument for* $\log R_t^{F,i}$ *is* $Exposure_{t_0} \times Post_t$ | | | | | |
| | | $-\sigma$ | $\kappa$ | | $-\sigma$ | $\kappa$ |
| $\log R_t^{F,i}$ | | -37.75*** | 21.97*** | | -42.65*** | 29.61*** |
| | | (8.74) | (6.88) | | (7.88) | (8.38) |
| Observations | 3587 | 3587 | 3587 | 3532 | 3532 | 3532 |
| 1st-Stage F-Stat | | 11.63 | 11.63 | | 15.19 | 15.19 |
| Controls | no | no | no | yes | yes | yes |

Panel A compares outcomes between 2006$Q1$ and 2007$Q2$, and 2008$Q3$ to 2011$Q4$ (post-period) for banks that were more or less indirectly exposed to the US financial crisis. Panel B presents the results of estimating equations 19 and 20 using 2SLS with $\text{Exposure}_{t_0} \times \text{Post}_t$ as an instrument. Initial asset exposure to lenders in the US market is taken in 2006$Q1$. Controls include direct asset exposure to the US, leverage, repayment rates on loans, and loan shares of non-MFI and household loans, each broken down into maturities of less than 1 year, between 1 and 5 years, and more than 5 years, as well as separate shares for secured and unsecured mortgages. All regressions include bank fixed-effects and quarter fixed-effects. Standard errors are clustered at the level of the bank group-quarter. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2011m12, own calculations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Panel A of Table 3 presents the first stages (loan rate on the LHS) and reduced forms (loans and own share on the LHS) without controls in columns 1-3 and with controls in columns 4-6. The inclusion of controls has little impact on the size of the coefficients, further supporting our assertion that indirect exposure represents exogenous variation from the perspective of German banks.

Panel B of Table 3 reports IV estimates of $\sigma$ in columns 2 and 5 and $\kappa$ in columns 3 and 6. The instrument is relevant, as demonstrated by the first-stage F-statistic of around 15 with controls. Our preferred estimate of loan demand elasticity $\sigma$ is 42.6 in column 5. Reassuringly, this value is similar to other estimates from the literature.[49] Regarding our IV estimates of $\kappa$, we find that the funding cost shock

[49] For example, Ulate (2021) estimates loan demand elasticities of 26.6 for the US, 37.4 for the Eurozone, and around 40 for the UK and Japan.

in the interbank market leads to a significant increase in banks' reliance on own funding sources, with an elasticity of 29.6 with controls and 22 without controls. For our subsequent model analysis, we choose $\kappa = 29.6$ as our preferred estimate for the interbank supply elasticity. Our estimated elasticities imply that the average bank in response to a funding cost shock on the interbank market and lower loan demand reorganizes its internal funding structure by reducing interbank borrowing more than twice as much as its own funds.[50]

In Table A.1, we present coefficients from a pre-trends test. We now include five quarters before 2006Q1 (when we construct the exposure measure) and define the Post dummy for the six quarters up to 2007Q2 (our pre-period in the main estimation). With this setup, we can evaluate whether indirect exposure can predict trends in our outcome variables for the pre-period relative to an earlier period. Our results suggest that there are no significant pre-trends in interest rates and weakly significant but small effects on loans and the own share.

Table A.2 shows several robustness checks. In columns 1-3, we construct our indirect exposure measure using all US assets (instead of only US MFI assets). We find similar but slightly smaller results for loans and own share, and stronger effects on interest rates, leading to smaller estimates of $\sigma$ and $\kappa$. In columns 4-6, we use the absolute exposure in terms of Euros as instrument. The difference-in-difference results are very similar evaluated at mean exposure, as are the estimates of the elasticities. In columns 7-9, we count lending to foreign MFIs as loans and borrowing from foreign MFIs as deposits instead of netting out the foreign interbank position as in the main specification. Estimates are broadly similar; however, the drop in loans is smaller, suggesting that banks reduced foreign less than German interbank activity. Finally, we interact the exposure measure after the crisis with a dummy for banks that are initially net borrowers in the interbank market. We expect net borrowers to be more affected by a funding cost shock, since their ability to cover structural deficits on the interbank market is now limited, in addition to short-term liquidity management. Indeed, we find larger effects for net borrowers across all outcomes.

For our final robustness check, we study the effect of indirect exposure of German banks to the Euro crisis which we date at 2009Q4-2012Q4 but keep the base period before the Great Recession (2006Q1-2008Q1). We describe this robustness check in more detail in Appendix 4.A. Table A.3 shows our estimates of $\sigma$ and $\kappa$ based on the same difference-in-difference approach as in the main specification. Our results are considerably noisier than the main specification, but the direction and magnitude of the effects are similar. In our welfare analysis below, we report results for alternative values of $\sigma$ and $\kappa$ based on these robustness checks.

### 6.4 Recovering Model "Wedges"

Having estimated the elasticities $\sigma$ and $\kappa$ in the previous subsection, we now turn to characterizing the "wedges" related to the financial shocks $\{a_t^n,\, T_t^n,\, d_t^{ni}\}_{\forall n,i,t}$. First, we use observed bank-level data on loan interest rates, $\left\{R_t^{F,n}\right\}_{\forall n,t}$, loans to the real economy, $\{L_t^n\}_{\forall n,t}$, shares of funding bank $i$ receives from bank $n$, $\{\lambda_t^{ni}\}_{\forall n,i,t}$, and the funding share of the central bank for each bank $n$, $\{\xi_t^{0n}\}_{\forall n,t}$, to recover estimates of

[50]By solving equation (19) for $\log R_t^{F,n}$ and plugging into equation (20), one can see that $\Delta \log M_t^{nn} \approx \left(1-\frac{\kappa}{\sigma}\right)\Delta \log L_t^n$. Our estimates imply that $\left(1-\frac{\kappa}{\sigma}\right) \approx .3$. Alternatively, the reduced forms in Panel A of Table 3 suggest a reduction in $M^{nn}$ of 2.8% per one percent of exposure ($.063-.091$), or roughly 30% of the reduction in loans.

the financial shocks for each quarter from the model's equilibrium relationships.
We back out estimates for $\{T_t^n\}_{\forall n,t}$ as

$$\hat{T}_t^n = \left(\lambda_t^{nn}\right)^{-1/\kappa}\left(1-\xi_t^{0n}\right)^{-1/\kappa}\left(\frac{\sigma}{\sigma-1}\right)^{-1} R_t^{F,n},$$

which is an expression that we obtain after combining equation (9) for the own trade share together with equations (5) and (8). Using the same set of equations but employing the formula for the bilateral trade share between any $\{n, i\}$ pair of banks, we obtain our estimates for $\{d_t^{ni}\}_{\forall n,i}$ as

$$\hat{d}_t^{ni} = \left(\hat{T}_t^n\right)^{-1}\left(\lambda_t^{ni}\right)^{-1/\kappa}\left(1-\xi_t^{0i}\right)^{-1/\kappa}\left(\frac{\sigma}{\sigma-1}\right)^{-1} R_t^{F,i}. \tag{21}$$

Finally, we recover the $\{a_t^n\}_{\forall n,t}$ shocks by using CES loan demand from equation (3).

For our model analysis, we calculate wedges for the around 1500 German banks and all their bilateral connections in each quarter. However, the interest rate sample (ZISTA) covers only between 200 and 240 banks per quarter, as discussed in Section 2.[51] We construct predicted interest rates for the remaining banks in the main sample with a regression of interest rates on observable bank characteristics and detailed balance sheet composition, which we observe for all banks and quarters in our main dataset. We test for sample selection bias and prediction performance by excluding the two smallest ZISTA banks of each stratum (approximately 35 banks per quarter). The R-squared of our prediction is 93%, and the average out-of-sample deviation of the predicted versus observed interest rates for the excluded sample is -1.7 basis points, suggesting that predicting using this selected sample does not create sizable bias for smaller banks outside the sample.

The size and nature of transaction costs $\hat{d}_t^{ni}$ are central in determining the openness and volatility of the interbank market and, consequently, the welfare effects of financial market integration. In Table A.4, we show several summary statistics of our recovered transaction costs from the previous steps, each broken down into the period before, during and after the financial crisis. Average transaction costs are around 1.3 with increases in levels and volatility after 2007Q3. The standard deviation conditional on a given lender is smaller than the overall standard deviation, suggesting that lenders engage in relatively uniform pricing vis-a-vis their borrowers (e.g., Begenau and Stafford (2023) for uniform pricing in deposits). The variation at the borrower level is the lowest, which implies that a given borrowing bank receives similar rates, but rates vary largely across borrowers, evidence that our recovered transaction costs capture borrower characteristics such as risk or quality of collateral. On the extensive margin, the average number of connections has decreased from 15 to 13 during and after the crisis. In the last panel, we calculate the transaction cost as a share of loan volume for each bank. On average, banks spend 1.2% of their funding needs on interbank transaction costs, ranging from .5% at the 25th percentile to 3.5% at the 95th percentile. Funding cost shares increase slightly during the crisis but are generally stable over the sample period due to two opposing forces: during and after the crisis banks rely more on own funds without

[51]The reported banks are selected through stratified sampling, which assigns banks to between 15 and 17 groups using a criterion that combines headquarter state and bank groups in order to capture regional and institutional heterogeneity. The largest banks within each stratum are then selected into the sample.

transaction costs but face higher costs on the remaining interbank borrowing.

Table 4: Correlations with Recovered Trade Wedges $\hat{d}_t^{ni}$

| | (1) | (2) | (3) |
|---|---|---|---|
| | | $\log \hat{d}_t^{ni}$ | |
| Log distance | 0.0022*** | 0.0018** | |
| | (0.0008) | (0.0007) | |
| Same state | 0.0208** | 0.0155* | |
| | (0.0089) | (0.0087) | |
| Same bank group | -0.0021 | -0.0034 | |
| | (0.0026) | (0.0026) | |
| Log size, Lender | | -0.0294*** | -0.0345*** |
| | | (0.0076) | (0.0091) |
| Log size, Borrower | | 0.0062 | -0.0024 |
| | | (0.0198) | (0.0208) |
| Log number connections, Lender | | 0.0061** | -0.0003 |
| | | (0.0031) | (0.0039) |
| Log number connections, Borrower | | 0.0001 | -0.0031 |
| | | (0.0054) | (0.0054) |
| Loan Default rate, Lender | | -0.0728 | -0.0391 |
| | | (0.0542) | (0.0272) |
| Loan Default rate, Borrower | | 0.1526 | 0.2463* |
| | | (0.1936) | (0.1395) |
| Observations | 366417 | 384530 | 381549 |
| R-squared | 0.7401 | 0.7265 | 0.8837 |
| Lender-Time & Borrower-Time FE | yes | no | no |
| Lender & Borrower FE | no | yes | no |
| Lender–Borrower FE | no | no | yes |

Table shows regressions of recovered trade wedges $\hat{d}_t^{ni}$ on various bilateral and bank level variables. Distance is calculated as the distance between the state capitals in which the banks' headquarters are located. Default rate calculated as one minus the share of loan repayment at the bank level. All regression include a quarter fixed effect. The process of how these wedges are recovered is described in Section 6.4. Standard errors clustered two-way at lender- and borrower-level. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

To investigate the origin of our recovered transaction costs, we regress $\log d_t^{ni}$ on several bilateral and lender/borrower-level variables. In column 1, we include lender-time and borrower-time fixed effects to assess the effect of bilateral variables (distance, state, same bank group) on the size of transaction costs. Transaction costs increase with physical distance, but are higher if lender and borrower have headquarters in the same state. In column 2, we include time, lender and borrower fixed effects to study how variables at the level of lender and borrower banks influence costs over time, while we include a time and a bilateral fixed effect in column 3 to assess how bank-level variables affect given bilateral connections over time. Lenders with large balance sheets have significantly lower transaction costs, which points to scale effects in screening and monitoring. Most importantly, when borrower banks exhibit higher risk of default in their loan portfolio, they transact at significantly higher costs. Consistent with our hypothesis that transaction costs are related to the monitoring of borrowers, the default rate of lenders themselves is not relevant to transaction costs.


Figure 6: Interbank Spread $\widetilde{R}^I_t$

Figure 7: Figure shows the evolution of the model-implied, aggregate gross interbank interest spread $\widetilde{R}^I_t$ in logs, normalized to zero at the event date 2007Q2. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations.

In Figure 6, we show log changes of the gross interbank interest spread $\widetilde{R}^I_t$ as implied by our recovered transaction costs, loan demand shifters $a^n_t$ and deposit supply shifters $T^n_i$.[52] With the collapse of Lehman Brothers the interest rate spread increases by 3%, or roughly 2% relative to the pre-crisis period average and remains elevated thereafter. The Euro-crisis around 2012 produces another spike in the interbank spread. Thus, the shock series recovered from the data and our model structure are consistent with key events such as the interbank market "freeze" in the financial crisis and show similar movements as the empirical credit spread in Figure 2a.

## 6.5 Stochastic Properties of Wedges

In the last step of our empirical analysis, we estimate the stochastic properties of our recovered wedges needed for our structural welfare formula in Section 5. Table 5 reports the estimated parameters, introduced in 5.1.3, underlying the stochastic processes in column 1 and the respective method used to obtain the parameters and standard errors in column 4. Since we find that shocks are fairly persistent with an auto-correlation of .7 to .8 for all three shocks, we assume for our welfare calculations a single autoregressive coefficient $\rho$ as the average of the three estimated parameters. The between-bank correlation coefficient $\zeta_{I,B}$ is particularly high, which implies that transaction cost shocks affecting a given borrower are correlated across the bank's lenders, much less so than the correlation of shocks affecting borrowers of

[52]The aggregate spread is calculated as $\widetilde{R}^I_t = \left[\sum_{i=1}^N a^i_t \left(\widetilde{R}^{I,i}_t\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$, where $\widetilde{R}^{I,i}_t \equiv \frac{R^{I,i}_t}{R^B_t} = \Phi^i_t\left(1-\xi^{0i}_t\right)^{1/\kappa}$ and $\Phi^i_t = \left[\sum_{n=1}^N \left(d^{ni}_t T^n_t\right)^{-\kappa}\right]^{-1/\kappa}$.

the same lender ($\zeta_{I,L}$). Transaction cost shocks have a standard deviation of .038 which is equivalent to approximately 30% of the unconditional variation in transaction costs in the panel (see Table A.4). The standard deviation of depositor preference is much smaller; however, those shocks spread with relatively large $\zeta_T$ between banks. Finally, the very large standard deviation of loan demand shocks ($\sigma_a$) requires additional explanation. Since the CES demand shifters $a_t^n$ add to one in every $t$ by assumption, we need to interpret $\sigma_a$ as the standard deviation of the odds-ratio. Assuming that the average $a_t^n$ is around $1/N$, the change in $a_t^n$ in response to a standard deviation shock $du_t^{a,n}$ is given by $da_t^n = a_t^n\left(1-a_t^n\right)du_t^{a,n}$. Thus, to get the standard deviation of $a_t^n$, we need to divide $\sigma_a$ by $N \approx 1500$.[53]

Table 5: Estimated Shock Parameters

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Parameter | Estimate | Standard Error | Variable | Method |
| $\boldsymbol{\rho}_T$ | 0.774 | (0.026) | $T^n$ | AR(1) on demeaned $\log T^n$ at bank-level, two-way clustered by bank and quarter |
| $\boldsymbol{\sigma}_T$ | 0.017 | (0.002) | $T^n$ | Delta-method transform |
| $\boldsymbol{\zeta}_T$ | 0.228 | (0.090) | $T^n$ | Quarter-collapsed off-diagonal correlation, Newey-West lag 4 |
| $\boldsymbol{\rho}_a$ | 0.822 | (0.031) | $a^n$ | AR(1) on cross-sectionally centered loan shares, two-way clustered by bank and quarter |
| $\boldsymbol{\sigma}_a$ | 0.337 | (0.043) | $a^n$ | Delta-method transform |
| $\boldsymbol{\zeta}_a$ | -0.001 | (0.000) | $a^n$ | Quarter-collapsed off-diagonal correlation, Newey-West lag 4 |
| $\boldsymbol{\rho}_I$ | 0.721 | (0.039) | $d^{ni}$ | AR(1) on dyad-level $\log d^{ni}$, clustered by lender, borrower and quarter |
| $\boldsymbol{\sigma}_I$ | 0.038 | (0.004) | $d^{ni}$ | Delta-method transform |
| $\boldsymbol{\zeta}_{I,B}$ | 0.507 | (0.142) | $d^{ni}$ | Quarter-collapsed same-destination correlation, Newey-West lag 4 |
| $\boldsymbol{\zeta}_{I,L}$ | 0.137 | (0.036) | $d^{ni}$ | Quarter-collapsed same-origin correlation, Newey-West lag 4 |
| $\boldsymbol{\zeta}_{I,X}$ | 0.008 | (0.004) | $d^{ni}$ | Quarter-collapsed different-origin-destination correlation, Newey-West lag 4 |
| $\boldsymbol{\rho}$ | 0.772 | | $T^n$, $a^n$, $d^{ni}$ | Average of $\rho_T$, $\rho_I$, $\rho_a$ |

Table shows estimates of stochastic properties of recovered shocks to transaction costs ($u_t^{I,ni}$), depositor preferences ($u_t^{T,n}$) and loan demand ($u_t^{a,n}$). Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations.

## 6.6 Calibrated Parameters

The fixed penalty rate $\varpi_1$ in the LoLR policy function (7) is calibrated to match the pre-crisis central bank trade share of 3.5% observed in the data, which we treat as an approximation of its steady-state value.

The remaining parameters are calibrated to values widely accepted in the literature. The subjective discount factor $\beta$ is set to align with a steady-state nominal risk-free rate $R^B$ of approximately four percent,

[53] We report estimated cross-correlations between shocks in Table A.5. Loan demand shocks and depositor preferences of banks are positively correlated since both capture bank size either on the loan or depositor side. Although depositor preferences of the lending bank and transaction costs are significantly correlated with around .26, this is due to the way $\hat{d}_t^{ni}$ are calculated in equation (21). Transaction cost shocks and depositor preferences of the borrower are uncorrelated, but loan demand shocks $a_t^n$ and transaction costs $d_t^{ni}$ are correlated which implies that large banks in terms of loan activity lend (borrow) at lower (higher) costs in the interbank market, consistent with our regression results in Table 4. However, we argue above that the volatility of $a_t^n$ is very small – i.e., bank sizes in the loan market are very stable – suggesting that our assumption of zero cross-correlation between $a_t^n$ and $d_t^{ni}$ is not biasing our welfare results.

consistent with historical data. The labor income share $(1-\alpha)$ is calibrated to sixty percent. The Frisch elasticity of labor supply $\eta$ is set to one, a midpoint between estimates in the micro and macro literature.[54] Steady-state trend inflation is set to zero, following standard practice in the literature (Woodford, 2003, Galí, 2015) to avoid the additional analytical complexity of welfare approximations under non-zero trend inflation.[55] Finally, the Calvo price stickiness coefficient $\theta$ and the Taylor rule parameters are set according to Coibion, Gorodnichenko, and Wieland (2012). Table A.6 reports all parameters.

# 7 Quantification

In this section, we combine the welfare formulas developed in Section 5, our estimated model parameters, and data on the German banking sector to quantify the welfare gains from interbank market integration. First, we evaluate the reduction in welfare and the associated channels for the scenario in which the current observed interbank market in Germany is no longer active. Second, we change the underlying structure of the interbank market in terms of bank concentration, stochastic processes, or the existence of the lender-of-last-resort to study how gains from trade depend on these market features.

## 7.1 Gains from Trade in the German Interbank Market

Our first set of results quantifies how the welfare of the representative household changes in a counterfactual economy in which banks cannot trade funds in the interbank market, so that all banks operate in autarky, relative to the observed interbank market. Section 5 presents two approaches to this exercise.

The first approach relies on the sufficient-statistic formula in equation (14), which approximates the gains from trade in interbank markets using a single observable moment: the aggregate own-funding share, $\lambda_t^{own}$. Given values for the elasticities $\kappa$ and $\alpha$, this expression can be applied in any setting in which the aggregate own-funding share is observed over time.[56] The first row of Table 6 reports the gains from trade implied by this statistic. According to this approach, moving from the observed interbank market to autarky reduces welfare by 0.71% of quarterly consumption. Its main advantage is that it does not require information on either the structure of the network or the stochastic processes affecting the interbank market. Its drawback is that it relies on the assumption that the aggregate own-funding share is only weakly correlated with the unobserved autarky credit spread, formally $\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\,\hat{\hat{r}}_t^{I,AU}\right] \approx 0$. More importantly, by construction, this formula provides limited information about the underlying sources of the gains from trade.

Our second approach addresses these limitations through the structural expression in equation (15), which allows us to decompose the gains from trade into economically meaningful components. The second row of Table 6 shows that, under this approach, the gains from trade in the German interbank market amount to 0.97% of quarterly consumption. This approach is more informative, but it also requires stronger structure: in particular, it requires us to take a stance on the mapping between the

[54] See Peterman (2016) for further discussion.

[55] As shown by Dordal i Carreras, Coibion, Gorodnichenko, and Wieland (2016), the approximation error remains relatively small for economies with stable, low inflation levels, such as Germany.

[56] In the absence of nominal rigidities, that is, in the RBC version of the model, the coefficient satisfies $\Lambda_{rr} = \left(\frac{\alpha}{1-\alpha}\right)^2$. With nominal rigidities, the corresponding expression for $\Lambda_{rr}$ is provided in Online Appendix 3.

trade shares and the interbank network through the underlying modeling assumptions, as well as on the variance-covariance structure of the shocks.

The remaining rows of Table 6 unpack these structural gains. Rows three and four show that 62.12% of the total gains come from first-order static gains, while the remaining 37.88% come from second-order moments, which we refer to as dynamic gains. This decomposition shows that a substantial share of the gains from interbank integration arises not only from the steady-state reduction in transaction frictions, but also from the way interbank integration interacts with fluctuations in banks' funding conditions over time.

Looking more closely at the dynamic component, the gains are driven primarily by the volatility of transaction-cost shocks, while the volatility of depositor-preference and firm-loan-demand shocks plays a much smaller role. Going one step further, the gains associated with transaction-cost volatility are themselves dominated by the *Openness* term introduced in Section 5.1.3, which accounts for 89.14% of that component. This term captures the welfare gains from banks being able to lower both the level and the volatility of borrowing costs across quarters by accessing the interbank market. The *Diversification* term explains an additional 6.32% of these gains, while the remaining share is accounted for by covariance terms across shocks.

Finally, the static gains in equation (12) are independent of the chosen approach and amount to about 0.6% of quarterly consumption. They therefore provide a natural lower bound for the gains from interbank trade. The total gains additionally include second-order effects, which likely place the overall magnitude in the 0.71%–0.97% range implied by the two approaches above.

Table 7 reports a robustness exercise on both the magnitude and the decomposition of the gains from trade under alternative calibrations of the loan-demand elasticity $\sigma$ and the deposit-preference shape parameter $\kappa$. The alternative values are chosen as the extremes of the point estimates of the robustness checks in Section 6, while the benchmark calibration is shown in bold in the center of the table.

Several patterns emerge. First, $\kappa$ is the main driver of the overall size of the gains from trade. When $\kappa$ is low, depositor preferences are more volatile, which creates larger fluctuations in banks' funding conditions and therefore more opportunities to reduce borrowing costs by accessing external funding sources through the interbank market. As a result, the gains from trade increase substantially as $\kappa$ falls, exceeding 2% of quarterly consumption when $\kappa = 12$.

By contrast, holding $\kappa$ fixed, variation in $\sigma$ has a more moderate effect on the overall magnitude of the gains. However, it has a large effect on their decomposition. When $\sigma$ is low, more than 60% of the gains are static. When $\sigma$ is high, the relative importance of dynamic and static gains can reverse, especially when $\kappa$ is also low. The reason is that a higher $\sigma$ makes firms' funding sources more substitutable, allowing them to take greater advantage of temporary fluctuations in funding costs across banks and over time. This increases the importance of the dynamic component of the gains from trade.

Finally, the decomposition of dynamic gains shows that the *Openness* term is the main source of dynamic gains in most calibrations. The main exception is the high-$\sigma$, low-$\kappa$ case, in which the three channels contribute in more similar proportions. The table also shows that the *Diversification* channel can become negative when the ratio $\left(\frac{\sigma-1}{\kappa}\right)$ is low. In those cases, the volatility costs associated with

Table 6: Welfare Gains from Interbank Trade

| Measure | Gains From Trade |
|---|---|
| **Sufficient Statistic Approach** in %, see equation (14) | 0.71 |
| **Structural Approximation** in %, see equation (15) | 0.97 |
| Share Static Gains in % | 62.12 |
| Share Dynamic Gains in % | 37.88 |
| Decomposition of Dynamic Gains by Shock, see equation (15) | |
| Share Loan Demand Shocks in % | 0.64 |
| Share Deposit Supply Shocks in % | -7.19 |
| Share Transaction Cost Shocks in % | 106.56 |
| Decomposition of Transaction Cost Shock, see equation (16) | |
| Share Openness in % | 89.14 |
| Share Diversification in % | 6.32 |
| Share Covariances of Shocks in % | 4.54 |

The table presents gains from interbank trade in Germany under main calibration of $\sigma$ and $\kappa$. Estimates are reported as percentages of steady-state consumption. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations.

transaction-cost shocks outweigh the benefits of broader interbank diversification, because firms find it difficult to substitute across funding sources and/or fluctuations in depositor preferences generate only limited movements in interbank funding costs.

## 7.2 Gains in Counterfactual Interbank Markets

Table 8 reports gains from interbank trade under a range of counterfactual market structures and stochastic environments. These exercises show how the gains from trade depend on bank heterogeneity, the level of bilateral interbank frictions, the macroeconomic environment, and the presence of lender-of-last-resort support.

We begin with two counterfactuals that isolate the role of cross-sectional bank heterogeneity. In CF1, we adjust the demand shifters $a^n$ to equalize each bank's share of lending to firms. This reduces heterogeneity on the lending side and lowers concentration in loan provision, as captured by the Herfindahl index $H^F$. The gains from trade rise only slightly, from 0.97% to 1.03% of quarterly consumption, suggesting that concentration in loan sizes is not a major determinant of the gains under the baseline calibration. By contrast, in CF2 we equalize depositor preferences across banks by setting the steady-state preference parameter $T^n$ to its cross-sectional average. This removes heterogeneity in banks' ability to attract deposits and raises the gains from trade much more substantially, to 1.62% of quarterly consumption. Taken together, these two exercises suggest that deposit-side heterogeneity matters more for the

Table 7: Welfare Gains from Interbank Trade, Robustness to Alternative Values of $\kappa$ and $\sigma$

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Robustness | Gains in % | Static Gains | Dynamic Gains | Openness | Diversification | Covariance |
| $\sigma = 14$ & $\kappa = 12$ | 2.48 | 64.60 | 35.40 | 98.05 | 1.20 | 0.75 |
| $\sigma = 42.6$ & $\kappa = 12$ | 2.14 | 48.05 | 51.95 | 56.27 | 25.35 | 18.38 |
| $\sigma = 80$ & $\kappa = 12$ | 2.36 | 30.37 | 69.63 | 35.41 | 37.82 | 26.76 |
| $\sigma = 14$ & $\kappa = 29.6$ | 1.11 | 69.33 | 30.67 | 123.35 | -12.98 | -10.37 |
| $\sigma = 42.6$ & $\kappa = 29.6$ | **0.97** | **62.12** | **37.88** | **89.14** | **6.32** | **4.54** |
| $\sigma = 80$ & $\kappa = 29.6$ | 0.88 | 53.77 | 46.23 | 66.08 | 19.97 | 13.95 |
| $\sigma = 14$ & $\kappa = 77$ | 0.46 | 66.27 | 33.73 | 133.47 | -18.71 | -14.76 |
| $\sigma = 42.6$ & $\kappa = 77$ | 0.43 | 64.22 | 35.78 | 116.36 | -9.11 | -7.25 |
| $\sigma = 80$ & $\kappa = 77$ | 0.39 | 61.94 | 38.06 | 99.94 | 0.21 | -0.15 |

The table presents gains from interbank trade in Germany under alternative calibrations of $\sigma$ and $\kappa$. Colums 2 to 6 are respective shares in %. Main calibration in bold. Estimates are reported as percentages of steady-state consumption. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations.

gains from interbank integration than concentration in firms' loan provision.

CF3 considers a free-trade benchmark in which all bilateral transaction costs are set to one, $d^{ni} = 1$. Under this counterfactual, gains from trade rise sharply to 14.70% of quarterly consumption. Nearly 90% of these gains are static, indicating that the main effect of removing bilateral frictions is a first-order improvement in the allocation of funds across banks. Online Appendix 5.A extends this exercise by tracing gains from trade as transaction costs vary from autarky to free trade. That exercise shows that dynamic gains account for a larger share of total gains at low levels of integration, whereas static gains dominate once transaction costs are already low, as in the free-trade case reported here. We therefore interpret this counterfactual as an upper bound on the welfare gains from eliminating interbank frictions altogether.

CF4 sets the Calvo parameter to zero and considers a flexible-price version of the model. The gains from trade remain essentially unchanged at 0.97%, with an almost identical decomposition into static and dynamic components. This confirms that the gains from interbank integration are not primarily driven by nominal rigidities, but instead reflect real funding-allocation effects and the interaction of market integration with banking-sector volatility.

CF5 and CF6 study the role of changes in the stochastic environment around the Great Recession. In CF5, we compute gains using the stochastic processes estimated from the pre-recession subsample, while in CF6 we use the corresponding post-recession processes. The resulting gains are 1.14% and 0.96% of quarterly consumption, respectively. Thus, the gains from interbank trade are lower under the post-recession environment, consistent with the persistent decline in interbank activity and the worsening of funding conditions documented in Section 3. At the same time, the decomposition remains broadly similar, suggesting that the post-crisis decline in gains is driven more by a deterioration in the overall environment than by a change in the relative importance of static and dynamic gains.

Finally, CF7 reports the gains from the observed lender-of-last-resort policy relative to a counterfactual economy without central-bank emergency credit. The gains amount to 2.46% of quarterly consumption and are entirely static in our approximation. This reflects the fact that discount-window access lowers the average cost of funds by placing a ceiling on borrowing costs, even if its effect on cyclical volatility is limited. Quantitatively, the gains from lender-of-last-resort support are larger than the gains from interbank integration itself, highlighting the importance of central-bank backstops in environments with occasionally severe funding stress.

Table 8: Welfare Gains from Interbank Trade, Counterfactual Interbank Markets

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Counterfactual | Gains in % | Static Gains | Dynamic Gains | Openness | Diverse | Covariance |
| Main Results | **0.97** | **62.12** | **37.88** | **0.89** | **0.06** | **0.05** |
| CF 1: Equal Loan Sizes | 1.03 | 64.33 | 35.67 | 0.90 | 0.08 | 0.01 |
| CF 2: Equal Depositor Preferences | 1.62 | 66.09 | 33.91 | 0.91 | 0.04 | 0.06 |
| CF 3: Free Trade (all $d^{ni} = 1$) | 14.70 | 89.93 | 10.07 | 0.89 | 0.00 | 0.11 |
| CF 4: Flex Price ($\theta == 0$) | 0.97 | 62.12 | 37.88 | 0.89 | 0.06 | 0.05 |
| CF 5: Pre-Recession Processes | 1.14 | 61.51 | 38.49 | | | |
| CF 6: Post-Recession Processes | 0.96 | 62.22 | 37.78 | | | |
| CF 7: LoLR relative to no-LoLR | 2.46 | 100 | 0 | | | |

The table presents gains from interbank trade in Germany under counterfactual scenarios, described in column 1. Colums 2 and 3 are respective shares of static and dynamic gains in %. Main gains from trade counterfactual in bold. Estimates are reported as percentages of steady-state consumption. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations.

## 8 Conclusion

In this paper, we study the welfare implications of interbank market integration and the role of lender-of-last-resort policy in a banking system with many heterogeneous institutions linked through a bilateral funding network. By incorporating a granular interbank market into a DSGE framework and combining it with proprietary data on the universe of German banks, we quantify the gains from interbank trade and examine how these gains depend on the structure of the banking system and on the nature of funding shocks.

Our results show that interbank market integration generates economically meaningful welfare gains. Under our benchmark calibration, moving from autarky to the observed level of interbank integration raises welfare by up to 0.97% of quarterly consumption. These gains reflect both first-order improvements in the allocation of funds across banks and second-order effects arising from the interaction of market integration with fluctuations in funding conditions. Quantitatively, 62.12% of the gains are static and 37.88% are dynamic. Within the dynamic component, transaction-cost shocks account for almost all of the gains, with the main contribution coming from the openness channel: access to the interbank market allows banks to substitute toward temporarily cheaper funding sources, lowering both the level and the volatility of borrowing costs over time.

We also show that the magnitude of the gains depends on the structure of the banking system. In particular, heterogeneity in banks' ability to attract deposits matters more for the gains from trade than concentration in loan provision to firms. Eliminating bilateral interbank frictions altogether implies much larger gains, while moving to a flexible-price economy leaves the benchmark gains essentially unchanged. This indicates that our results are driven mainly by real improvements in the allocation of liquidity across banks, rather than by the interaction of banking-sector shocks with nominal rigidities.

Finally, we quantify the role of lender-of-last-resort policy. Relative to a counterfactual without emergency central-bank credit, the observed policy raises welfare by 2.46% of quarterly consumption, underscoring the importance of central-bank backstops in environments with occasionally severe financial disruptions.

**Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**:

During the preparation of this work the author(s) used Cursor in order to improve writing and streamline coding. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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# Online Appendix (Not For Publication)

## Appendix 1 Additional Figures & Tables

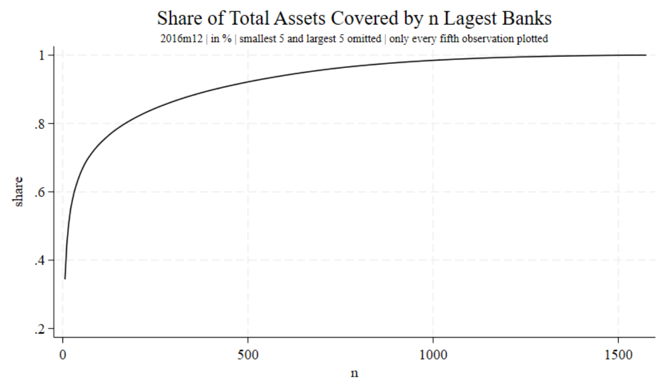
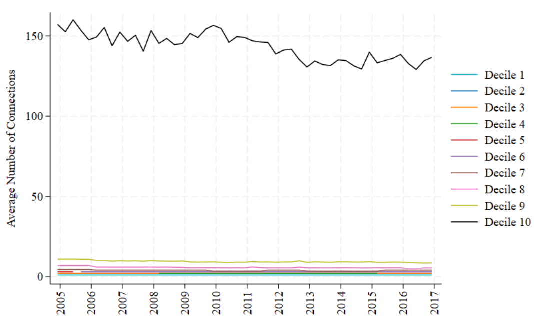
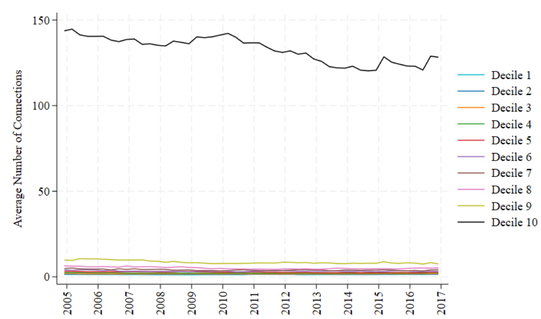



Figure A.1: Cumulative share of total MFI assets by the $n$ largest MFIs. The smallest and largest three MFIs are omitted, and only every third observation is plotted due to confidentiality requirements. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, BISTA, 2016m12, own calculations.

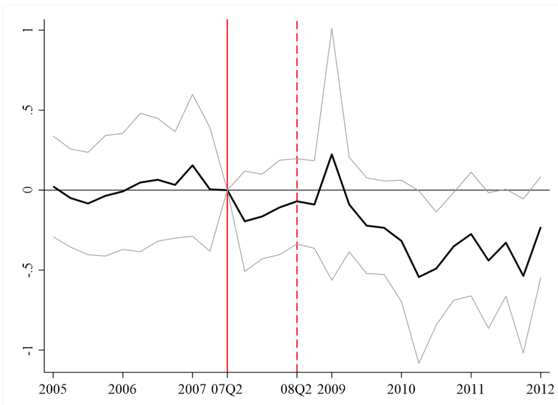



(a) By number of connections

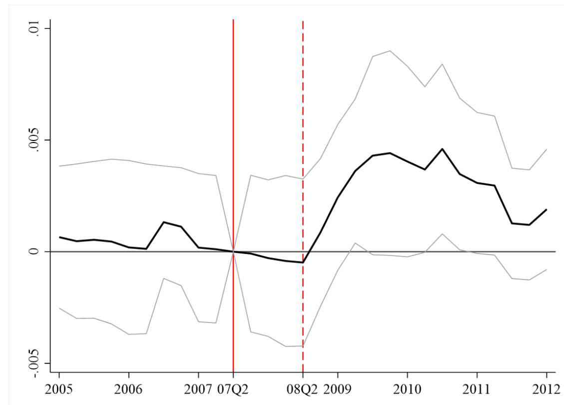



(b) By bank size

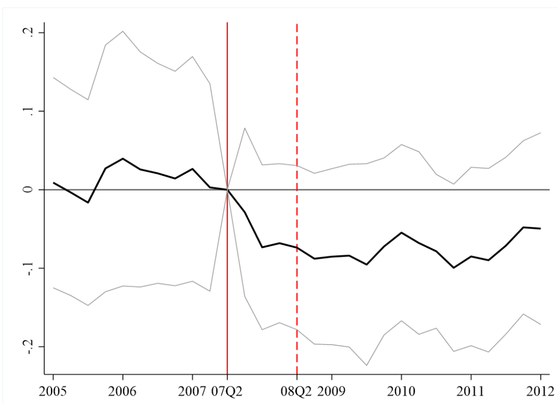
Figure A.2: Average number of distinct interbank funding sources, by deciles. Figure A.2a constructs deciles based on the number of distinct interbank funding sources. Figure A.2b defines deciles with respect to the total asset size of the MFIs. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, BISTA and Credit Registry, 2004m12 - 2018m12, own calculations.



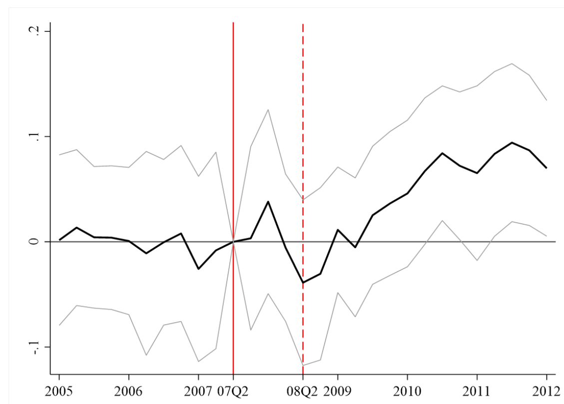

(a) Interbank borrowing




(b) Interest rate on outstanding final loans

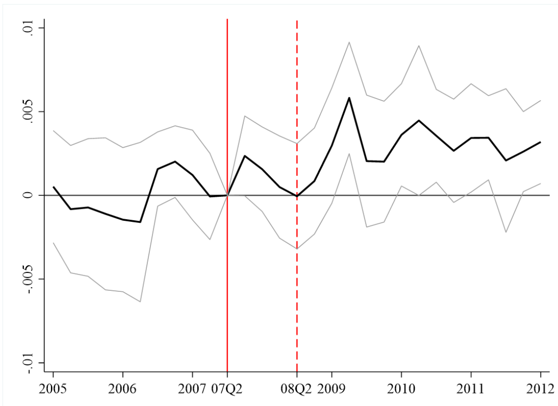



(c) Outstanding final loans

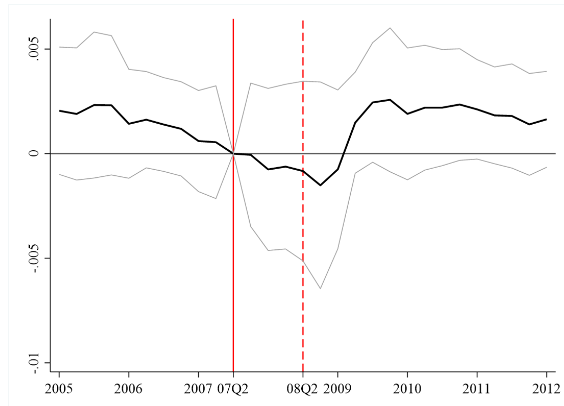



(d) Own share

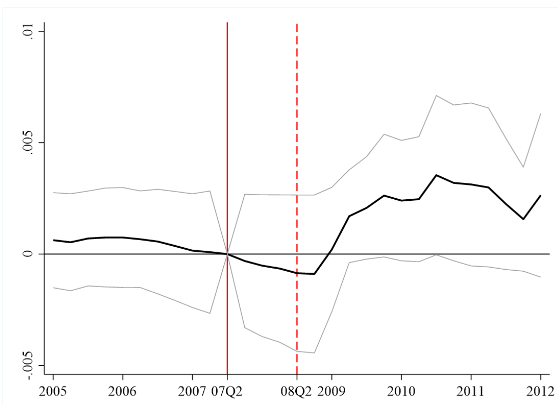
Figure A.3: This event-study examines the impact of indirect exposure to the US financial crisis on borrowing from other German MFIs (upper left), interest rates of final loans (upper right), the quantity of final loans (lower left), and the share of funding from own sources (lower right). Each figure presents coefficients on $Exposure^{US}_{2006Q1} \times Quarter - FE$ and 95% confidence intervals. Solid red vertical lines indicate 2007$Q$2, the event quarter immediately preceding the crisis, while dashed red lines represent 2008$Q$2, just before the collapse of Lehman Brothers. The regression incorporates quarter fixed effects and bank fixed effects. Robust standard errors are clustered at the bank group-quarter level. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m1 - 2012m1, own calculations.



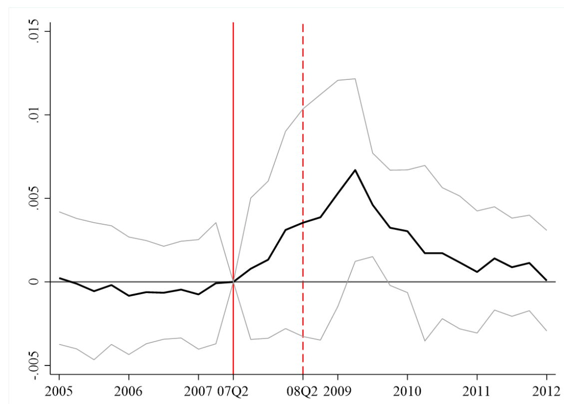

(a) Interest rate on loans with maturity below 1 year




(b) Interest rate on loans between 1 and 5 years

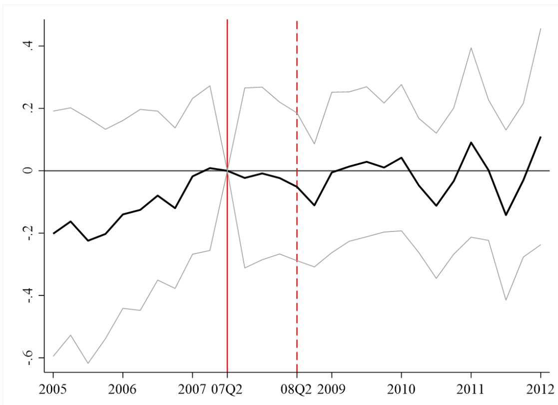



(c) Interest rate on loans above 5 years

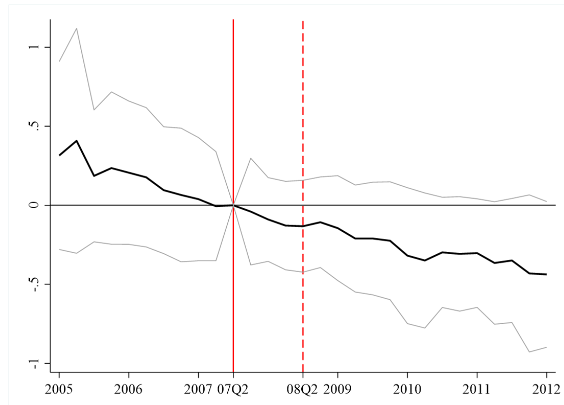



(d) Interest rate on outstanding deposits

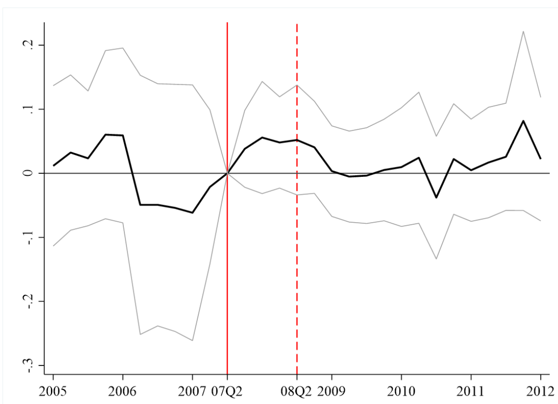
Figure A.4: This event-study examines the impact of indirect exposure to the US financial crisis on interest rates on outstanding loans with maturity below 1 year (upper left), between 1 and 5 years (upper right), above 5 years (lower left), and the interest rate on outstanding deposits (lower right). Each figure presents coefficients on $Exposure^{US}_{2006Q1} \times Quarter - FE$ and 95% confidence intervals. Solid red vertical lines indicate 2007$Q$2, the event quarter immediately preceding the crisis, while dashed red lines represent 2008$Q$2, just before the collapse of Lehman Brothers. Regressions include quarter fixed effects and bank fixed effects and controls for leverage and repayment rates.Robust standard errors are clustered at the bank group-quarter level. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m1 - 2012m1, own calculations.



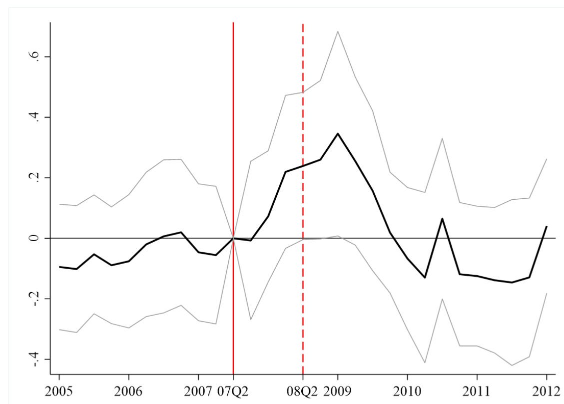

(a) Loans with maturity below 1 year




(b) Loans with maturity between 1 and 5 years




(c) Loans with maturity above 5 years




(d) Volume of outstanding deposits

Figure A.5: This event-study examines the impact of indirect exposure to the US financial crisis on volumes of loans with maturity below 1 year (upper left), between 1 and 5 years (upper right), above 5 years (lower left), and outstanding deposits (lower right). Each figure presents coefficients on $Exposure^{US}_{2006Q1} \times Quarter-FE$ and 95% confidence intervals. Solid red vertical lines indicate 2007$Q2$, the event quarter immediately preceding the crisis, while dashed red lines represent 2008$Q2$, just before the collapse of Lehman Brothers. Regressions include quarter fixed effects and bank fixed effects and controls for leverage and repayment rates. Robust standard errors are clustered at the bank group-quarter level. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m1 - 2012m1, own calculations.

Table A.1: Pre-trends for Estimation of $\sigma$ and $\kappa$

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Loan Rate | Loans | Own Share | Loan Rate | Loans | Own Share |
| $Exposure_{t_0} \times Post_{t_0}$ | 0.000 | 0.006 | -0.016 | -0.000 | -0.017* | -0.022* |
| | (0.000) | (0.015) | (0.011) | (0.000) | (0.009) | (0.013) |
| Observations | 1805 | 1805 | 1805 | 1794 | 1794 | 1794 |
| R-squared | 0.954 | 0.995 | 0.949 | 0.964 | 0.996 | 0.950 |
| Mean Exposure | 0.757 | 0.757 | 0.757 | 0.768 | 0.768 | 0.768 |
| Controls | no | no | no | yes | yes | yes |

The regression compares outcomes between $2004Q4$ to $2005Q4$ and the pre-period in the main regression ($2006Q1$ until $2007Q2$) for banks with varying degrees of indirect exposure to the US financial crisis. The initial asset exposure to lenders in the US market is taken in $2006Q1$. Controls include direct asset exposure to the US, leverage, repayment rates on loans, and loan shares of non-MFI and household loans, each broken down into maturities of less than 1 year, between 1 and 5 years, and more than 5 years, as well as separate shares for secured and unsecured mortgages. All regressions include bank fixed-effects and quarter fixed-effects. Standard errors are clustered at the level of the bank group-quarter. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2007m6, own calculations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.2: Estimation of $\sigma$ and $\kappa$

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Panel A: | Loan Rate | Loans | Own Share | Loan Rate | Loans | Own Share | Loan Rate | Loans | Own Share | Loan Rate | Loans | Own Share |
| $Exposure_{t_0} \times Post_t$ | 0.000*** | -0.007*** | 0.006*** | 0.000*** | -0.020*** | 0.014*** | 0.002*** | -0.057*** | 0.101*** | 0.002*** | -0.092*** | 0.044*** |
| | (0.000) | (0.002) | (0.002) | (0.000) | (0.003) | (0.004) | (0.001) | (0.016) | (0.024) | (0.000) | (0.019) | (0.015) |
| $Exposure_{t_0} \times Post_t \times NetBorrower$ | | | | | | | | | | 0.001*** | 0.003 | 0.081*** |
| | | | | | | | | | | (0.000) | (0.011) | (0.014) |
| R-squared | 0.940 | 0.994 | 0.914 | 0.938 | 0.994 | 0.915 | 0.938 | 0.995 | 0.854 | 0.939 | 0.994 | 0.917 |
| Mean Exposure | 4.944 | 4.944 | 4.944 | 2.318 | 2.318 | 2.318 | 0.767 | 0.767 | 0.767 | 0.768 | 0.768 | 0.768 |
| Panel B: | *2SLS: Instrument for* $\log R_t^{F,i}$ *is* $Exposure_{t_0} \times Post_t$ | | | | | | | | | | | |
| | | $-\sigma$ | $\kappa$ | | $-\sigma$ | $\kappa$ | | $-\sigma$ | $\kappa$ | | $-\sigma$ | $\kappa$ |
| $\log R_t^{F,i}$ | | -14.05*** | 12.24*** | | -39.88*** | 29.13*** | | -25.56*** | 45.57*** | | -37.02*** | 34.57*** |
| | | (3.55) | (4.71) | | (7.82) | (8.45) | | (7.05) | (11.60) | | (8.16) | (8.24) |
| $\log R_t^{F,i} \times NetBorrower$ | | | | | | | | | | | -11.46** | -10.08*** |
| | | | | | | | | | | | (5.73) | (2.66) |
| Observations | 3532 | 3532 | 3532 | 3532 | 3532 | 3532 | 3543 | 3543 | 3543 | 3532 | 3532 | 3532 |
| 1st-Stage F-Stat | | 31.88 | 31.88 | | 23.99 | 23.99 | | 16.15 | 16.15 | | 8.25 | 8.25 |
| Controls | yes | yes | yes | yes | yes | yes | yes | yes | yes | yes | yes | yes |
| Robustness | All US Assets | All US Assets | All US Assets | Exposure in Euro | Exposure in Euro | Exposure in Euro | Foreign as Loans | Foreign as Loans | Foreign as Loans | Net Borrower | Net Borrower | Net Borrower |

Panel A compares outcomes between 2006$Q$1 and 2007$Q$2, and 2008$Q$3 to 2011$Q$4 (post-period) for banks that were more or less indirectly exposed to the US financial crisis. Panel B presents the results of estimating equations 19 and 20 using 2SLS with Exposure$_{t_0}$ × Post$_t$ as an instrument. Initial asset exposure to lenders in the US market is taken in 2006$Q$1. Controls include direct asset exposure to the US, leverage, repayment rates on loans, and loan shares of non-MFI and household loans, each broken down into maturities of less than 1 year, between 1 and 5 years, and more than 5 years, as well as separate shares for secured and unsecured mortgages. All regressions include bank fixed-effects and quarter fixed-effects. Standard errors are clustered at the level of the bank group-quarter. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2011m12, own calculations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.3: Estimation of $\sigma$ and $\kappa$ with indirect exposure to Euro Crisis

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Panel A: | Loan Rate | Loans | Own Share | Loan Rate | Loans | Own Share |
| $Exposure_{t_0} \times Post_t$ | 0.000** | -0.010*** | 0.013*** | 0.000** | -0.006*** | 0.012*** |
| | (0.000) | (0.003) | (0.002) | (0.000) | (0.002) | (0.002) |
| R-squared | 0.938 | 0.989 | 0.903 | 0.945 | 0.992 | 0.909 |
| Mean Exposure | 3.094 | 3.094 | 3.094 | 3.150 | 3.150 | 3.150 |
| Panel B: | *2SLS: Instrument for* $\log R_t^{F,i}$ *is* $Exposure_{t_0} \times Post_t$ | | | | | |
| | | $-\sigma$ | $\kappa$ | | $-\sigma$ | $\kappa$ |
| $\log R_t^{F,i}$ | | -52.18* | 68.79** | | -39.49** | 77.12** |
| | | (28.05) | (31.87) | | (18.90) | (33.80) |
| Observations | 4094 | 4094 | 4094 | 4021 | 4021 | 4021 |
| 1st-Stage F-Stat | | 6.06 | 6.06 | | 6.05 | 6.05 |
| Controls | no | no | no | yes | yes | yes |

Panel A compares outcomes between 2006$Q$1 and 2008$Q$1, and 2009$Q$4 to 2012$Q$4 (post-period) for banks that were more or less indirectly exposed to the Euro Crisis. Instrument is based on exposure to non-MFI assets in Portugal, Ireland, Italy, Greece and Spain (PIIGS) relative to all assets in 2006Q1. Panel B presents the results of estimating equations 19 and 20 using 2SLS with $\text{Exposure}_{t_0} \times \text{Post}_t$ as an instrument. Controls include direct asset exposure to PIIGS, leverage, repayment rates on loans, and loan shares of non-MFI and household loans, each broken down into maturities of less than 1 year, between 1 and 5 years, and more than 5 years, as well as separate shares for secured and unsecured mortgages. All regressions include bank fixed-effects and quarter fixed-effects. Standard errors are clustered at the level of the bank group-quarter. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2012m12, own calculations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.4: Summary Statistics of $d_t^{ni}$

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| | Period | Mean | Median | Std | p(5) | p(25) | p(75) | p(95) | Obs |
| Interbank Frictions $d_t^{ni}$ | | | | | | | | | |
| $d_t^{ni}, i \neq n$ | Pre-Crisis | 1.288 | 1.300 | 0.148 | 1.031 | 1.186 | 1.377 | 1.536 | 292706 |
| $d_t^{ni}, i \neq n$ | Crisis | 1.298 | 1.309 | 0.139 | 1.049 | 1.213 | 1.388 | 1.507 | 196337 |
| $d_t^{ni}, i \neq n$ | Post-Crisis | 1.305 | 1.306 | 0.159 | 1.045 | 1.214 | 1.381 | 1.575 | 480249 |
| Std Deviation of $d_t^{in}$ by Lender | | | | | | | | | |
| $std(d_t^{in})$ | Pre-Crisis | 0.095 | 0.091 | 0.039 | 0.046 | 0.073 | 0.111 | 0.156 | 17922 |
| $std(d_t^{in})$ | Crisis | 0.086 | 0.081 | 0.037 | 0.041 | 0.066 | 0.100 | 0.148 | 12539 |
| $std(d_t^{in})$ | Post-Crisis | 0.112 | 0.107 | 0.060 | 0.028 | 0.070 | 0.146 | 0.215 | 33852 |
| Std Deviation of $d_t^{ni}$ by Borrower | | | | | | | | | |
| $std(d_t^{ni})$ | Pre-Crisis | 0.057 | 0.056 | 0.034 | 0.007 | 0.035 | 0.075 | 0.112 | 12926 |
| $std(d_t^{ni})$ | Crisis | 0.057 | 0.053 | 0.039 | 0.006 | 0.029 | 0.077 | 0.127 | 8410 |
| $std(d_t^{ni})$ | Post-Crisis | 0.062 | 0.058 | 0.045 | 0.006 | 0.029 | 0.085 | 0.139 | 22116 |
| Number of Connections by Lender | | | | | | | | | |
| $N(d_t^{in})$ | Pre-Crisis | 15.150 | 11.000 | 37.306 | 1.000 | 5.000 | 17.000 | 29.000 | 19320 |
| $N(d_t^{in})$ | Crisis | 14.534 | 11.000 | 34.183 | 1.000 | 5.000 | 17.000 | 28.000 | 13509 |
| $N(d_t^{in})$ | Post-Crisis | 13.234 | 9.000 | 32.895 | 1.000 | 4.000 | 16.000 | 27.000 | 36290 |
| Number of Connections by Borrower | | | | | | | | | |
| $N(d_t^{ni})$ | Pre-Crisis | 14.424 | 2.000 | 79.423 | 0.000 | 1.000 | 4.000 | 17.000 | 20293 |
| $N(d_t^{ni})$ | Crisis | 13.975 | 2.000 | 80.324 | 0.000 | 1.000 | 4.000 | 15.000 | 14049 |
| $N(d_t^{ni})$ | Post-Crisis | 12.819 | 2.000 | 73.532 | 0.000 | 1.000 | 3.000 | 16.000 | 37464 |
| Funding Costs as Share of Total Funds | | | | | | | | | |
| $\sum \lambda_t^{ni} d_t^{ni} - 1$ | Pre-Crisis | 0.012 | 0.007 | 0.014 | 0.000 | 0.004 | 0.014 | 0.038 | 20293 |
| $\sum \lambda_t^{ni} d_t^{ni} - 1$ | Crisis | 0.013 | 0.011 | 0.013 | 0.000 | 0.008 | 0.016 | 0.036 | 14049 |
| $\sum \lambda_t^{ni} d_t^{ni} - 1$ | Post-Crisis | 0.011 | 0.008 | 0.012 | 0.000 | 0.005 | 0.013 | 0.030 | 37464 |

Table shows summary statistics of recovered trade wedges $d_t^{ni}$. The process of how these wedges are recovered is described in Section 6.4. Pre-crisis refers to the period 2004Q1-2007Q2; crisis to the period 2007Q3-2011Q4; post-crisis to the period 2012Q1-2016Q4. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations.

Table A.5: Cross-Correlations of Banking Sector Shocks

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| corr $\left(u_t^{T,n}, u_t^{a,n}\right)$ | corr $\left(u_t^{I,ni}, u_t^{T,n}\right)$ | corr $\left(u_t^{I,ni}, u_t^{T,i}\right)$ | corr $\left(u_t^{I,ni}, u_t^{a,n}\right)$ | corr $\left(u_t^{I,ni}, u_t^{a,i}\right)$ |
| 0.382 | -0.257 | -0.023 | -0.244 | 0.151 |
| (0.059) | (0.041) | (0.038) | (0.024) | (0.028) |

Table shows cross-correlations of recovered shocks to transaction costs ($u_t^{I,ni}$), depositor preferences ($u_t^{T,n}$) and loan demand ($u_t^{a,n}$). Standard errors clustered at bank and time; at lender and borrower and time when correlation involves $u_t^{I,ni}$. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2016m12, own calculations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.6: Parameter Calibration

| Parameter | Value | Description | References |
|---|---|---|---|
| $\eta$ | 1 | Frisch labor supply elasticity | See Peterman (2016) |
| $\beta$ | 0.99 | Discount factor | Annualized steady-state nominal risk-free rate of 4% |
| $\epsilon$ | 7 | Elasticity of substitution intermediate output | Target steady-state mark-up of 16.7%. See Galí (2015) |
| $\theta$ | 0.55 | Calvo price stickiness | See Coibion, Gorodnichenko, and Wieland (2012) |
| $\alpha$ | 0.4 | Capital share in production | Aggregate data historical approximation |
| $\sigma$ | 42.6 | Firm's loan elasticity of demand | Estimation, Sec. 6 |
| $\kappa$ | 29.6 | Interbank loan elasticity of demand | Estimation, Sec. 6 |
| $\Pi$ | 1 | Target gross inflation rate | See Galí (2015) |
| $\gamma_\pi$ | 2.5 | Taylor rule inflation response | See Coibion, Gorodnichenko, and Wieland (2012) |
| $\gamma_y$ | 1.5 | Taylor rule output gap response | See Coibion, Gorodnichenko, and Wieland (2012) |
| $\varpi_1$ | 0.12 | Fixed penalty rate | Match 3.5% pre-crisis central bank trade share |
| $\rho_T$ | 0.77 | Persistence depositor shocks | Estimation, see Sec. 6 |
| $\rho_I$ | 0.77 | Persistence interbank shocks | Estimation, see Sec. 6 |
| $\rho_a$ | 0.77 | Persistence loan demand shocks | Estimation, see Sec. 6 |
| $\zeta_T$ | 0.228 | Correlation depositor preferences shock | Estimation, see Sec. 6 |
| $\zeta_{I,B}$ | 0.507 | Pairwise correlation of interbank transaction shocks with the same borrower | Estimation, see Sec. 6 |
| $\zeta_{I,L}$ | 0.137 | Pairwise correlation of interbank transaction shocks with the same lender | Estimation, see Sec. 6 |
| $\zeta_{I,X}$ | 0.008 | Pairwise correlation of interbank transaction shocks with different lender and borrower | Estimation, see Sec. 6 |
| $\zeta_a$ | -0.0008 | Correlation loan demand shock | Estimation, see Sec. 6 |
| $\sigma_T$ | 0.028 | Standard deviation depositor preferences shock | Estimation, see Sec. 6 |
| $\sigma_I$ | 0.04 | Standard deviation interbank transactions shock | Estimation, see Sec. 6 |
| $\sigma_a$ | 0.337 | Standard deviation loan demand shock | Estimation, see Sec. 6 |

The table presents the baseline parameter values used to calibrate the model in Section 6. The last column indicates the reference source for parameter calibration.

# Appendix 2 Detailed Model Derivation

**Representative Household** As in Section 4.2, the household's period utility is:

$$U_t = \log(X_t) - \left(\frac{\eta}{\eta+1}\right)\int_0^1 N_{t,\tau}^{1+1/\eta}\, \mathrm{d}\tau\,.$$

Here, $X_t$ is consumption of the composite good and $N_{t,\tau}$ is labor supply at instant $\tau$. The household's budget constraint at $(t,\tau)$ is:

$$v_{t,\tau}: \qquad C_{t,\tau} + \frac{\sum_{n=1}^{N} D_{t,\tau}^{n}}{P_t} + \frac{B_{t,\tau}}{P_t} = \frac{\sum_{n=1}^{N}(1+\varsigma_D)R_{t-1,\tau}^{D,n}D_{t-1,\tau}^{n}}{P_t} + \frac{R_{t-1}^{B}B_{t-1,\tau}}{P_t} + \int_0^1 \frac{W_t(\nu)N_{t,\tau}(\nu)}{P_t}\,\mathrm{d}\nu + \frac{\Upsilon_{t,\tau}}{P_t}\,. \tag{A.1}$$

In (A.1), $B_{t,\tau}$ is the one-period government bond and $R_t^B$ is its gross return. Bank $n$ pays the gross deposit rate $R_{t,\tau}^{D,n}$. The wage in industry $\nu$ is $W_t(\nu)$ and $P_t$ is the aggregate price index. Profits (from firms and banks) are rebated lump-sum as $\Upsilon_{t,\tau}$. The parameter $\varsigma_D$ denotes a subsidy to deposits, and $v_{t,\tau}$ is the Lagrange multiplier on the budget constraint.

Maximizing expected discounted utility subject to (A.1) yields the first-order conditions:

$$C_{t,\tau}: \qquad v_{t,\tau} = X_t^{-1}\,, \tag{A.2}$$

$$N_{t,\tau}(\nu): \qquad N_{t,\tau}(\nu)^{1/\eta} = v_t\frac{W_t(\nu)}{P_t}\,, \qquad \forall \nu\,, \tag{A.3}$$

$$B_{t,\tau}: \qquad 1/R_t^B = \beta\mathbb{E}_t\left[\frac{v_{t+1,\tau}}{v_{t,\tau}\Pi_{t+1}}\right], \tag{A.4}$$

$$D_{t,\tau}^n: \qquad 1/R_{t,\tau}^{D,n} = (1+\varsigma_D)\,\beta\mathbb{E}_t\left[\frac{v_{t+1,\tau}}{T_t^n z_{t,\tau}^n v_{t,\tau}\Pi_{t+1}}\right], \qquad \forall n\,. \tag{A.5}$$

By combining equations (A.4) and (A.5), we derive the following relationship between the interest rates of deposits and bonds:

$$R_{t,\tau}^{D,n} = (1+\varsigma_D)^{-1}\, T_t^n z_{t,\tau}^n R_t^B\,. \tag{A.6}$$

**Firms** Under complete depreciation and immediate capital accumulation (Section 4.3), capital of type $n$ satisfies $K_{t,\tau}^n(\nu) = I_{t,\tau}^n(\nu)/P_t$ for all $n$, where $I_{t,\tau}^n(\nu)$ is investment in capital $n$. At each instant $\tau$, firms finance investment through bank credit. There are $N$ banks, each specializing in loans $L_{t,\tau}^n(\nu)$ tied to a distinct type of capital. Loans are repaid after one quarter at gross rate $R_t^{F,n}$, and firms face the constraint $I_{t,\tau}^n(\nu) \le L_{t,\tau}^n(\nu)$ for all $n, t, \tau$.

A representative, perfectly competitive firm combines intermediate products into a final good according to the following equation:

$$Y_t = \left[\int_0^1 Y_t(\nu)^{\left(\frac{\epsilon-1}{\epsilon}\right)}\,\mathrm{d}\nu\right]^{\left(\frac{\epsilon}{\epsilon-1}\right)},$$

where $\epsilon > 1$ denotes the elasticity of substitution between varieties. Individual demand for intermediates is given by:

$$Y_t(v) = \left(\frac{P_t(v)}{P_t}\right)^{-\epsilon} Y_t \,,$$

with $P_t(v)$ representing the price of intermediate $v$ and $P_t = \left[\int_0^1 P_t(v)^{1-\epsilon}\,\mathrm{d}v\right]^{\frac{1}{1-\epsilon}}$ as the aggregate price index.

Intermediate producers set prices subject to Calvo rigidity (Calvo 1983): at the start of each quarter, they reset with probability $1-\theta$. In equilibrium, all resetting firms choose the same optimal price $P_t^*$. The resulting price index recursion is:

$$P_t^{1-\epsilon} = (1-\theta)\left(P_t^*\right)^{1-\epsilon} + \theta\left(P_{t-1}\right)^{1-\epsilon} \,. \tag{A.7}$$

Intermediate firm $v$ aims to maximize the following present discounted stream of profits:

$$\sum_{j=0}^{\infty} \mathbb{E}_t \left[ Q_{t,t+j} \int_0^1 (1+\varsigma_F) P_{t+j}(v) Y_{t+j,\tau}(v) - W_{t+j}(v) N_{t+j,\tau}(v) - \sum_{n=1}^{N} R_{t+j-1}^{F,n} L_{t+j-1}^{n}(v)\,\mathrm{d}\tau \right] \,,$$

where $Q_{t,t+j} = \beta^j \dfrac{X_t}{X_{t+j}\Pi_{t+j}}$ represents the firm's stochastic discount factor between periods $t$ and $t+j$, and $\varsigma_F$ is a government production subsidy. By minimizing firm $v$'s production costs with respect to labor and loans, we obtain the following demand for inputs:

$$
\begin{aligned}
N_t(v) &= (1-\alpha) Y_t(v) \left(\frac{\widetilde{R}_t^F}{W_t(v)/P_t}\right)^{\alpha} \,, && \text{(A.8)}\\
\frac{L_t(v)}{P_t} &= \alpha Y_t(v) \left(\frac{\widetilde{R}_t^F}{W_t(v)/P_t}\right)^{-(1-\alpha)} \,, \\
L_t^n(v) &= a_t^n \left(\frac{R_t^{F,n}}{R_t^F}\right)^{-\sigma} L_t(v) \,,
\end{aligned}
$$

where $\alpha$ is the share of capital in the production function and $\widetilde{R}_t^{F,n} = \mathbb{E}_t\left[Q_{t,t+1}\right] R_t^{F,n}$ is the expected discounted gross rate on a loan from bank $n$ and $R_t^F = \left[\sum_{n=1}^N a_t^n \left(R_t^{F,n}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$ is the aggregate gross rate index. Maximizing the present discounted stream of profits also yields the optimal reset price at time $t$, $P_t^*$:

$$\frac{P_t^*}{P_t} = \frac{\mathbb{E}_t\left[\sum_{j=0}^{\infty} \theta^j Q_{t,t+j} \left(\frac{P_{t+j}}{P_t}\right)^{\epsilon+1} Y_{t+j} \left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1}\right)\left(\frac{MC_{t+j|t}(v)}{P_{t+j}}\right)\right]}{\mathbb{E}_t\left[\sum_{j=0}^{\infty} \theta^j Q_{t,t+j} \left(\frac{P_{t+j}}{P_t}\right)^{\epsilon} Y_{t+j}\right]} \,, \tag{A.9}$$

where subindex $t+j|t$ denotes the value of a variable conditional on the firm having last reset its price at period $t$, and $MC_{t+j|t}(v)/P_t = \left(\widetilde{R}_{t+j}^F\right)^{\alpha}\left(\frac{W_{t+j|t}(v)}{P_{t+j}}\right)^{1-\alpha}$ is the real marginal cost of production, with $\widetilde{R}_t^F =$

$\mathbb{E}_t\left[Q_{t,t+1}\right] R_t^F$. Over the period, the aggregate firm profits are given by:

$$\Upsilon_t^F = (1+\varsigma_F) P_t Y_t - \int_0^1 W_t(\nu) N_t(\nu)\, \mathrm{d}\nu - \sum_{n=1}^{N} R_{t-1}^{F,n} L_{t-1}^n \,.$$

**Banks** Each bank performs three activities: it raises deposits from the household, extends credit to firms, and trades funds in the interbank market. For exposition, we split each bank into two divisions. The Loan Division originates loans and obtains funding either internally or via interbank borrowing. The Deposit Division collects deposits and allocates them across internal funding and interbank lending.
Loan Division: Loan Division $n$ seeks to maximize expected profits:

$$\begin{aligned} &\int_0^1 (1+\varsigma_B) R_t^{F,n} L_{t,\tau}^n - R_{t,\tau}^{I,n} M_{t,\tau}^n \,\mathrm{d}\tau \,, \\ \text{where:} \quad & R_{t,\tau}^{I,n} = \min_i \left\{ R_{t,\tau}^{I,in} \right\} \,, \\ & M_{t,\tau}^n = M_{t,\tau}^{in}, \quad i_{t,\tau}(n) = \arg_j \min \left\{ R_{t,\tau}^{I,jn} \right\} \,, \end{aligned} \tag{A.10}$$

subject to the constraints in (4). The parameter $\varsigma_B$ is a subsidy to bank lending. The variable $R_{t,\tau}^{I,n}$ is the (gross) funding rate bank $n$ faces at instant $\tau$, i.e., the minimum over available counterparties. Banks take the aggregate gross rate index $R^F$ as given, but set $R_t^{F,n}$ under monopolistic competition. We assume firm loan rates are reset only at the beginning of each period (so they are constant within $t$), while interbank rates are fully flexible within the period. The optimal firm loan rate is therefore a constant markup over the average cost of funds:

$$R_t^{F,n} = \left( \frac{(1+\varsigma_B)^{-1}\sigma}{\sigma - 1} \right) R_t^{I,n} \,, \qquad R_t^F = \left( \frac{(1+\varsigma_B)^{-1}\sigma}{\sigma - 1} \right) R_t^I \,, \tag{A.11}$$

where we define $R_t^{I,n} \equiv \int_0^1 R_{t,\tau}^{I,n}\, \mathrm{d}\tau$ and $R_t^I = \left[ \sum_{n=1}^N a_t^n \left( R_t^{I,n} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$.
Deposit Division: Deposit Divisions acquire deposits from the representative household and transform them into internal funding or interbank loans for other banks. The quantity of funds that bank $n$ can provide is expressed as:

$$\sum_{i=1}^N d_t^{ni} M_{t,\tau}^{ni} = D_{t,\tau}^n \,, \tag{A.12}$$

subject to $M_{t,\tau}^{ni} \geq 0$ and $D_{t,\tau}^n \geq 0$ for all $n, i, t, \tau$. The terms $d_t^{ni} \geq 1$ are transaction costs for transferring funds from bank $n$ to $i$ (e.g., screening, enforcement, or other intermediation costs). We define the functional form of transaction costs shocks as

$$d_t^{ni} = \begin{cases} \left(d^{ni}\right)^{\varrho} exp(u_t^{I,ni}) & \text{, if } i \neq n \,, \\ 1 & \text{, otherwise.} \end{cases} \,, \qquad u_t^{I,ni} = \rho_I u_{t-1}^{I,ni} + \varepsilon_t^{I,ni} \,, \quad \forall n, i \,, \tag{A.13}$$

where $\varepsilon_t^{I,ni}$ are mean-zero, exogenous (but potentially correlated) stochastic shocks. Parameter $\varrho$ will allow us to modify the size of transaction costs once we look at the additional banking system integration counterfactual presented in Online Appendix 5.A. In the main text, $\varrho$ is implicitely assumed to be equal to one. The shock processes for loan demand $a_t^i$ and depositor preferences $T_t^i$ follow a similar autoregressive structure as in equation A.13 with stochastic shocks $\varepsilon_t^{a,i}$ and $\varepsilon_t^{T,i}$. Although our empirical estimation of the transaction costs in Section 6.4 does not require the specification of a particular process for the $d_t^{ni}$'s, assuming a flexible stationary process as in (A.13) facilitates the computation of the solution to the log-linearized system in Section 4.6 and the derivation of the welfare formulas presented in Section 5.

The markets for interbank loans and deposits are perfectly competitive, with banks acting as price takers. The expected profits of Deposit Division $n$ are given by:

$$\int_0^1 \sum_{i=1}^{N} R_{t,\tau}^{I,ni} M_{t,\tau}^{ni} - R_{t,\tau}^{D,n} D_{t,\tau}^{n} \, \mathrm{d}\tau \, .$$

Upon solving the optimization problem, the interest rate charged by bank $n$ to $i$ at moment $\tau$ is:

$$R_{t,\tau}^{I,ni} = d_t^{ni} R_{t,\tau}^{D,n} = (1+\varsigma_D)^{-1} d_t^{ni} T_t^n z_{t,\tau}^n R_t^B \, , \tag{A.14}$$

where equation (A.6) is employed to obtain the final equality.

**Central Bank** As outlined in Section 4.5, the central bank determines the policy and lending facility rates. Here we assume and extended version of the penalty rate with the capacity to respond to business cycle fluctuations. Formally, let the lending facility rate be

$$R_{t,\tau}^{I,0n} = \text{penalty}_{t,\tau}^{n} \Phi_t^n R_t^B \, , \tag{A.15}$$

where $\text{penalty}_{t,\tau}^{n}$ is the penalty rate. We study different lending policies by assigning a flexible functional form to the penalty rate:

$$\text{penalty}_{t,\tau}^{n} = e^{\varpi_1} \underbrace{\left(\frac{\Phi_t^n}{\Phi^n}\right)^{-\varpi_2} z_{t,\tau}^0}_{\text{variable component}} \, , \tag{A.16}$$

where $\varpi_1$ is a parameter that controls the steady state size of the penalty, and $\varpi_2$ its response to steady state deviations of the bank's borrowing spread. $z_{t,\tau}^0$ is a policy shock, which we introduce for analytical convenience and assume to be distributed Weibull with mean one and shape parameter $\kappa$. Online Appendix 5.B provides a quantitative overview of how gains from trade vary across different calibrations of $\{\varpi_1, \varpi_2\}$.

Furthermore, we assume that any profits generated by the central bank's lending facility are returned to the representative household through a lump-sum transfer:

$$\Upsilon_t^{CB} = \sum_{n=1}^{N} \int_0^1 R_{t-1,\tau}^{I,0n} M_{t-1,\tau}^{0n} \, d\tau \, .$$

**Banking sector aggregation** By incorporating equations (A.14) and (A.15) into equation (A.10), we derive the distribution of the interbank rate paid by bank $n$:

$$\widetilde{R}^{I,n}_{t,\tau} \sim W\left(\frac{\widetilde{R}^{I,n}_t}{\Gamma(1+1/\kappa)}, \kappa\right), \quad \widetilde{R}^{I,n}_t = \Phi^n_t \left[1 + e^{-\kappa\varpi_1}\left(\frac{\Phi^n_t}{\Phi^n}\right)^{\kappa\varpi_2}\right]^{-1/\kappa}, \quad \Phi^n_t = \left[\sum_{i=1}^{N}\left((1+\varsigma_D)^{-1} d^{in}_t T^i_t\right)^{-\kappa}\right]^{-1/\kappa}, \tag{A.17}$$

where we use the fact that the minimum of Weibull-distributed draws is itself Weibull.

**Transaction volumes and Deposits** We introduce $\xi^{0i}_t$ to represent the proportion of funding that bank $i$ acquires from the central bank:

$$\xi^{0i}_t = \left[1 + e^{\kappa\varpi_1}\left(\frac{\Phi^i_t}{\Phi^i}\right)^{-\kappa\varpi_2}\right]^{-1}. \tag{A.18}$$

The transaction volume between any pair of banks can be expressed as:

$$M^{ni}_t = \begin{cases} \xi^{0i}_t L^i_t, & \text{if } n = 0, \\ \left(1-\xi^{0i}_t\right)\lambda^{ni}_t L^i_t, & \text{otherwise,} \end{cases} \quad \text{where: } \lambda^{ni}_t = \left(\frac{(1+\varsigma_D)^{-1} d^{ni}_t T^n_t}{\Phi^i_t}\right)^{-\kappa}. \tag{A.19}$$

The share $\lambda^{ni}_t$ is the fraction of bank $i$'s non-central-bank funding obtained from bank $n$. A detailed interpretation is provided in Section 4.6.

By integrating equation (A.12) along the continuum and across banks, we derive the following expressions for aggregate bank deposits:

$$D^n_t = \sum_{i=1}^{N} d^{ni}_t M^{ni}_t,$$
$$D_t + \sum_{n=1}^{N} M^{0n}_t = \sum_{n=1}^{N} L^n_t + \sum_{n=1}^{N}\sum_{i=1}^{N}\left(d^{ni}_t - 1\right) M^{ni}_t.$$

The second line shows that aggregate deposits and central bank money equal total lending plus interbank transaction costs. Aggregate banking profits in period $t$ are:

$$\Upsilon^B_t = \sum_{n=1}^{N} R^{F,n}_{t-1} L^n_{t-1} - \sum_{n=1}^{N}\int_0^1 R^{D,n}_{t-1,\tau} D^n_{t-1,\tau}\,\mathrm{d}\tau - \sum_{n=1}^{N}\int_0^1 R^{I,0n}_{t-1,\tau} M^{0n}_{t-1,\tau}\, d\tau.$$

Combining (A.18) and (A.17) yields:

$$\widetilde{R}^{I,n}_t = \Phi^n_t (1-\xi^{0n}_t)^{1/\kappa}. \tag{A.20}$$

From (A.19), we obtain:

$$\Phi_t^n = (\lambda_t^{nn})^{1/\kappa}\left(\frac{T_t^n}{1+\varsigma_D}\right) \quad \text{and} \quad \Phi_t^{n,AU} = \left(\frac{T_t^n}{1+\varsigma_D}\right), \tag{A.21}$$

where $\Phi_t^{n,AU}$ denotes the average credit spread of bank $n$ under autarky, i.e., when $\lambda_t^{in} = +\infty$ for $i \neq n$. In this case, $\Phi_t^{n,AU}$ corresponds to the average cost of raising deposits.

Equations (A.20) and (A.21) imply the following expression for the ratio of the autarky credit spread to its interbank-lending counterpart:

$$\frac{\widetilde{R}_t^{I,n,AU}}{\widetilde{R}_t^{I,n}} = (\lambda_t^{nn})^{-1/\kappa}\left(\frac{1-\xi_t^{0n}}{1-\xi_t^{0n,AU}}\right)^{-1/\kappa}. \tag{A.22}$$

Using the definition of the aggregate credit spread, $\widetilde{R}_t^I = \left[\sum_{n=1}^N a_t^n \left(\widetilde{R}_t^{I,n}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$, we obtain an expression for the ratio of the aggregate credit spread under autarky to its interbank-lending counterpart:

$$\frac{\widetilde{R}_t^{I,AU}}{\widetilde{R}_t^I} = \left[\sum_{n=1}^N s_t^n \left(\frac{\widetilde{R}_t^{I,n,AU}}{\widetilde{R}_t^{I,n}}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}. \tag{A.23}$$

where $s_t^n \equiv a_t^n \left(\frac{\widetilde{R}_t^{I,n}}{\widetilde{R}_t^I}\right)^{1-\sigma}$ as defined in (10).

Combining (A.22) and (A.23), we obtain the following relationship between the aggregate credit spread and its autarky counterpart:

$$\widetilde{R}_t^I = (\lambda_t^{Own})^{1/\kappa}(E_t)^{1/\kappa}\widetilde{R}_t^{I,AU}, \tag{A.24}$$

where $\lambda_t^{Own} = \left[\sum_{n=1}^N s_t^n \left(\lambda_t^{nn}\right)^{\frac{\sigma-1}{\kappa}}\right]^{\frac{\kappa}{\sigma-1}}$ is defined in (10), and $E_t$ is a CES index capturing the average relative reliance on central bank funding in the current state relative to autarky, defined as:

$$E_t = \left[\sum_{n=1}^N \gamma_t^n \left(\frac{1-\xi_t^{0n}}{1-\xi_t^{0n,AU}}\right)^{\frac{\sigma-1}{\kappa}}\right]^{\frac{\kappa}{\sigma-1}}, \quad \text{where } \gamma_t^n \equiv \frac{s_t^n \left(\lambda_t^{nn}\right)^{\frac{\sigma-1}{\kappa}}}{\sum_{i=1}^N s_t^i \left(\lambda_t^{ii}\right)^{\frac{\sigma-1}{\kappa}}}. \tag{A.25}$$

**Government** The government provides subsidies to firms, banks, and depositors, financed by lump-sum taxes on the representative household. Government transfers are therefore:

$$\Upsilon_t^G = -\left[\varsigma_F P_t Y_t + \varsigma_B \sum_{n=1}^N R_{t-1}^{F,n} L_{t-1}^n + \varsigma_D \sum_{n=1}^N R_{t-1}^{D,n} D_{t-1}^n\right].$$

Section 5 derives the welfare approximation under the assumption that these subsidies eliminate real distortions and are financed by lump-sum taxes.[1,2] The production subsidy offsets inefficient underproduction due to monopolistic pricing by intermediate producers. The banking subsidy offsets distortions from monopolistic competition in loan provision that would otherwise reduce capital accumulation. Finally, the deposit subsidy offsets a steady-state distortion to household saving induced by the central bank's lending facility: when banks substitute away from household deposits toward central bank funding, the deposit rate paid to households is depressed absent an offsetting transfer.[3]

**Market Clearing** Total transfers to the household are:

$$\Upsilon_t \equiv \Upsilon_t^F + \Upsilon_t^B + \Upsilon_t^{CB} + \Upsilon_t^G = P_t Y_t - \int_0^1 W_t(\nu) N_t(\nu)\, \mathrm{d}\nu - (1+\varsigma_D) \sum_{n=1}^{N} \int_0^1 R_{t-1,\tau}^{D,n} D_{t-1,\tau}^{n}\, \mathrm{d}\tau\,.$$

Aggregating (A.1) over $\tau$ and using the expression above yields the aggregate resource constraint:

$$C_t + \frac{D_t}{P_t} = Y_t\,. \tag{A.26}$$

**Aggregation** Using (A.2), (A.3), (A.8), and (A.11), we can express firm-level marginal costs as a function of aggregate variables:

$$\frac{MC_{t+j|t}(\nu)}{P_{t+j}} = (1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)} \left(X_{t+j}\right)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} \left(Y_{t+j}\right)^{\frac{1-\alpha}{\eta+\alpha}} \left(\widetilde{R}_{t+j}^{I}\right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)} \left(\frac{P_t^*}{P_{t+j}}\right)^{-\left(\frac{\epsilon(1-\alpha)}{\eta+\alpha}\right)}. \tag{A.27}$$

In a similar manner, loan and labor demand can be integrated across the continuum of firms to obtain:

$$\begin{aligned} \frac{L_t}{P_t} &= \alpha(1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}} \left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha}\right)} (X_t)^{\frac{\eta(1-\alpha)}{\eta+\alpha}} (Y_t)^{\frac{\eta+1}{\eta+\alpha}} \left(\widetilde{R}_t^I\right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha}\right)} \Delta_t\,, \\ N_t &= (1-\alpha)^{\left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{\alpha\left(\frac{\eta}{\eta+\alpha}\right)} (X_t)^{(1-\alpha)\left(\frac{\eta}{\eta+\alpha}\right)} \left(\frac{X_t}{Y_t}\right)^{-\left(\frac{\eta}{\eta+\alpha}\right)} \left(\widetilde{R}_t^I\right)^{\alpha\left(\frac{\eta}{\eta+\alpha}\right)} \Delta_t^{\frac{\eta}{\eta+1}}\,, \end{aligned} \tag{A.28}$$

where $\Delta_t$ denotes a measure of price dispersion that can be recursively defined as:

$$\Delta_t = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{-\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)} + \theta \Pi_t^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)} \Delta_{t-1}\,.$$

[1] This assumption greatly simplifies the derivation and is standard in the macroeconomics literature; see, for example, Woodford (2003).

[2] Subsidies are not required to derive the equilibrium conditions. For simplicity, we present the main analytical expressions under zero subsidies unless otherwise stated.

[3] The smoothness induced by extreme value liquidity shocks $z_t^n$ implies that banks borrow a non-zero amount from the lending facility in steady state. While steady-state facility borrowing is small under reasonable calibrations, we offset the associated distortion to maintain analytical tractability in Section 5. Empirically, this assumption is consistent with the absence of evidence that LoLR interventions depress the steady-state deposit rate.

Substitute equation (A.27) and the expressions for $Q_{t+j}$ into the optimal resetting price equation (A.9):

$$\left(\frac{P_t^*}{P_t}\right)^{1+\epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)} = \frac{\mathbb{E}_t\left[\Sigma_{j=0}^{\infty}(\theta\beta)^j\,(1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}}\left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1}\right)\left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)}\left(X_{t+j}\right)^{-\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)}\left(Y_{t+j}\right)^{\frac{\eta+1}{\eta+\alpha}}\left(\frac{P_{t+j}}{P_t}\right)^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)}\left(\widetilde{R}^I_{t+j}\right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)}\right]}{\mathbb{E}_t\left[\Sigma_{j=0}^{\infty}(\theta\beta)^j\left(\frac{P_{t+j}}{P_t}\right)^{\epsilon-1}\left(X_{t+j}\right)^{-1}\left(Y_{t+j}\right)\right]}.$$

This expression can be simplified as:

$$\frac{P_t^*}{P_t} = \left(\frac{F_t}{H_t}\right)^{1/\left[1+\epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]},$$

where

$$F_t = (1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}}\left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1}\right)\left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)}(X_t)^{-\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)}(Y_t)^{\frac{\eta+1}{\eta+\alpha}}\left(\widetilde{R}^I_t\right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)} + \theta\beta\mathbb{E}_t\left[\Pi_{t+1}^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)}F_{t+1}\right] \tag{A.29}$$

$$H_t = \left(\frac{X_t}{Y_t}\right)^{-1} + \theta\beta\mathbb{E}_t\left[\Pi_{t+1}^{\epsilon-1}H_{t+1}\right].$$

Applying equation (A.7) to the previous equations yields the following equilibrium conditions:

$$\frac{F_t}{H_t} = \left(\frac{1-\theta}{1-\theta\Pi_t^{\epsilon-1}}\right)^{\left(\frac{1}{\epsilon-1}\right)\left[1+\epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]},$$

$$\Delta_t = (1-\theta)\left(\frac{1-\theta\Pi_t^{\epsilon-1}}{1-\theta}\right)^{\left(\frac{\epsilon}{\epsilon-1}\right)\left(\frac{\eta+1}{\eta+\alpha}\right)} + \theta\Pi_t^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)}\Delta_{t-1}.$$

Utilizing equation (A.26), we derive an expression for the composite good as:

$$X_t = Y_t - \sum_{n=1}^{N}\int_0^1 T_t^n z_{t,\tau}^n \frac{D_{t,\tau}^n}{P_t}\,\mathrm{d}\tau.$$

In the previous equation, the deposits expression is related to the model's aggregate variables as follows:

$$\sum_{n=1}^{N}\int_0^1 T_t^n z_{t,\tau}^n \frac{D_{t,\tau}^n}{P_t}\,\mathrm{d}\tau = \frac{1}{(1+\varsigma_D)^{-1}}\sum_{n=1}^{N}\sum_{i=1}^{N}\int_0^1 \widetilde{R}_{t,\tau}^{I,ni}\frac{M_{t,\tau}^{ni}}{P_t}\,\mathrm{d}\tau = \frac{1-\xi_t^0}{(1+\varsigma_D)^{-1}}\widetilde{R}_t^I\frac{L_t}{P_t},$$

where $\xi_t^0 = \sum_{i=1}^N s_t^i \xi_t^{0i}$ denotes a weighted share of the funding obtained from the central bank lending facility, and $s_t^i = a_t^i\left(\widetilde{R}_t^{I,i}/\widetilde{R}_t^I\right)^{1-\sigma}$ represents the share of loans to firms supplied by bank $i$. By incorporating the aggregate loan demand in equation (A.28) and the previous expressions, we obtain:

$$\frac{X_t}{Y_t} = 1-\alpha(1-\alpha)^{\frac{1-\alpha}{\eta+\alpha}}\left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{-\left(\frac{\eta(1-\alpha)}{\eta+\alpha}\right)}\left(\frac{X_t}{Y_t}\right)^{-\left(\frac{1-\alpha}{\eta+\alpha}\right)}(X_t)^{(1-\alpha)\left(\frac{\eta+1}{\eta+\alpha}\right)}\left(\widetilde{R}_t^I\right)^{\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)}\left(\frac{1-\xi_t^0}{(1+\varsigma_D)^{-1}}\right)\Delta_t. \tag{A.30}$$

The real money balances lent by the central bank to bank $i$ are:

$$\frac{M_t^{0i}}{P_t} = \xi_t^{0i}\frac{L_t^i}{P_t} = s_t^i\xi_t^{0i}\frac{L_t}{P_t}.$$

The total real money balances lent by the central bank are:

$$\frac{M_t^0}{P_t} = \sum_{i=1}^{N} \frac{M_t^{0i}}{P_t} = \xi_t^0 \frac{L_t}{P_t} \,.$$

**Flexible Price Equilibrium** We define the flexible equilibrium of the economy as the one in which all firms can reset prices every quarter ($\theta = 0$). By combining equations (A.29)-(A.30), we derive the following expressions:

$$X_t^n = (1-\alpha)^{-\left(\frac{1}{\eta+1}\right)} \left[1-\alpha\left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1}\right)^{-1}\left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{-1}\left(\frac{1-\xi_t^0}{(1+\varsigma_D)^{-1}}\right)\right]^{\frac{1}{\eta+1}} \left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1}\right)^{-\left(\frac{1}{1-\alpha}\right)\left(\frac{\eta+\alpha}{\eta+1}\right)} \left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{-\left(\frac{\alpha}{1-\alpha}\right)} \left(\widetilde{R}_t^I\right)^{-\left(\frac{\alpha}{1-\alpha}\right)} ,$$
$$\frac{X_t^n}{Y_t^n} = 1-\alpha\left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1}\right)^{-1}\left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{-1}\left(\frac{1-\xi_t^0}{(1+\varsigma_D)^{-1}}\right),$$

where the index $n$ denotes a variable under flexible prices. Furthermore, we can express the latter equation as:

$$\frac{\widetilde{X}_t}{\widetilde{Y}_t} = \left[1-\alpha\left(\frac{(1+\varsigma_F)^{-1}\epsilon}{\epsilon-1}\right)^{-1}\left(\frac{(1+\varsigma_B)^{-1}\sigma}{\sigma-1}\right)^{-1}\left(\frac{1-\xi_t^0}{(1+\varsigma_D)^{-1}}\right)\right]^{-1}\frac{X_t}{Y_t} \,,$$

where $\widetilde{X}_t = X_t / X_t^n$ and $\widetilde{Y}_t = Y_t / Y_t^n$ represent the composite good and output gaps, respectively.

**Equilibrium Conditions Summary** We provide a summary of the equilibrium conditions under the assumption of zero trend inflation ($\Pi = 1$) and optimal government subsidies to firms, banks, and depositors ($\varsigma_F^* = \frac{1}{\epsilon-1}$; $\varsigma_B^* = \frac{1}{\sigma-1}$; $\varsigma_D^* = \frac{\xi^0}{1-\xi^0}$) to offset steady-state real distortions arising from monopolistic mark-ups and central bank intervention.

$$F_t = \left[1-\alpha\frac{1-\xi_t^0}{1-\xi^0}\right]^{-1}\left(\frac{\widetilde{X}_t}{\widetilde{Y}_t}\right)^{-\left(\frac{\eta+1}{\eta+\alpha}\right)}\widetilde{X}_t^{(1-\alpha)\left(\frac{\eta+1}{\eta+\alpha}\right)} + \theta\beta\mathbb{E}_t\left[\Pi_{t+1}^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)}F_{t+1}\right], \tag{A.31}$$

$$H_t = \left[1-\alpha\frac{1-\xi_t^0}{1-\xi^0}\right]^{-1}\left(\frac{\widetilde{X}_t}{\widetilde{Y}_t}\right)^{-1} + \theta\beta\mathbb{E}_t\left[\Pi_{t+1}^{\epsilon-1}H_{t+1}\right],$$

$$\frac{F_t}{H_t} = \left(\frac{1-\theta}{1-\theta\Pi_t^{\epsilon-1}}\right)^{\left(\frac{1}{\epsilon-1}\right)\left[1+\epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]}, \tag{A.32}$$

$$\Delta_t = (1-\theta)\left(\frac{1-\theta\Pi_t^{\epsilon-1}}{1-\theta}\right)^{\left(\frac{\epsilon}{\epsilon-1}\right)\left(\frac{\eta+1}{\eta+\alpha}\right)} + \theta\Pi_t^{\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)}\Delta_{t-1},$$

$$\left[1-\alpha\frac{1-\xi_t^0}{1-\xi^0}\right]\frac{\widetilde{X}_t}{\widetilde{Y}_t} = 1-\alpha\left(\frac{1-\xi_t^0}{1-\xi^0}\right)\left(\frac{\widetilde{X}_t}{\widetilde{Y}_t}\right)^{-\left(\frac{1-\alpha}{\eta+\alpha}\right)}\widetilde{X}_t^{(1-\alpha)\left(\frac{\eta+1}{\eta+\alpha}\right)}\Delta_t, \tag{A.33}$$

$$\frac{1}{R_t^B} = \beta\mathbb{E}_t\left[\frac{\widetilde{X}_t}{\widetilde{X}_{t+1}\Pi_{t+1}}\frac{X_t^n}{X_{t+1}^n}\right], \tag{A.34}$$

$$R_t^B = R^B\,\Pi_t^{\gamma_\pi}\,\widetilde{Y}_t^{\gamma_y}, \tag{A.35}$$

$$N_t = (1-\alpha)^{\frac{\eta}{\eta+1}}\left[1-\alpha\frac{1-\xi_t^0}{1-\xi^0}\right]^{-\frac{\eta}{\eta+1}}\left(\frac{\widetilde{X}_t}{\widetilde{Y}_t}\right)^{-\left(\frac{\eta}{\eta+\alpha}\right)}\widetilde{X}_t^{(1-\alpha)\left(\frac{\eta}{\eta+\alpha}\right)}\Delta_t^{\frac{\eta}{\eta+1}}, \tag{A.36}$$

$$X_t^n = \left[1-\alpha\frac{1-\xi_t^0}{1-\xi^0}\right]^{\frac{1}{\eta+1}}(1-\alpha)^{-\frac{1}{\eta+1}}\left(\widetilde{R}_t^I\right)^{-\left(\frac{\alpha}{1-\alpha}\right)}, \tag{A.37}$$

$$\xi_t^0 = \sum_{i=1}^N s_t^i\left[1+e^{\kappa\varpi_1}\left(\frac{\Phi_t^i}{\Phi^i}\right)^{-\kappa\varpi_2}\right]^{-1}, \tag{A.38}$$

$$s_t^i = a_t^i\left(\frac{\widetilde{R}_t^{I,i}}{\widetilde{R}_t^I}\right)^{1-\sigma},$$

$$\widetilde{R}_t^I = \left[\sum_{i=1}^N a_t^i\left(\widetilde{R}_t^{I,i}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}, \tag{A.39}$$

$$\widetilde{R}_t^{I,i} = \Phi_t^i\left[1+e^{-\kappa\varpi_1}\left(\frac{\Phi_t^i}{\Phi^i}\right)^{\kappa\varpi_2}\right]^{-1/\kappa}, \tag{A.40}$$

$$\Phi_t^i = \left[\sum_{n=1}^N\left(\left(1-\xi^0\right)d_t^{ni}T_t^n\right)^{-\kappa}\right]^{-\frac{1}{\kappa}}, \tag{A.41}$$

$$
\begin{aligned}
T_t^n &= T^n exp\left(u_t^{T,n}\right), \\
d_t^{ni} &= \left(d^{ni}\right)^{\varrho} exp\left(u_t^{I,ni}\right), \\
a_t^n &= \frac{a^n exp\left(u_t^{a,n}\right)}{\sum_{j=1}^{N} a^j exp\left(u_t^{a,j}\right)}, &\text{(A.42)}\\
u_t^{T,n} &= \rho_I u_{t-1}^{T,n} + \varepsilon_t^{T,n}, &\text{(A.43)}\\
u_t^{a,n} &= \rho_I u_{t-1}^{a,n} + \varepsilon_t^{a,n}, \\
u_t^{I,ni} &= \rho_I u_{t-1}^{I,ni} + \varepsilon_t^{I,ni}.
\end{aligned}
$$

**Steady State Summary** These are the steady-state values of the variables, assuming zero trend inflation and the implementation of optimal government subsidies.

$$
\begin{aligned}
F &= \frac{1}{(1-\alpha)(1-\theta\beta)}, \\
H &= \frac{1}{(1-\alpha)(1-\theta\beta)}, \\
\Delta &= 1, \\
\widetilde{X} &= 1, \\
\widetilde{Y} &= 1, \\
\Phi^i &= \left[\sum_{n=1}^{N}\left(\left(1-\xi^0\right) d^{ni} T^n\right)^{-\kappa}\right]^{-\frac{1}{\kappa}}, \\
\widetilde{R}^I &= \left[\sum_{i=1}^{N} a^i \left(1-\xi^{0i}\right)^{\frac{1-\sigma}{\kappa}} \left(\Phi^i\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}, \\
\frac{M}{P} &= \xi^0 \left(\frac{\alpha}{1-\alpha}\right)\left(\widetilde{R}^I\right)^{-\frac{1}{1-\alpha}},
\end{aligned}
$$

$$
\begin{aligned}
X^n &= \left(\widetilde{R}^I\right)^{-\left(\frac{\alpha}{1-\alpha}\right)}, \\
u^{T,n} &= 0, \\
u^{a,n} &= 0, \\
u^{I,ni} &= 0, \\
N &= 1, \\
R^B &= \beta^{-1}, \\
\xi^0 &= \left[1+e^{\kappa\varpi_1}\right]^{-1}.
\end{aligned}
$$

The steady-state aggregate interbank rate can be further expressed as:

$$
\widetilde{R}^I = \left(\lambda^{Own}\right)^{1/\kappa} \widetilde{R}^{I,AU},
$$

which follows from equations (A.24) and (A.25), noting that $E = 1$ in steady state since $\xi^{0n} = \xi^{0n,AU}$ for all $n$. Steady-state utility at the efficient allocation is then:

$$
U = \log\left(X^n\right) - \left(\frac{\eta}{\eta+1}\right)\int_0^1 N_\tau^{1+1/\eta}\, \mathrm{d}\tau \propto -\frac{1}{\kappa}\left(\frac{\alpha}{1-\alpha}\right)\log\left(\lambda^{Own}\right) - \left(\frac{\alpha}{1-\alpha}\right)\log\left(\widetilde{R}^{I,AU}\right). \tag{A.44}
$$

## Appendix 2.A Linearized Model Solution

This section presents a first-order log-linear approximation of the model's equilibrium conditions and derives its solution. Throughout, hats denote log-deviations from steady state, and lowercase letters represent logarithms of the corresponding variables.

Approximating equations (A.41) and (A.42), we obtain:

$$\hat{\hat{r}}_t^I = \sum_{i=1}^N s^i \left[ \left(1-\varpi_2 \xi^0\right) \sum_{n=1}^N \lambda^{ni} \left[ \hat{u}_t^{T,n} + \hat{u}_t^{I,ni} \right] - \frac{\widehat{\log\left(a_t^i\right)}}{\sigma - 1} \right] . \tag{A.45}$$

Combining (A.43)-(A.45), we obtain the following expression:

$$\begin{aligned}
\hat{\hat{r}}_t^I &= \rho_I \hat{\hat{r}}_{t-1}^I + (1-\varpi_2\xi^0)\left[\hat{\varepsilon}_t^{T,r} + \hat{\varepsilon}_t^{I,r}\right] - \frac{\hat{\varepsilon}_t^a}{\sigma-1} , \qquad (A.46)\\
\text{where:} \quad \hat{\varepsilon}_t^a &= \sum_{i=1}^N \left(s^i - a^i\right)\hat{\varepsilon}_t^{a,i} , \quad \hat{\varepsilon}_t^{T,r} = \sum_{i=1}^N \left[\frac{1-\varpi_2\xi^{0i}}{1-\varpi_2\xi^0}\right] s^i \sum_{n=1}^N \lambda^{ni}\hat{\varepsilon}_t^{T,n} ,\\
\hat{\varepsilon}_t^{I,r} &= \sum_{i=1}^N \left[\frac{1-\varpi_2\xi^{0i}}{1-\varpi_2\xi^0}\right] s^i \sum_{n=1}^N \lambda^{ni}\hat{\varepsilon}_t^{I,ni} .
\end{aligned}$$

Similarly, we obtain a log-linear approximation of equations (A.33) and (A.38) as:

$$\begin{aligned}
&\hat{\hat{x}}_t - \hat{\hat{y}}_t = -\alpha\left(\frac{\eta+1}{\eta}\right)\hat{\hat{x}}_t , \qquad (A.47)\\
&\widehat{\log\left(\xi_t^0\right)} = \rho_I \widehat{\log\left(\xi_{t-1}^0\right)} + \kappa\varpi_2(1-\xi^0)\left[\hat{\varepsilon}_t^{T,\xi} + \hat{\varepsilon}_t^{I,\xi}\right] ,\\
&\text{where:} \quad \hat{\varepsilon}_t^a = \sum_{i=1}^N \left(s^i - a^i\right)\hat{\varepsilon}_t^{a,i}; \quad \hat{\varepsilon}_t^{T,\xi} = \sum_{i=1}^N \left[\frac{1-\xi^{0i}}{1-\xi^0}\right] s^i \sum_{n=1}^N \lambda^{ni}\hat{\varepsilon}_t^{T,n} ,\\
&\qquad \hat{\varepsilon}_t^{I,\xi} = \sum_{i=1}^N \left[\frac{1-\xi^{0i}}{1-\xi^0}\right] s^i \sum_{n=1}^N \lambda^{ni}\hat{\varepsilon}_t^{I,ni} .
\end{aligned}$$

The approximations of equations (A.31) - (A.32) can be expressed as follows:

$$\begin{aligned}
\hat{f}_t &= (1-\theta\beta)\left[-\left(\frac{\eta+1}{\eta+\alpha}\right)(\hat{\hat{x}}_t - \hat{\hat{y}}_t) + (1-\alpha)\left(\frac{\eta+1}{\eta+\alpha}\right)\hat{\hat{x}}_t - \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\log\left(\xi_t^0\right)\right] + \theta\beta\left[\epsilon\left(\frac{\eta+1}{\eta+\alpha}\right)\mathbb{E}_t\left[\hat{\pi}_{t+1}\right] + \mathbb{E}_t\left[\hat{f}_{t+1}\right]\right] ,\\
\hat{h}_t &= -(1-\theta\beta)\left[(\hat{\hat{x}}_t - \hat{\hat{y}}_t) + \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\log\left(\xi_t^0\right)\right] + \theta\beta\left[(\epsilon-1)\mathbb{E}_t\left[\hat{\pi}_{t+1}\right] + \mathbb{E}_t\left[\hat{h}_{t+1}\right]\right] ,\\
\hat{f}_t - \hat{h}_t &= \left[1+\epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]\left(\frac{\theta}{1-\theta}\right)\hat{\pi}_t . \qquad (A.48)
\end{aligned}$$

By combining equations (A.47)-(A.48), we derive the New-Keynesian Phillips Curve:

$$\hat{\pi}_t = \Omega\hat{\hat{y}}_t + \beta\mathbb{E}_t\left[\hat{\pi}_{t+1}\right] , \tag{A.49}$$

where $\Omega = (1-\alpha)\left(\frac{\eta+1}{\eta}\right)\left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-1}\left[1+\epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]^{-1}\left(\frac{(1-\theta)(1-\theta\beta)}{\theta}\right)$. The log-linear approximations of equations (A.34) and (A.35) are as follows:

$$\hat{r}_t^B = \gamma_\pi\hat{\pi}_t + \gamma_y\hat{\hat{y}}_t , \tag{A.50}$$

$$-\hat{r}_t^B = \left[\hat{\hat{x}}_t - \mathbb{E}_t\left[\hat{\hat{x}}_{t+1}\right]\right] - \mathbb{E}_t\left[\hat{\pi}_{t+1}\right] - \left(\frac{\alpha}{1-\alpha}\right)\left[\hat{\hat{r}}_t^I - \mathbb{E}_t\left[\hat{\hat{r}}_{t+1}^I\right]\right] + \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{1}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left[\widehat{\log\left(\xi_t^0\right)} - \mathbb{E}_t\left[\widehat{\log\left(\xi_{t+1}^0\right)}\right]\right] . \tag{A.51}$$

By combining equations (A.46), (A.47), (A.50), and (A.51), we derive the Dynamic IS Equation:

$$\hat{\hat{y}}_t = -\left[1+\alpha\left(\frac{\eta}{\eta+1}\right)\right]\left[r_t^B - \mathbb{E}_t\left[\pi_{t+1}\right] - r_t^n\right] + \mathbb{E}_t\left[\hat{\hat{y}}_{t+1}\right], \tag{A.52}$$

where the natural rate of interest $r_t^n$ is given by

$$r_t^n = -\log(\beta) + \left(\frac{1}{1+\eta}\right)\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\mathbb{E}_t\left[\log\left(\xi_{t+1}^0\right) - \log\left(\xi_t^0\right)\right] - \left(\frac{\alpha}{1-\alpha}\right)\mathbb{E}_t\left[\tilde{r}_{t+1}^I - \tilde{r}_t^I\right].$$

Hence, expected movements in the central bank funding share $\xi_t^0$ and the aggregate interbank spread $\tilde{r}_t^I$ shift $r_t^n$ and, in turn, the IS wedge. When $\varpi_2 = 0$, LoLR lending shares $\xi_t^0$ are constant and we recover the expression presented in the main text.

**First-Order Model Solution** The first-order model solution consists of equations (A.49) and (A.52), which form a system of stochastic differential equations. These can be summarized in matrix form as follows:

$$A_0\begin{bmatrix}\hat{\hat{y}}_t\\ \hat{\pi}_t\end{bmatrix} = B_0\mathbb{E}_t\begin{bmatrix}\hat{\hat{y}}_{t+1}\\ \hat{\pi}_{t+1}\end{bmatrix} + C_0\begin{bmatrix}\hat{\tilde{r}}_t^I\\ \widehat{\log\left(\xi_t^0\right)}\end{bmatrix} \quad \text{and} \quad \begin{bmatrix}\hat{\tilde{r}}_t^I\\ \widehat{\log\left(\xi_t^0\right)}\end{bmatrix} = G_0\begin{bmatrix}\hat{\tilde{r}}_{t-1}^I\\ \widehat{\log\left(\xi_{t-1}^0\right)}\end{bmatrix} + G_1\begin{bmatrix}\varepsilon_t^{T,r}\\ \varepsilon_t^{I,r}\\ \varepsilon_t^{T,\xi}\\ \varepsilon_t^{I,\xi}\\ \varepsilon_t^{a}\end{bmatrix},$$

$$\text{where:}\quad A_0 = \begin{bmatrix}-\Omega & 1\\ \left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-1}+\gamma_y & \gamma_\pi\end{bmatrix}, \quad B_0 = \begin{bmatrix}0 & \beta\\ \left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-1} & 1\end{bmatrix},$$

$$C_0 = \begin{bmatrix}0 & 0\\ (1-\rho_I)\left(\frac{\alpha}{1-\alpha}\right) & -(1-\rho_I)\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{1}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\end{bmatrix},$$

$$G_1 = \begin{bmatrix}\left(1-\varpi_2\xi^0\right) & \left(1-\varpi_2\xi^0\right) & 0 & 0 & -\frac{1}{\sigma-1}\\ 0 & 0 & \kappa\varpi_2(1-\xi^0) & \kappa\varpi_2(1-\xi^0) & 0\end{bmatrix},$$

$$G_0 = diag(\rho_I, \rho_I)$$

Solving forward the system of equations:

$$\begin{bmatrix}\hat{\hat{y}}_t\\ \hat{\pi}_t\end{bmatrix} = \underbrace{\left[\sum_{s=0}^{\infty}\left(A_0^{-1}B_0\right)^s\left(A_0^{-1}C_0\right)G_0^s\right]}_{\equiv\Psi}\begin{bmatrix}\hat{\tilde{r}}_t^I\\ \widehat{\log\left(\xi_t^0\right)}\end{bmatrix}. \tag{A.53}$$

If the matrix $\left(I - G^T \otimes \left(A_0^{-1}B_0\right)\right)$ is invertible, a solution for $\Psi$ can be obtained as follows:

$$vec(\Psi) = \left(I - G_0^T \otimes \left(A_0^{-1}B_0\right)\right)^{-1} vec(A_0^{-1}C_0),$$

where $vec()$ denotes the matrix vectorization operation.

# Appendix 3 Welfare

We define the static gains from trade as the steady-state change in utility relative to autarky. Using (A.44), we obtain:

$$\mathbb{J}^{ss} \equiv \frac{U - U^{AU}}{U_x X} = -\frac{\Lambda_r}{\kappa} \log\left(\lambda^{Own}\right), \quad \text{where } \Lambda_r \equiv \left(\tfrac{\alpha}{1-\alpha}\right).$$

A second-order expansion of representative-household utility around the efficient steady state (with zero trend inflation, $\Pi = 1$) is:

$$U_t - U = \hat{x}_t - \frac{1}{2}\hat{x}_t^2 - \left[\hat{n}_t + \frac{1}{2\eta}\hat{n}_t^2\right] + \text{h.o.t.}, \tag{A.54}$$

where h.o.t. denotes higher-order terms omitted from the approximation. Using the second-order approximations to $\hat{x}_t$ and $\hat{n}_t$ collected in Appendix Appendix 3.A, expected per-period utility can be written as:

$$\mathbb{E}\left[\frac{U_t - U}{U_x X}\right] = -\Lambda_r \mathbb{E}\left[\hat{\hat{r}}_t^I\right] + \Lambda_\xi \mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}\right] - \frac{\Lambda_{rr}}{2}\mathbb{E}\left[\left(\hat{\hat{r}}_t^I\right)^2\right] + \frac{\Lambda_{\xi\xi}}{2}\mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}^2\right] + \Lambda_{r\xi}\mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}\hat{\hat{r}}_t^I\right] + \text{t.i.p.} + \text{h.o.t.}, \tag{A.55}$$

where $\Lambda_r$, $\Lambda_\xi$, $\Lambda_{\xi\xi}$, and $\Lambda_{r\xi}$ are constants (defined in (A.68)) that summarize the economy's sensitivity to credit-spread shocks and central bank lending. The term t.i.p. collects components that are independent of policy.

Using (A.55), the stochastic gains from trade are given by:

$$\begin{aligned}\mathbb{J} \equiv \mathbb{E}\left[\frac{U_t - U_t^{AU}}{U_x X}\right] = \; & \mathbb{J}^{ss} - \Lambda_r \mathbb{E}\left[\hat{\hat{r}}_t^I - \hat{\hat{r}}_t^{I,AU}\right] + \Lambda_\xi \mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)} - \widehat{\log\left(\xi_t^{0,AU}\right)}\right] \\ & - \frac{\Lambda_{rr}}{2}\mathbb{E}\left[\left(\hat{\hat{r}}_t^I\right)^2 - \left(\hat{\hat{r}}_t^{I,AU}\right)^2\right] + \frac{\Lambda_{\xi\xi}}{2}\mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}^2 - \widehat{\log\left(\xi_t^{0,AU}\right)}^2\right] \\ & + \Lambda_{r\xi}\mathbb{E}\left[\hat{\hat{r}}_t^I \widehat{\log\left(\xi_t^0\right)} - \hat{\hat{r}}_t^{I,AU}\widehat{\log\left(\xi_t^{0,AU}\right)}\right] + \text{t.i.p.} + \text{h.o.t.}.\end{aligned} \tag{A.56}$$

Substituting (A.24) into (A.56) to replace the aggregate credit spread $\hat{\hat{r}}_t^I$ yields:

$$\mathbb{J} = -\left[\frac{\Lambda_r}{\kappa}\mathbb{E}\left[\log\left(\lambda_t^{Own}\right)\right] + \frac{\Lambda_{rr}}{2\kappa^2}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}^2\right] + \frac{\Lambda_{rr}}{\kappa}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\hat{\hat{r}}_t^{I,AU}\right]\right] + \mathbb{J}^{CB} + \text{t.i.p.} + \text{h.o.t.}, \tag{A.57}$$

where $\mathbb{J}^{CB}$ captures the central bank's lender-of-last-resort contribution to the gains from trade and is defined as:

$$\begin{aligned}\mathbb{J}^{CB} = \; & \Lambda_\xi \mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)} - \widehat{\log\left(\xi_t^{0,AU}\right)}\right] - \frac{\Lambda_r}{\kappa}\mathbb{E}\left[\hat{e}_t\right] - \frac{\Lambda_{rr}}{2\kappa^2}\mathbb{E}\left[\hat{e}_t^2\right] - \frac{\Lambda_{rr}}{\kappa^2}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\hat{e}_t\right] \\ & - \frac{\Lambda_{rr}}{\kappa}\mathbb{E}\left[\hat{\hat{r}}_t^{I,AU}\hat{e}_t\right] + \frac{\Lambda_{\xi\xi}}{2}\mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}^2 - \widehat{\log\left(\xi_t^{0,AU}\right)}^2\right] \\ & + \Lambda_{r\xi}\mathbb{E}\left[\hat{\hat{r}}_t^{I,AU}\left(\widehat{\log\left(\xi_t^0\right)} - \widehat{\log\left(\xi_t^{0,AU}\right)}\right)\right] + \frac{\Lambda_{r\xi}}{\kappa}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\hat{e}_t\right] + \frac{\Lambda_{r\xi}}{\kappa}\mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}\hat{e}_t\right].\end{aligned}$$

When the lender-of-last-resort parameter governing responses to deviations in the credit spread is set to zero as in the main text, that is, when $\varpi_2 = 0$, the central bank trade share is constant, $\xi_t^0 = \xi^0$ for all $t$. This results in the utility component associated with the central bank satisfying $\mathbb{J}^{CB} = 0$, which yields the expression for $\mathbb{J}$ reported in Section 5.1.2.

**Stochastic gains sufficient statistic approximation** The expression in (A.57) is not, in general, directly estimable from the data. It involves counterfactual objects—most notably the credit spread under autarky—and therefore requires additional structure linking observables to the model's fundamental shocks, together with distributional assumptions on those shocks. A tractable approximation obtains under the assumption:

$$\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\hat{\hat{r}}_t^{I,AU}\right] \approx 0,$$

together with the assumption that the central bank does not intervene in the interbank market, $\xi^0 \approx 0$ (or alternatively, that it does not target the credit spread, i.e. $\varpi_2 = 0$). Under these assumptions, (A.57) simplifies to:

$$\mathbb{J} = -\left[\frac{\Lambda_r}{\kappa}\mathbb{E}\left[\log\left(\lambda_t^{Own}\right)\right] + \frac{\Lambda_{rr}}{2\kappa^2}\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}^2\right]\right] + \text{t.i.p.} + \text{h.o.t.}.$$

Regarding the former assumption, equations (A.20) and (A.21) imply that, under autarky, bank credit spreads primarily fluctuate with depositor-preference shocks $T_t^n$ (for all $n$). Thus, $\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}\hat{\hat{r}}_t^{I,AU}\right] \approx 0$ amounts to assuming that the main drivers of the own trade share $\lambda_t^{Own}$ are other shocks (e.g., shocks to interbank transaction costs), so that $\lambda_t^{Own}$ is approximately orthogonal to the autarky credit spread. By further approximating $\mathbb{E}\left[\widehat{\log\left(\lambda_t^{Own}\right)}^2\right] \approx Var\left(\log\left(\lambda_t^{Own}\right)\right)$, we recover the expression in the main body.

Appendix 3.A provides second-order approximations of the variables in (A.56) as functions of the fundamental shocks. Appendix 3.B combines these approximations with distributional assumptions on the shocks to derive tractable expressions for the stochastic gains from trade and to decompose these gains into the contributions of individual shock types.

## Appendix 3.A Second-Order Approximations

This section collects the second-order log-linear approximations used to derive the welfare expressions above.

**Price dispersion** Following the standard New Keynesian model (see, e.g., Galí 2015), we express second-order price dispersion as a function of inflation:

$$\mathbb{E}\left[\widehat{\log(\Delta_t)}\right] = \frac{\theta\epsilon}{2(1-\theta)(1-\theta\beta)\Theta}\mathbb{E}\left[\hat{\pi}_t^2\right], \quad \text{where: } \Theta = \left(\frac{\eta+1}{\eta+\alpha}\right)^{-1}\left[1+\epsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]^{-1}. \tag{A.58}$$

**Aggregate variables** By implicit differentiation of (A.33), we obtain a second-order approximation of $\hat{\hat{x}}_t - \hat{\hat{y}}_t$ as a function of $\hat{\hat{x}}_t$ and $\widehat{\log(\Delta_t)}$.

$$\begin{aligned}\hat{\hat{x}}_t - \hat{\hat{y}}_t = & -\alpha\left(\frac{\eta+1}{\eta}\right)\hat{\hat{x}}_t - \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta+\alpha}{\eta}\right)\widehat{\log(\Delta_t)} - \frac{\alpha}{2}\left(\frac{\eta+1}{\eta}\right)^2\left(\frac{\eta+\alpha}{\eta}\right)\hat{\hat{x}}_t^2 \\ & + \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta+1}{\eta}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left(\frac{\eta+\alpha}{\eta}\right)\hat{\hat{x}}_t\widehat{\log(\xi_t^0)}.\end{aligned} \tag{A.59}$$

A second-order approximation of equations (A.36) and (A.37) is:

$$
\begin{aligned}
\hat{n}_t =& -\left(\frac{\eta}{\eta+\alpha}\right)(\hat{\hat{x}}_t - \hat{\hat{y}}_t) + (1-\alpha)\left(\frac{\eta}{\eta+\alpha}\right)\hat{\hat{x}}_t - \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\widehat{\log\left(\xi_t^0\right)} \\
&+ \left(\frac{\eta}{\eta+1}\right)\widehat{\log\left(\Delta_t\right)} - \frac{1}{2}\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left(\frac{1-\alpha-\xi^0}{(1-\alpha)(1-\xi^0)}\right)\widehat{\log\left(\xi_t^0\right)}^2 ,
\end{aligned} \tag{A.60}
$$

$$
\begin{aligned}
\hat{x}_t^n =& -\left(\frac{\alpha}{1-\alpha}\right)\hat{\hat{r}}_t^I + \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{1}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\widehat{\log\left(\xi_t^0\right)} \\
&+ \frac{1}{2}\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{1}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left(\frac{1-\alpha-\xi^0}{(1-\alpha)(1-\xi^0)}\right)\widehat{\log\left(\xi_t^0\right)}^2 .
\end{aligned} \tag{A.61}
$$

Plugging equation (A.59) into (A.60) and rearranging, we obtain:

$$
\begin{aligned}
\hat{n}_t =& \hat{\hat{x}}_t - \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\widehat{\log\left(\xi_t^0\right)} + \left[\left(\frac{\alpha}{1-\alpha}\right) + \left(\frac{\eta}{\eta+1}\right)\right]\widehat{\log\left(\Delta_t\right)} \\
&+ \frac{\alpha}{2}\left(\frac{\eta+1}{\eta}\right)^2\left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-2}(\hat{\hat{y}}_t)^2 - \frac{1}{2}\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left[1-\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\right]\widehat{\log\left(\xi_t^0\right)}^2 \\
&- \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta+1}{\eta}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-1}\widehat{\log\left(\xi_t^0\right)}\hat{\hat{y}}_t ,
\end{aligned} \tag{A.62}
$$

where the output gap expression in the second order terms follows from $\hat{\hat{x}}_t = \left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-1}\hat{\hat{y}}_t$ up to a first order, which also implies:

$$
(\hat{\hat{x}}_t)^2 = \left[1+\alpha\left(\tfrac{\eta+1}{\eta}\right)\right]^{-2}(\hat{\hat{y}}_t)^2 . \tag{A.63}
$$

We can obtain an expression for the second-order approximation to the squares of the remaining variables as:

$$
\begin{aligned}
(\hat{n}_t)^2 =& \left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-2}(\hat{\hat{y}}_t)^2 + \left[\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\right]^2\widehat{\log\left(\xi_t^0\right)}^2 \\
&- 2\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\eta}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left[1+\alpha\left(\tfrac{\eta+1}{\eta}\right)\right]^{-1}\widehat{\log\left(\xi_t^0\right)}\hat{\hat{y}}_t ,
\end{aligned} \tag{A.64}
$$

$$
\begin{aligned}
(\hat{x}_t^n)^2 =& \left(\frac{\alpha}{1-\alpha}\right)^2(\hat{\hat{r}}_t^I)^2 + \left[\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{1}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\right]^2\widehat{\log\left(\xi_t^0\right)}^2 \\
&- 2\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{1}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left(\frac{\alpha}{1-\alpha}\right)\widehat{\log\left(\xi_t^0\right)}\hat{\hat{r}}_t^I ,
\end{aligned} \tag{A.65}
$$

$$
\begin{aligned}
\hat{\hat{x}}_t\hat{x}_t^n =& -\left(\frac{\alpha}{1-\alpha}\right)\left[1+\alpha\left(\tfrac{\eta+1}{\eta}\right)\right]^{-1}\hat{\hat{y}}_t\hat{\hat{r}}_t^I \\
&+ \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{1}{\eta+1}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left[1+\alpha\left(\tfrac{\eta+1}{\eta}\right)\right]^{-1}\widehat{\log\left(\xi_t^0\right)}\hat{\hat{y}}_t .
\end{aligned} \tag{A.66}
$$

We can now rearrange and take expectations of the second-order approximation to utility (A.54) to obtain:

$$
\mathbb{E}\left[\frac{U_t - U}{U_x X}\right] = \mathbb{E}\left[\hat{\hat{x}}_t + \hat{x}_t^n - \frac{1}{2}\left[(\hat{\hat{x}}_t)^2 + (\hat{x}_t^n)^2 + 2\hat{\hat{x}}_t\hat{x}_t^n\right] - \left[\hat{n}_t + \frac{1}{2\eta}\hat{n}_t^2\right]\right] + \text{h.o.t.} , \tag{A.67}
$$

where in equilibrium $U_x X = 1$.

Combining the linearized model solution for the output gap ($\hat{\hat{y}}_t$) and inflation ($\hat{\pi}_t$) from (A.53), along with equations (A.58) and (A.61)–(A.66), into (A.67), we obtain:

$$\mathbb{E}\left[\frac{U_t-U}{U_x X}\right] = -\Lambda_r \mathbb{E}\left[\hat{r}_t^I\right] + \Lambda_\xi \mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}\right] - \frac{\Lambda_{rr}}{2}\mathbb{E}\left[(\hat{r}_t^I)^2\right] + \frac{\Lambda_{\xi\xi}}{2}\mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}^2\right] + \Lambda_{r\xi}\mathbb{E}\left[\widehat{\log\left(\xi_t^0\right)}\hat{r}_t^I\right], \tag{A.68}$$

where the constants are defined as:

$$\Xi_1 = \left(\frac{\varepsilon}{1-\alpha}\right)\left[1+\varepsilon\left(\frac{1-\alpha}{\eta+\alpha}\right)\right]\left(\frac{\theta}{(1-\theta)(1-\theta\beta)}\right),$$

$$\Xi_2 = \left[1+\alpha\left(\frac{\eta+1}{\eta}\right)^2\right]\left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-2},$$

$$\Xi_3 = \left(\frac{\alpha}{1-\alpha}\right)\left[1+\alpha\left(\frac{\eta+1}{\eta}\right)\right]^{-1},$$

$$\Xi_4 = \left(\frac{\eta+1}{\eta}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\Xi_3,$$

$$\Xi_5 = \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\left[1-\left(1+\frac{1}{1+\eta}\right)\left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\xi^0}{1-\xi^0}\right)\right],$$

$$\Xi_6 = \left(\frac{\alpha}{1-\alpha}\right)^2\left(\frac{1}{1+\eta}\right)\left(\frac{\xi^0}{1-\xi^0}\right),$$

$$\Lambda_r = \left(\frac{\alpha}{1-\alpha}\right),$$

$$\Lambda_\xi = \left(\frac{\alpha}{1-\alpha}\right)\left(\frac{\xi^0}{1-\xi^0}\right),$$

$$\Lambda_{rr} = \left(\frac{\alpha}{1-\alpha}\right)^2 + \Xi_1(\Psi_{21})^2 + \Xi_2(\Psi_{11})^2 - 2\Xi_3\Psi_{11},$$

$$\Lambda_{\xi\xi} = \Xi_5 + 2\Xi_4\Psi_{12} - \left[\Xi_1(\Psi_{22})^2 + \Xi_2(\Psi_{12})^2\right],$$

$$\Lambda_{r\xi} = \Xi_6 + \Xi_3\Psi_{12} + \Xi_4\Psi_{11} - \left[\Xi_1\Psi_{21}\Psi_{22} + \Xi_2\Psi_{11}\Psi_{12}\right].$$

**Banking variables** A second-order approximation of equations (A.39) and (A.40) is:

$$\begin{aligned}\hat{r}_t^I = {} & \sum_{i=1}^N s^i\left[\hat{r}_t^{I,i} - \frac{\widehat{\log(a_t^i)}}{\sigma-1}\right] - \left(\frac{\sigma-1}{2}\right)\sum_{i=1}^N s^i\left[\left(\frac{1}{\sigma-1}\right)^2\widehat{\log(a_t^i)}^2 + \left(\hat{r}_t^{I,i}\right)^2 - 2\left(\frac{1}{\sigma-1}\right)\widehat{\log(a_t^i)}\hat{r}_t^{I,i}\right] \\ & + \left(\frac{\sigma-1}{2}\right)\sum_{i=1}^N\sum_{j=1}^N s^i s^j\left[\left(\frac{1}{\sigma-1}\right)^2\widehat{\log(a_t^i)}\widehat{\log(a_t^j)} + \hat{r}_t^{I,i}\hat{r}_t^{I,j} - 2\left(\frac{1}{\sigma-1}\right)\widehat{\log(a_t^i)}\hat{r}_t^{I,j}\right],\end{aligned} \tag{A.69}$$

$$\hat{r}_t^{I,i} = \left(1-\varpi_2\xi^{0i}\right)\hat{\phi}_t^i - \frac{1}{2}\kappa(\varpi_2)^2\xi^{0i}\left(1-\xi^{0i}\right)\left(\hat{\phi}_t^i\right)^2. \tag{A.70}$$

By combining equations (A.69) and (A.70), we obtain:

$$\begin{aligned}\hat{r}_t^I = {} & \sum_{i=1}^N s^i\left[\left(1-\varpi_2\xi^{0i}\right)\hat{\phi}_t^i - \frac{\widehat{\log\left(a_t^i\right)}}{\sigma-1}\right] - \left(\frac{\sigma-1}{2}\right)\sum_{i=1}^N s^i\left[\frac{\kappa}{\sigma-1}(\varpi_2)^2\xi^{0i}\left(1-\xi^{0i}\right) + (1-\varpi_2\xi^{0i})^2\right]\left(\hat{\phi}_t^i\right)^2 \\ & - \left(\frac{\sigma-1}{2}\right)\sum_{i=1}^N s^i\left[\left(\frac{1}{\sigma-1}\right)^2\widehat{\log\left(a_t^i\right)}^2 - 2\left(1-\varpi_2\xi^{0i}\right)\hat{\phi}_t^i\frac{\widehat{\log\left(a_t^i\right)}}{\sigma-1}\right] \\ & + \left(\frac{\sigma-1}{2}\right)\sum_{i=1}^N\sum_{j=1}^N s^i s^j\left[\left(1-\varpi_2\xi^{0i}\right)\left(1-\varpi_2\xi^{0j}\right)\hat{\phi}_t^i\hat{\phi}_t^j + \left(\frac{1}{\sigma-1}\right)^2\widehat{\log\left(a_t^i\right)}\widehat{\log\left(a_t^j\right)} - 2\left(1-\varpi_2\xi^{0i}\right)\phi_t^i\frac{\widehat{\log\left(a_t^j\right)}}{\sigma-1}\right].\end{aligned} \tag{A.71}$$

It follows from the previous expressions that

$$\left(\hat{r}_t^I\right)^2 = \sum_{i=1}^N\sum_{n=1}^N s^i s^n\left[\left(1-\varpi_2\xi^{0i}\right)\left(1-\varpi_2\xi^{0n}\right)\hat{\phi}_t^i\hat{\phi}_t^n + \frac{\widehat{\log\left(a_t^i\right)}\widehat{\log\left(a_t^n\right)}}{(\sigma-1)^2} - 2\left(1-\varpi_2\xi^{0n}\right)\hat{\phi}_t^n\frac{\widehat{\log\left(a_t^i\right)}}{\sigma-1}\right]$$

The second-order approximations for $\xi_t^{0i}$, $\xi_t^0$, and $s_t^i$ are:

$$\widehat{\log\left(\xi_t^{0i}\right)} = \kappa\varpi_2\left(1-\xi^{0i}\right)\hat{\phi}_t^i - \frac{1}{2}\kappa^2(\varpi_2)^2\xi^{0i}\left(1-\xi^{0i}\right)\left(\hat{\phi}_t^i\right)^2,$$

$$\begin{aligned}\widehat{\log\left(\xi_t^{0}\right)} = &\sum_{i=1}^{N} s^i\left(\frac{\xi^{0i}}{\xi^0}\right)\left[\widehat{\log\left(s_t^i\right)}+\widehat{\log\left(\xi_t^{0i}\right)}\right]+\frac{1}{2}\sum_{i=1}^{N} s^i\left(\frac{\xi^{0i}}{\xi^0}\right)\left[\widehat{\log\left(s_t^i\right)}^2+\widehat{\log\left(\xi_t^{0i}\right)}^2+2\widehat{\log\left(s_t^i\right)}\widehat{\log\left(\xi_t^{0i}\right)}\right]\\&-\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} s^i\left(\frac{\xi^{0i}}{\xi^0}\right)s^j\left(\frac{\xi^{0j}}{\xi^0}\right)\left[\widehat{\log\left(\xi_t^{0i}\right)}\widehat{\log\left(\xi_t^{0j}\right)}+\widehat{\log\left(s_t^i\right)}\widehat{\log\left(s_t^j\right)}+2\widehat{\log\left(\xi_t^{0i}\right)}\widehat{\log\left(s_t^j\right)}\right],\end{aligned}$$

$$\widehat{\log\left(s_t^i\right)} = (\sigma-1)\left[\hat{\bar{r}}_t^I - \left[\hat{\bar{r}}_t^{I,i} - \frac{\widehat{\log\left(a_t^i\right)}}{\sigma-1}\right]\right]. \tag{A.72}$$

Additionally, the second-order approximations of equations (A.41) and (A.42) are:

$$\begin{aligned}\hat{\phi}_t^i = &\sum_{n=1}^{N}\lambda^{ni}\left[\hat{u}_t^{T,n}+\hat{u}_t^{I,ni}\right]+\frac{\kappa}{2}\sum_{n=1}^{N}\sum_{j\neq n}\lambda^{ni}\lambda^{ji}\left[\hat{u}_t^{T,n}\hat{u}_t^{T,j}+\hat{u}_t^{I,ni}\hat{u}_t^{I,ji}+2\hat{u}_t^{T,j}\hat{u}_t^{I,ni}\right] \\&-\frac{\kappa}{2}\sum_{n=1}^{N}\lambda^{ni}(1-\lambda^{ni})\left[\left(\hat{u}_t^{T,n}\right)^2+\left(\hat{u}_t^{I,ni}\right)^2+2\hat{u}_t^{T,n}\hat{u}_t^{I,ni}\right],\end{aligned} \tag{A.73}$$

$$\widehat{\log\left(a_t^n\right)} = \hat{u}_t^{a,n} - \sum_{i=1}^{N} a^i\hat{u}_t^{a,i} - \frac{1}{2}\sum_{i=1}^{N} a^i\left(1-a^i\right)\left(\hat{u}_t^{a,i}\right)^2 + \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i} a^i a^j\hat{u}_t^{a,i}\hat{u}_t^{a,j}.$$

It follows from the previous expressions that:

$$\hat{\phi}_t^i\hat{\phi}_t^n = \sum_{j=1}^{N}\sum_{k=1}^{N}\lambda^{ji}\lambda^{kn}\left[\hat{u}_t^{T,j}\hat{u}_t^{T,k}+\hat{u}_t^{I,ji}\hat{u}_t^{I,kn}+\hat{u}_t^{T,j}\hat{u}_t^{I,kn}+\hat{u}_t^{T,k}\hat{u}_t^{I,ji}\right],$$

$$\widehat{\log\left(a_t^i\right)}\widehat{\log\left(a_t^n\right)} = \hat{u}_t^{a,i}\hat{u}_t^{a,n} - \sum_{j=1}^{N} a^j\hat{u}_t^{a,n}\hat{u}_t^{a,j} - \sum_{k=1}^{N} a^k\hat{u}_t^{a,i}\hat{u}_t^{a,k} + \sum_{j=1}^{N}\sum_{k=1}^{N} a^j a^k\hat{u}_t^{a,j}\hat{u}_t^{a,k}. \tag{A.74}$$

By combining equations (A.68) and (A.71)-(A.72), we derive an expression for the household's utility:

$$\begin{aligned}\mathbb{E}\left[\frac{U_t-U}{U_xX}\right] = &-\sum_{i=1}^{N} s^i\aleph_{0,i}\mathbb{E}\left[\hat{\phi}_t^i\right]+\sum_{i=1}^{N} s^i\aleph_{1,i}\mathbb{E}\left[\widehat{\log(a_t^i)}\right]+\frac{1}{2}\sum_{i=1}^{N} s^i\aleph_{2,i}\mathbb{E}\left[\left(\hat{\phi}_t^i\right)^2\right]-\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} s^i s^j\aleph_{3,ij}\mathbb{E}\left[\hat{\phi}_t^i\hat{\phi}_t^j\right]\\&+\frac{1}{2}\sum_{i=1}^{N} s^i\aleph_4\mathbb{E}\left[\widehat{\log(a_t^i)}^2\right]-\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} s^i s^j\aleph_{5,ij}\mathbb{E}\left[\widehat{\log(a_t^i)}\widehat{\log(a_t^j)}\right]\\&-\sum_{i=1}^{N} s^i\aleph_{6,i}\mathbb{E}\left[\hat{\phi}_t^i\widehat{\log(a_t^i)}\right]+\sum_{i=1}^{N}\sum_{j=1}^{N} s^i s^j\aleph_{7,ij}\mathbb{E}\left[\hat{\phi}_t^i\widehat{\log(a_t^j)}\right],\end{aligned}$$

where the constants are defined as

$$\aleph_{0,i} = \Lambda_r(1-\varpi_2\xi^{0i})-(\sigma-1)\Lambda_\xi\left[\left(1-\frac{\xi^{0i}}{\xi^0}\right)(1-\varpi_2\xi^{0i})+\left(\frac{\xi^{0i}}{\xi^0}\right)\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0i})\right],$$

$$\aleph_{1,i} = \frac{\Lambda_r}{\sigma-1}-\Lambda_\xi\left(1-\frac{\xi^{0i}}{\xi^0}\right),$$

$$\begin{aligned}\aleph_{2,i} = &(\sigma-1)\Lambda_r\left[\frac{\kappa(\varpi_2)^2}{\sigma-1}\xi^{0i}(1-\xi^{0i})+(1-\varpi_2\xi^{0i})\right]\\&-(\sigma-1)^2\Lambda_\xi\left[\left(\frac{\xi^{0i}}{\xi^0}\right)\left[2\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0i})(1-\varpi_2\xi^{0i})-\frac{\kappa^2(\varpi_2)^2}{(\sigma-1)^2}(1-\xi^{0i})(1-2\xi^{0i})\right]+\left(1-\frac{\xi^{0i}}{\xi^0}\right)\frac{\kappa(\varpi_2)^2}{\sigma-1}\xi^{0i}(1-\xi^{0i})\right],\end{aligned}$$

$$\aleph_{3,ij} = [(\sigma-1)\Lambda_r + \Lambda_{rr}](1-\varpi_2\xi^{0i})(1-\varpi_2\xi^{0j}) - (\sigma-1)^2\Lambda_\xi\left[1-\left(\frac{\xi^{0i}}{\xi^0}\right)\left(\frac{\xi^{0j}}{\xi^0}\right)\right](1-\varpi_2\xi^{0i})(1-\varpi_2\xi^{0j})$$
$$- (\sigma-1)^2\Lambda_\xi\left(\frac{\xi^{0i}}{\xi^0}\right)\left(\frac{\xi^{0j}}{\xi^0}\right)\left[2\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0i})(1-\varpi_2\xi^{0j}) - \frac{\kappa^2(\varpi_2)^2}{(\sigma-1)^2}(1-\xi^{0i})(1-\xi^{0j})\right]$$
$$- (\sigma-1)^2\Lambda_{\xi\xi}\left[\left(1-\frac{\xi^{0i}}{\xi^0}\right)(1-\varpi_2\xi^{0i}) + \left(\frac{\xi^{0i}}{\xi^0}\right)\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0i})\right]\left[\left(1-\frac{\xi^{0j}}{\xi^0}\right)(1-\varpi_2\xi^{0j}) + \left(\frac{\xi^{0j}}{\xi^0}\right)\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0j})\right]$$
$$- 2(\sigma-1)\Lambda_{r\xi}\left[\left(1-\frac{\xi^{0i}}{\xi^0}\right)(1-\varpi_2\xi^{0i}) + \left(\frac{\xi^{0i}}{\xi^0}\right)\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0i})\right]\left(1-\varpi_2\xi^{0j}\right),$$
$$\aleph_4 = \frac{\Lambda_r}{\sigma-1},$$
$$\aleph_{5,ij} = \left(\frac{1}{\sigma-1}\right)\left[\Lambda_r + \frac{\Lambda_{rr}}{\sigma-1}\right] - \Lambda_{\xi\xi}\left(1-\frac{\xi^{0i}}{\xi^0}\right)\left(1-\frac{\xi^{0j}}{\xi^0}\right) - 2\frac{\Lambda_{r\xi}}{\sigma-1}\left(1-\frac{\xi^{0i}}{\xi^0}\right) - \Lambda_\xi\left[1-\left(\frac{\xi^{0i}}{\xi^0}\right)\left(\frac{\xi^{0j}}{\xi^0}\right)\right],$$
$$\aleph_{6,i} = \Lambda_r(1-\varpi_2\xi^{0i}) - \Lambda_\xi\left(\frac{\xi^{0i}}{\xi^0}\right)\kappa\varpi_2(1-\xi^{0i}),$$
$$\aleph_{7,ij} = (\sigma-1)\Lambda_\xi\left[\left(1-\frac{\xi^{0i}}{\xi^0}\right)(1-\xi^{0i})\frac{\kappa\varpi_2}{\sigma-1} - \left[1-\left(\frac{\xi^{0i}}{\xi^0}\right)\left(\frac{\xi^{0j}}{\xi^0}\right)\right](1-\varpi_2\xi^{0i})\right] + \Lambda_r(1-\varpi_2\xi^{0i})$$
$$+ \Lambda_{rr}\left(\frac{1-\varpi_2\xi^{0i}}{\sigma-1}\right) - (\sigma-1)\Lambda_{\xi\xi}\left[\left(1-\frac{\xi^{0i}}{\xi^0}\right)(1-\varpi_2\xi^{0i}) + \left(\frac{\xi^{0i}}{\xi^0}\right)\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0i})\right]\left(1-\frac{\xi^{0j}}{\xi^0}\right)$$
$$- \Lambda_{r\xi}\left[\left(2-\frac{\xi^{0i}}{\xi^0}-\frac{\xi^{0j}}{\xi^0}\right)(1-\varpi_2\xi^{0i}) + \left(\frac{\xi^{0i}}{\xi^0}\right)\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^{0i})\right].$$

In the baseline case, where all banks have equal access to the central bank, $\xi^{0i} = \xi^0$, $\forall j$, the above expression simplifies as follows:

$$\mathbb{E}\left[\frac{U_t - U}{U_x X}\right] = -\aleph_0\sum_{i=1}^N s^i\mathbb{E}\left[\hat{\phi}_t^i\right] + \aleph_1\sum_{i=1}^N s^i\mathbb{E}\left[\widehat{\log(a_t^i)}\right] + \frac{\aleph_2}{2}\sum_{i=1}^N s^i\mathbb{E}\left[\left(\hat{\phi}_t^i\right)^2\right] - \frac{\aleph_3}{2}\sum_{i=1}^N\sum_{j=1}^N s^i s^j\mathbb{E}\left[\hat{\phi}_t^i\hat{\phi}_t^j\right] + \frac{\aleph_4}{2}\sum_{i=1}^N s^i\mathbb{E}\left[\widehat{\log(a_t^i)}^2\right]$$
$$- \frac{\aleph_5}{2}\sum_{i=1}^N\sum_{j=1}^N s^i s^j\mathbb{E}\left[\widehat{\log(a_t^i)}\widehat{\log(a_t^j)}\right] - \aleph_6\sum_{i=1}^N s^i\mathbb{E}\left[\hat{\phi}_t^i\widehat{\log(a_t^i)}\right] + \aleph_7\sum_{i=1}^N\sum_{j=1}^N s^i s^j\mathbb{E}\left[\hat{\phi}_t^i\widehat{\log(a_t^j)}\right], \tag{A.75}$$

where the constants are defined as

$$\aleph_0 = \Lambda_r(1-\varpi_2\xi^0) - \Lambda_\xi\kappa\varpi_2(1-\xi^0),$$
$$\aleph_1 = \frac{\Lambda_r}{\sigma-1},$$
$$\aleph_2 = (\sigma-1)\Lambda_r\left[\frac{\kappa(\varpi_2)^2}{\sigma-1}\xi^0(1-\xi^0) + (1-\varpi_2\xi^0)\right] - (\sigma-1)^2\Lambda_\xi\left[2\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^0)(1-\varpi_2\xi^0) - \frac{\kappa^2(\varpi_2)^2}{(\sigma-1)^2}(1-\xi^0)(1-2\xi^0)\right],$$
$$\aleph_3 = [(\sigma-1)\Lambda_r + \Lambda_{rr}](1-\varpi_2\xi^0)^2 - \Lambda_{\xi\xi}\kappa^2(\varpi_2)^2(1-\xi^0)^2 - 2\Lambda_{r\xi}\kappa\varpi_2(1-\xi^0)(1-\varpi_2\xi^0)$$
$$- (\sigma-1)^2\Lambda_\xi\left[2\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^0)(1-\varpi_2\xi^0) - \frac{\kappa^2(\varpi_2)^2}{(\sigma-1)^2}(1-\xi^0)^2\right],$$
$$\aleph_4 = \frac{\Lambda_r}{\sigma-1},$$
$$\aleph_5 = \left(\frac{1}{\sigma-1}\right)\left[\Lambda_r + \frac{\Lambda_{rr}}{\sigma-1}\right],$$
$$\aleph_6 = \Lambda_r(1-\varpi_2\xi^0) - \Lambda_\xi\kappa\varpi_2(1-\xi^0),$$
$$\aleph_7 = \left[\Lambda_r + \frac{\Lambda_{rr}}{\sigma-1}\right](1-\varpi_2\xi^0) - \Lambda_{r\xi}\frac{\kappa\varpi_2}{\sigma-1}(1-\xi^0).$$

## Appendix 3.B Welfare Decomposition

**Additional Assumptions** To derive a tractable welfare expression in (A.56), we impose the following assumptions, which can also be found in Section 6.4.

*1. CES firm weights:*

$$\mathbb{E}\left[u_t^{a,i}u_t^{a,n}\right] = \begin{cases} \sigma_a^2 & \text{if } n = i\,, \\ \zeta_a\sigma_a^2 & \text{otherwise}\,. \end{cases}$$

*2. Depositor Preferences:*

$$\mathbb{E}\left[u_t^{T,i}u_t^{T,n}\right] = \begin{cases} \sigma_T^2 & \text{if } n = i\,, \\ \zeta_T\sigma_T^2 & \text{otherwise}\,. \end{cases}$$

*3. Bilateral Transaction Costs:*

$$\mathbb{E}\left[u_t^{I,ji}u_t^{I,kn}\right] = \begin{cases} 0 & \text{if } j = i \text{ or } k = n\,, \\ \sigma_I^2 & \text{if } k = j,\ n = i\,, \\ \zeta_{I,B}\sigma_I^2 & \text{if } k \neq j,\ n = i\,, \\ \zeta_{I,L}\sigma_I^2 & \text{if } k = j,\ n \neq i\,, \\ \zeta_{I,X}\sigma_I^2 & \text{otherwise}\,. \end{cases}$$

*4. Zero Cross-Correlation:*

$$\begin{aligned} \mathbb{E}\left[u_t^{I,ji}u_t^{a,k}\right] &= \mathbb{E}\left[u_t^{I,ji}u_t^{T,k}\right] \\ &= \mathbb{E}\left[u_t^{a,j}u_t^{T,k}\right] = 0\,, \quad \forall j,i,k\,. \end{aligned}$$

**Welfare Expectations** The expectations of (A.73)-(A.74) are:

$$\mathbb{E}\left[\hat{\phi}_t^i\right] = -\frac{\kappa}{2}\left[\sigma_T^2(1-\zeta_T)\left[1-H^{I,i}\right]+\sigma_I^2\left[(1-\lambda^{ii})-\left[H^{I,i}-(\lambda^{ii})^2\right]\right]-\zeta_{I,B}\sigma_I^2\left[H^{O,i}-H^{I,i}\right]\right], \tag{A.76}$$

$$\mathbb{E}\left[\widehat{\log\left(a_t^i\right)}\right] = -\frac{\sigma_a^2}{2}(1-\zeta_a)\left[1-H^a\right],$$

$$\mathbb{E}\left[\hat{\phi}_t^i\hat{\phi}_t^n\right] = \begin{cases} \sigma_T^2(1-\zeta_T)\left[\dfrac{\zeta_T}{1-\zeta_T}+H^{I,i}\right]+\sigma_I^2\left[H^{I,i}-\left(\lambda^{ii}\right)^2\right]+\zeta_{I,B}\sigma_I^2\left[H^{O,i}-H^{I,i}\right], & \text{if } n = i\,, \\ \sigma_T^2(1-\zeta_T)\left[\dfrac{\zeta_T}{1-\zeta_T}+\sum_{j=1}^N\lambda^{ji}\lambda^{jn}\right]+\zeta_{I,L}\sigma_I^2\sum_{j\neq\{i,n\}}\lambda^{ji}\lambda^{jn}+\zeta_{I,X}\sigma_I^2\left[\left(1-\lambda^{ii}\right)(1-\lambda^{nn})-\sum_{j\neq i}\lambda^{ji}\lambda^{jn}\right], & \text{otherwise}\,, \end{cases}$$

$$\mathbb{E}\left[\widehat{\log\left(a_t^i\right)}\widehat{\log\left(a_t^n\right)}\right] = \begin{cases} \sigma_a^2(1-\zeta_a)\left[1+H^a-2a^i\right], & \text{if } n = i\,, \\ \sigma_a^2(1-\zeta_a)\left[H^a-\left[a^i+a^n\right]\right], & \text{otherwise}\,, \end{cases} \tag{A.77}$$

where $H^a = \sum_{n=1}^N (a^n)^2$ is the Herfindahl index for concentration in firm loan demand, $H^{I,i} = \sum_{j=1}^N \left(\lambda^{ji}\right)^2$ is the Herfindahl index for concentration in bank $i$'s funding sources, and $H^{O,i} = \left(\lambda^{ii}\right)^2 + \left(1-\lambda^{ii}\right)^2$ is the

Herfindahl index for bank $i$'s own versus outside funding. Plugging (A.76)-(A.77) into (A.75), we obtain:

$$
\begin{aligned}
\mathbb{E}\left[\frac{U_t-U}{U_x X}\right] &= \frac{\sigma_a^2}{2}(1-\zeta_a)\left[\aleph_5\left[1-H^F\right]-[\aleph_1+\aleph_4-\aleph_5]\left[1-H^a\right]-2(\aleph_5-\aleph_4)\left[1-\sum_{i=1}^N s^i a^i\right]\right] \\
&+\frac{\sigma_T^2}{2}(1-\zeta_T)\left[\frac{\aleph_2-\zeta_T\aleph_3}{1-\zeta_T}-\aleph_3 H^F+\left[(\aleph_0\kappa-\aleph_2)+\aleph_3 H^F\right]\left[1-H^I\right]-\aleph_3\sum_{i=1}^N\sum_{n\neq i}\sum_{j=1}^N s^i s^n\lambda^{ji}\lambda^{jn}\right] \\
&+\frac{\sigma_I^2}{2}\left[\aleph_0\kappa\left(1-\lambda^{Avg}\right)-\left[(\aleph_0\kappa-\aleph_2)+\aleph_3 H^F\right]\sqsupset^I H^I\right] \\
&-\frac{\sigma_I^2}{2}\zeta_{I,B}\left[(\aleph_0\kappa-\aleph_2)+\aleph_3 H^F\right]\left[H^O-H^I\right] \\
&-\frac{\sigma_I^2}{2}\zeta_{I,L}\aleph_3\sum_{i=1}^N\sum_{n\neq i}\sum_{j\neq\{i,n\}} s^i s^n\lambda^{ji}\lambda^{jn} \\
&-\frac{\sigma_I^2}{2}\zeta_{I,X}\aleph_3\sum_{i=1}^N\sum_{n\neq i} s^i s^n\left[(1-\lambda^{ii})(1-\lambda^{nn})-\sum_{j\neq i}\lambda^{ji}\lambda^{jn}\right],
\end{aligned}
$$

$$
\text{where:}\quad \omega^i=\frac{(\aleph_0\kappa-\aleph_2)s^i+\aleph_3(s^i)^2}{(\aleph_0\kappa-\aleph_2)+\aleph_3 H^F},\quad H^I=\sum_{i=1}^N\omega^i H^{I,i},\quad H^O=\sum_{i=1}^N\omega^i H^{O,i},
$$

$$
\sqsupset^I=\sum_{i=1}^N\omega^i\left(\frac{H^{I,i}}{H^I}\right)\left(\frac{H^{I,i}-\left(\lambda^{ii}\right)^2}{H^{I,i}}\right),\quad \lambda^{Avg}=\sum_{i=1}^N s^i\lambda^{ii}.
$$

The variable $\sqsupset^I$ represents the share of a bank's funding concentration attributed to outside sources. Finally, we derive an expression for the dynamic gains from trade as follows:

$$
\begin{aligned}
&\mathbb{J}=\mathbb{J}^{ss}+\frac{1}{2}\left[\sigma_a^2\mathfrak{I}^a+\sigma_T^2\mathfrak{I}^T+\sigma_I^2\mathfrak{I}^I\right], \\
\text{where:}\quad &\mathfrak{I}^a=(1-\zeta_a)\left[\aleph_5\left[H^{F,AU}-H^F\right]-2(\aleph_5-\aleph_4)\sum_{i=1}^N\left(s^{i,AU}-s^i\right)a^i\right], \\
&\mathfrak{I}^T=(1-\zeta_T)\left[\aleph_3\left[H^{F,AU}-H^F\right]+\left[(\aleph_0\kappa-\aleph_2)+\aleph_3 H^F\right]\left[1-H^I\right]-\aleph_3\sum_{i=1}^N\sum_{n\neq i}\sum_{j=1}^N s^i s^n\lambda^{ji}\lambda^{jn}\right], \\
&\mathfrak{I}^I=\aleph_0\kappa\left(1-\lambda^{Avg}\right)-\left[(\aleph_0\kappa-\aleph_2)+\aleph_3 H^F\right]\sqsupset^I H^I \\
&\qquad-\zeta_{I,B}\left[(\aleph_0\kappa-\aleph_2)+\aleph_3 H^F\right]\left[H^O-H^I\right] \\
&\qquad-\zeta_{I,L}\aleph_3\sum_{i=1}^N\sum_{n\neq i}\sum_{j\neq\{i,n\}} s^i s^n\lambda^{ji}\lambda^{jn} \\
&\qquad-\zeta_{I,X}\aleph_3\sum_{i=1}^N\sum_{n\neq i} s^i s^n\left[(1-\lambda^{ii})(1-\lambda^{nn})-\sum_{j\neq i}\lambda^{ji}\lambda^{jn}\right].
\end{aligned}
$$

Under the assumptions of no central bank response to cyclical credit spread fluctuations, $\varpi_2=0$, the expression for multiplier $\mathfrak{J}^I$ becomes the one featured in the main text:

$$\begin{aligned}
\mathfrak{J}^I = & \overbrace{\Theta_0 \sum_{n=1}^{N} s^n \left[1-\lambda^{nn}\right]}^{\text{Openness}} + \overbrace{\Theta_0 \left[\left(\frac{\sigma-1}{\kappa}\right) - \left[1+\Theta_1 H^F\right]\right] \sum_{n=1}^{N} \omega^n \left[H^{I,n} - \left(\lambda^{nn}\right)^2\right]}^{\text{Exposure/Diversification}} \\
& \left.\begin{aligned}
&+ \zeta_{I,B}\Theta_0 \left[\left(\frac{\sigma-1}{\kappa}\right) - \left[1+\Theta_1 H^F\right]\right] \sum_{n=1}^{N} \omega^n \left[(1-\lambda^{nn})^2 - \left[H^{I,n} - (\lambda^{nn})^2\right]\right] \\
&- \zeta_{I,L}\Theta_0\Theta_1 \sum_{i=1}^{N} \sum_{n\neq i} \sum_{\substack{j\neq i \\ j\neq n}} s^i s^n \lambda^{ji}\lambda^{jn} \\
&- \zeta_{I,X}\Theta_0\Theta_1 \sum_{i=1}^{N} \sum_{n\neq i} s^i s^n \left[(1-\lambda^{ii})(1-\lambda^{nn}) - \sum_{j\neq i} \lambda^{ji}\lambda^{jn}\right]
\end{aligned}\right\} \text{covariance terms}\,.
\end{aligned}$$

where the constants are defined as:

$$\begin{aligned}
\Theta_0 &= \Lambda_r \kappa, \\
\Theta_1 &= \left(\frac{\sigma-1}{\kappa}\right) + \frac{\Lambda_{rr}}{\Theta_0}.
\end{aligned}$$

# Appendix 4 Estimation Appendix

## Appendix 4.A Exposure to Euro Crisis

We conduct an additional robustness check for out main specification by constructing an indirect exposure measure of German bank to the Euro crisis. We define our exposure measure similar to equation (18), according to

$$Exposure_{t_0}^{PIIGS,n} = \frac{1}{L_{t_0}^n + \sum_{n \neq i} M_{t_0}^{ni}} \sum_{i \neq n}^{N} \frac{M_{t_0}^{in}}{\sum_{i' \neq n}^{N} M_{t_0}^{i'n}} \mathscr{M}_{t_0}^{i,PIIGS}$$

$\mathscr{M}_{t_0}^{i,PIIGS}$ represents the value of assets that bank $n$'s lender $i$ reports with households and firms in the so-called PIIGS countries (Portugal, Ireland, Italy, Greece and Spain) in $t_0$ in the External Position of MFIs (AUSTA) dataset. As the base period $t_0$, we choose 2006Q1, – the same period as the main specification – six quarters before the first signs of the financial crisis in 2007Q2 and ten quarters before the collapse of Lehman Brothers in 2008Q3. We drop the Financial Crisis period 2008Q4–2009Q3. Our post-crisis period is then 2009Q4-2012Q4 which ends shortly after Mario Draghi's famous speech.

The second row of Table 2 summarizes the Euro Crisis exposure measure. The average exposure among the 165 banks in the sample is around 3%, 1.7% at the 25th percentile, and over 7% at the 95th percentile.

As reported in columns 1 and 4 Table A.3, interest rates of banks with average exposure increase by around 6 basis points, while final loans contract by around 3.5% in column 5. The own share increases by around 4.5% as shown in column 6. The corresponding estimates of $\sigma$ and $\kappa$ are around 60; however, the first stage is weak with an F-Statistic of 7.3. Thus, we need to regard the relatively high estimates of the elasticities with caution. Table A.7 confirms that there are no significant pre-trends in all three variables when controls are included in columns 4-6.

Table A.7: Pre-trends for Estimation of $\sigma$ and $\kappa$ with indirect Euro Crisis Exposure

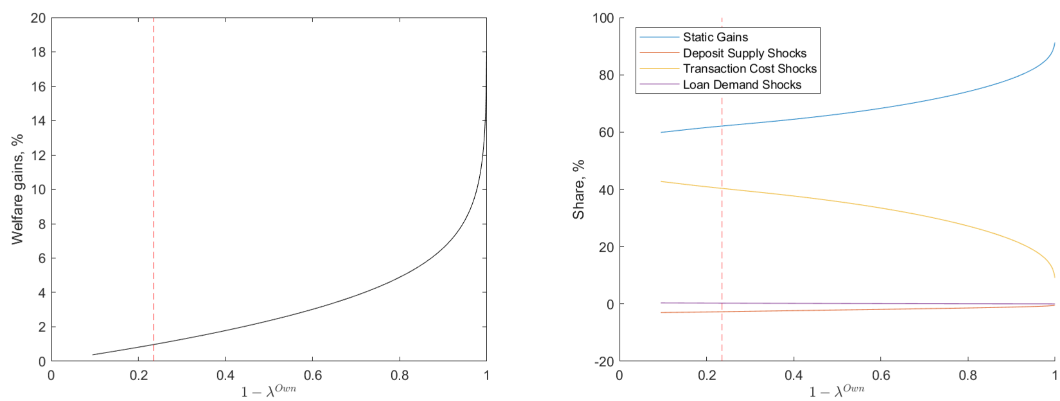
| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Loan Rate | Loans | Own Share | Loan Rate | Loans | Own Share |
| $Exposure_{t_0} \times Post_{t_0}$ | 0.000 | -0.002 | 0.002* | 0.000 | -0.001 | 0.001 |
| | (0.000) | (0.003) | (0.001) | (0.000) | (0.004) | (0.001) |
| Observations | 2297 | 2297 | 2297 | 2283 | 2283 | 2283 |
| R-squared | 0.940 | 0.994 | 0.943 | 0.952 | 0.995 | 0.944 |
| Mean Exposure | 3.094 | 3.094 | 3.094 | 3.150 | 3.150 | 3.150 |
| Controls | no | no | no | yes | yes | yes |

The regression compares outcomes between 2004$Q4$ to 2005$Q4$ and the pre-period in the main regression (2006$Q1$ until 2008$Q1$) for banks that were more or less indirectly exposed to the Euro Crisis. Instrument is based on exposure to non-MFI assets in Portugal, Ireland, Italy, Greece and Spain (PIIGS) relative to all assets in 2006Q1. Controls include direct non-MFI asset exposure to PIIGS, leverage, repayment rates on loans, and loan shares of non-MFI and household loans, each broken down into maturities of less than 1 year, between 1 and 5 years, and more than 5 years, as well as separate shares for secured and unsecured mortgages. All regressions include bank fixed-effects and quarter fixed-effects. Standard errors are clustered at the level of the bank group-quarter. Source: Research Data and Service Centre (RDSC) of Deutsche Bundesbank, AUSTA, BISTA, VJKRE, ZISTA, 2004m12 - 2008m3, own calculations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

# Appendix 5 Quantification Appendix

## Appendix 5.A Counterfactual trade openess

Figure A.6 presents gains from trade across counterfactual levels of interbank market openness. We compute these gains by progressively and proportionaly scaling interbank transaction costs $d^{ni}$ up or down within the range $[1, +\infty)$. The endpoints correspond to zero transaction costs and infinite transaction costs (autarky), respectively.[4] Panel (a) shows that gains from trade range from 0.97% of consumption per quarter under the current regime to a theoretical maximum of 14% when trade costs are fully eliminated ($d^{ni} = 1$ for all $n$, $i$). Panel (b) plots the share of dynamic gains explained by each component of equation (15). Steady-state gains and lower interbank volatility are the two main sources of welfare, accounting for about 60% and 40% of total gains, respectively. Gains from lower volatility decline rapidly as banks increase interbank participation, indicating that most diversification benefits are realized at early stages of integration.




(a) Gains from trade

(b) Decomposition of gains

Figure A.6: Gains from interbank trade. Panel A.6a presents the gains from trade under the assumption of a proportional reduction or increase in transaction costs between banks. Panel A.6b shows the contribution to the gains from trade of the different components in equation (15). A dashed red line marks the current level of interbank integration.

## Appendix 5.B Lender-of-Last-Resort and the Business Cycle

The gains from the extended lender-of-last-resort (LoLR) policy introduced in equations (A.15) and (A.16) can be decomposed into gains from short-term liquidity provision (arising from idiosyncratic intra-quarter funding shocks, as in the main text, see Table 8) and gains associated with the cyclical fluctuations in funding costs (as captured by the banks' credit spreads across quarters) when $\varpi_2 \neq 0$.

Figure A.7 quantifies the gains associated with a cyclical response in liquidity provision within the

[4] Welfare gains under alternative non-proportional integration patterns are not considered in this exercise but can be computed with minor adjustments to the model.

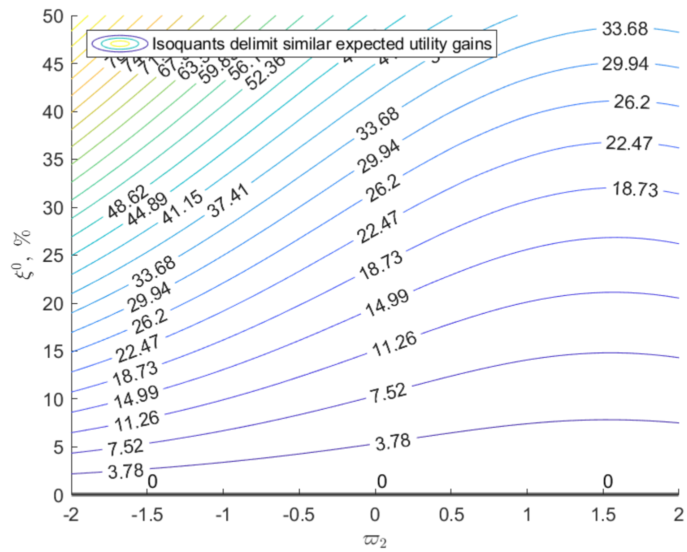



Figure A.7: Gains from trade for alternative lender-of-last-resort (LoLR) policy parameter calibrations, own calculations. The y-axis displays the percentage participation of the central bank in the interbank market, while the x-axis represents the responsiveness of the penalty rate to deviations of funding costs from the steady state. Isoquants exhibit constant levels of welfare gains (in percentage) across the parameter space.

policy parameter space delineated by the pairs $\{\xi^0, \varpi_2\}$. In this context, the isoquants demarcate equivalent welfare gains for different parameter combinations. Two significant observations emerge from the figure. First, welfare gains exhibit a monotonic increase with the share of liquidity provided by the lender-of-last-resort (LoLR), $\xi^0$.[5] Second, welfare is enhanced (albeit mildly) when the borrowing penalty imposed by the LoLR exhibits a positive correlation with the credit spread of the banks, as captured by a negative $\varpi_2$. Although this result appears counterintuitive, it follows from interbank transaction shocks functioning as negative supply shocks to the economy's capacity to produce loans (and hence, capital) within the context of our model. Hence, the interaction between interbank transaction shocks and price rigidity creates a positive deviation of the output gap that the lending facility can optimally combat by raising the borrowing penalty. It is worth noting that a more sophisticated model with built-in amplification mechanisms should potentially be able to reverse this result and produce beneficial countercyclical responses to the credit spread, provided it can generate negative output gap deviations in response to negative supply shocks.[6]

[5]As previously discussed in Section 5.2, it is crucial to interpret this result cautiously, however, as the model does not incorporate any costs arising from the provision of liquidity by the central bank (e.g., resource misallocation, monitoring costs, moral hazard, etc.). While such costs may be relatively small at lower levels of intervention, they are likely to be relevant in determining the true gains at some of the more extreme points considered in the figure.

[6]The reference point chosen for the second-order welfare approximation is also important. Throughout this paper, we maintained a neutral stance regarding the nature of interbank market shocks, assuming that they constitute a component of the "efficient" economy and, consequently, affect the natural level of output, denoted as $y_t^n$. Conversely, if these shocks were excluded from the efficient economy (for instance, by attributing their origins to market inefficiencies or irrational behavior), it would be possible to generate negative deviations in the output gap without further modifications to the model.

## Appendix 5.C Lender-of-Last-Resort Expansion

This section presents a counterfactual exercise that studies how the welfare gains from LoLR policies vary with the set of banks granted discount window access. Our procedure is as follows. We first rank the monetary financial institutions (MFIs) in our sample by balance sheet size. We then compute households' utility under the counterfactual in which no MFI has access to central bank credit. Next, we gradually expand discount window access in batches of roughly forty banks, proceeding either from the smallest to the largest banks or in the reverse order.

As shown in Figure A.8a, extending discount window access to a broader set of banks always raises welfare. However, most of the gains from LoLR policies come from granting access to the largest MFIs in the sample: about 75% of the total gains are accounted for by the first one hundred largest MFIs. This finding is consistent with the well-known idea that some institutions are "too big to fail." Changes in the interbank borrowing conditions of the largest banks can generate aggregate fluctuations and therefore account for most of the welfare gains associated with the central bank's lender-of-last-resort policy.

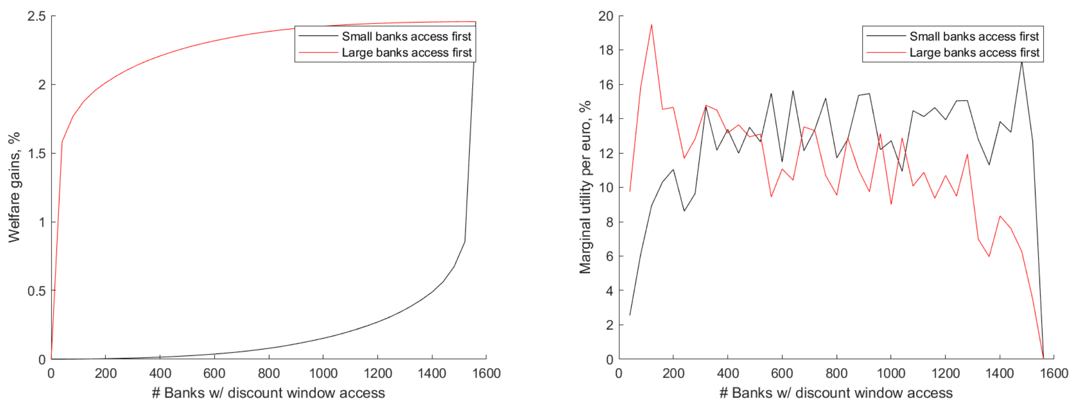
One caveat is that larger banks also tend to require more liquidity when they access the discount window. A more informative metric for evaluating the effectiveness of liquidity provision across participants is therefore the marginal welfare gain per euro of liquidity provided. We define this object as

$$\mathbb{W}^{(j)} \equiv \frac{\mathbb{G}^{(j)} - \mathbb{G}^{(j-1)}}{M^{0(j)} - M^{0(j-1)}} \,,$$

where $\mathbb{G}^{(j)}$ and $M^{0(j)}$ denote, respectively, the expected gains from LoLR and the total steady-state liquidity provided by the central bank after granting discount window access up to, and including, batch $j$ of banks. Figure A.8b reports a normalized version of this measure, $\frac{\mathbb{W}^{(j)}}{\mathbb{W}^{(\text{all})}} - 1$, which expresses the marginal return per euro of batch $j$ relative to the marginal return of the final batch that completes coverage of all banks, denoted by the superscript (all).

The red line in Figure A.8b shows that the marginal welfare gain per euro of liquidity provided to the largest banks is up to 20% higher than that of the final batch of smaller banks. Under the alternative expansion order, shown by the black line, the first small banks to gain access generate marginal welfare returns around 2% above those of the full-sample benchmark. These gains relative to the benchmark reflect the advantage of being early entrants. As progressively larger banks are added, the marginal gains from liquidity provision rise further, reaching up to 14% above the benchmark. These results are broadly consistent with the central role of large banks in the interbank network, as discussed earlier in relation to the number of unique connections shown in Figure A.2. At the same time, they suggest that the order of entry, and more generally the breadth of LoLR coverage, also matters. Because banks compete with one another in supplying credit to the private sector, the first institutions to receive LoLR credit can generate larger marginal welfare gains per euro regardless of their size.

More speculatively, our findings raise the possibility that extending discount window access to financial institutions that have traditionally been excluded, such as investment funds and insurers, could generate nontrivial welfare gains. We view this as a broader conjecture suggested by the mechanisms in our model, and leave a careful evaluation of it to future research.




(a) Gains from LoLR

(b) Marginal Gains

Figure A.8: Welfare gains from expanded discount window access, own calculations. The x-axis reports the number of banks with access to the discount window in the counterfactual scenario. Banks are ordered by balance sheet size, with smaller banks receiving access either first (black line) or last (red line). Due to confidentiality constraints, the discount window is expanded in increments of approximately forty banks at a time. Panel A.8a shows welfare gains, in percent, relative to the no-access counterfactual. Panel A.8b reports the relative marginal utility per euro, in percent, compared with the final group of banks granted access.